

Protein

INTRODUCTION

Dietary crude protein (CP) is defined as nitrogen (N) $\times 6.25$, assuming feedstuffs average 16 g of N per 100 g of true protein (TP). Chemistry of feed CP is detailed in NRC (2001) and Chapter 18. Feedstuffs vary widely in their relative proportions of protein and nonprotein N (NPN), in the rate and extent of ruminal degradation of protein, and in the intestinal digestibility and amino acid (AA) composition of rumen-undegradable protein (RUP). Metabolizable protein (MP) supply, defined as TP digested postruminally, should be used to describe and assess protein nutrition. However, intestinally absorbed AAs, and not protein per se, are metabolized. Since the NRC (2001) derivation of optimal lysine (Lys) and methionine (Met) concentrations in MP, an effort has been made to develop models to balance dairy rations for AA, as done for swine and poultry. Absorbed AAs are vital to the maintenance, growth, reproduction, and lactation of dairy cattle. Although AAs are used principally as building blocks of proteins, they are also involved in many other metabolic functions. For example, AAs other than leucine (Leu) and Lys can serve as precursors for gluconeogenesis, and all can be converted to fatty acids (FAs) or serve as sources of metabolic energy when oxidized to CO_2 ; Met is converted to S-adenosylmethionine, which is directly involved in hundreds of reactions. Because of the central role of proteins in cellular function, protein synthesis is regulated, and AAs have been shown to exert control through intracellular signaling pathways, including the mammalian target of rapamycin (mTOR) and integrated stress response pathways (Arriola Apelo et al., 2014b).

The main goals of protein and AA nutrition are to characterize feeds for rumen-degradable protein (RDP) and digestible RUP, predict the conversion of RDP into ruminal microbial CP (MCP), and account for how efficiently the resultant MP and metabolizable AA are used under different animal and dietary conditions. Responses to individual AAs can differ from a weighted mean response (i.e., protein); further research

is needed to fully replace MP as a primary nutrient with metabolizable AA. When RDP or metabolizable RUP is in excess or when metabolizable AAs are not properly balanced, their carbon skeletons are metabolized by the microbes or the animal for energy; however, the excess N poses an environmental challenge (see Chapter 14). Therefore, improved prediction equations are needed to relate dietary inputs of AA and N to ruminal and animal responses so that animal performance can be optimized while minimizing environmental impact.

MAJOR DIFFERENCES FROM THE

PREVIOUS VERSION

Metabolizable Protein

Many factors influencing ruminal degradation of different dietary protein sources and the importance of ruminal degradability on the efficiency of microbial protein synthesis (EMPS) were discussed in the previous version (NRC, 2001). In particular, the 2001 update highlighted an improved library to estimate RDP and the intestinal digestibility of RUP, and it included an improved prediction equation to estimate MCP flow from the rumen. The current committee started with the NRC (2001) components and updated the database to include more studies using exclusively dairy cattle that reported duodenal or omasal flows of microbial N, nonammonia nonmicrobial N (NANMN), and AA. The updated data were used to evaluate NRC (2001) and, when compared to observed data, they compared favorably with other models, but all models evaluated by Pacheco et al. (2012) exhibited room for improved accuracy or precision. Predictions of both NANMN flows based on duodenal (Bateman et al., 2005) or omasal (Broderick et al., 2010) sampling were biased when the NRC (2001) model was used. Therefore, the committee deemed that protein supply should start with a fresh appraisal of data reporting measured ruminal outflow of N from dairy cattle. Because most flow studies did not independently derive RUP of feeds consistent with

NRC (2001) and did not measure intestinal digestibility, the current feed library must be used to extend microbial N and NANMN flows to an MP basis.

NRC (2001) estimated RDP based on kinetics of N disappearance of feedstuffs in situ. The N that washes out of bags prior to incubation and the N that remains in bags regardless of length of incubation are the A and C fractions, respectively. The B fraction is computed by difference of A and C from the total N. The degradation rate (k_d) of the potentially degraded B fraction is derived from in situ incubations. The in situ information can subsequently be used with rate of passage (k_p) in a first-order model to predict the extent of ruminal degradation. Because of limited N kinetic data, some feeds were poorly represented in the feed library (NRC, 2001). The NRC (2001) feed library also had limited estimates of intestinal digestibility for RUP (dRUP) for many feeds, prompting the use of increments of 5 percentage units to document data limitations. The current publication has a much stronger representation among feeds for the components used to derive RUP and dRUP.

In contrast with improved feed library values for N disappearance and for dRUP, the k_p of feed components has received little research attention since the NRC (2001) publication. One of the three k_p equations in NRC (2001) had a coefficient error corrected by Seo et al. (2006), and those authors noted a reasonably unbiased fit of the equations to the data, which were primarily derived from the use of rare earth markers. The committee considered whether ruminal in situ kinetics and k_p equations should be retained based on a complete reassessment of equations compared to a database with postruminal flows (White et al., 2017b). In that report, neither k_d nor k_p accounted for significant variation in ruminal NANMN outflows. However, the equations from that paper were for classes of feeds, and distinguishing RDP based on the A, B, and C fractions within feed classes was deemed insufficient by the committee to yield accurate differences in RUP among feeds processed to increase RUP. Consequently, a system was devised to retain elements of the NRC (2001) system (A, B, C, and k_d) but to re-derive static k_p values for forages and concentrates to remove the bias compared to NANMN flows (see subsequent section).

The NRC (2001) model estimated duodenal supply of MCP using intake of total digestible nutrients (TDN), which was discounted for negative associative effects. The equation was biased (St-Pierre, 2003; White et al., 2017b) and incorrect mechanistically because TDN includes postruminal digestion, which would not influence ruminally synthesized MCP. Therefore, the committee assessed a more mechanistic prediction of MCP that integrated responses to RDP and rumen-degraded starch and neutral detergent fiber (NDF).

NRC (1989) utilized an efficiency of RDP transfer to microbial N of 0.9, which would yield an RDP requirement of $1.11 \times \text{MCP}$ (i.e., $1/0.9=1.11$). Because the mean RDP/MCP was 1.18 in their data set, NRC (2001) revised this efficiency to 1.18; that is, if RDP intake was <118 percent of MCP, then MCP was limited to 85 percent (i.e., 100 percent /

118 percent $\times 100$) of RDP intake. The beef NASEM (2016) maintained its previous conversion efficiency of 100 percent, assuming blood urea nitrogen (BUN) transfer into the rumen would provide the shortfall needed for N assimilation into rumen MCP. Dairy cattle have different feeding conditions than grazing or feedlot beef cattle, and completely relying on BUN would not compensate for incomplete efficiency of transfer of RDP to MCP. Li et al. (2019) observed that the efficiency of transfer of ruminal ammonia N to microbial N was less than 70 percent under normal dairy feeding conditions. However, this efficiency is variable and declines as RDP and CP intake increases. Thus, the committee did not attempt to capture the potential use of BUN in support of microbial N synthesis directly and adopted an approach to predict MCP as a function of both RDP and rumen-degraded carbohydrate (starch and NDF), which is similar to that described by White et al. (2017b). While not direct, this approach inherently accommodated BUN transfer as it was derived from in vivo observations where such transfer is occurring.

In the current version, the committee retained the definition of net MP supply as the sum of MP from RUP and MCP. Duodenal flow of endogenous protein is not included in the net supply because it is synthesized from previously absorbed AA and, as such, does not represent a new source of MP and AA to the animal. To predict the requirements, estimations of the endogenous urinary loss and metabolic fecal output have been revisited. The current version also corrects the erroneous assumption that the transfer coefficient for MP to milk protein and other protein secretions is constant under different dietary and physiological conditions (Hanigan et al., 1998; Arriola Apelo et al., 2014b; Patton et al., 2014). Gestation requirements have been adjusted to utilize the model of Koong et al. (1975) fitted to the data of Bell et al. (1995), using estimated calf birth weight as an input. The efficiency of utilization of MP to proteins exported and protein accretion during growth is assumed to vary except for endogenous urinary loss. The urinary loss being a nonprotein fraction of end products of metabolic pathways, its efficiency is assumed to be 100 percent. The requirements for growth were derived as net protein (NP) gain, which was calculated as a function of user-specified live weight gain. The efficiency of conversion of MP to NP for growth is now calculated as a function of body weight (BW) relative to mature weight (see Chapter 11). Although requirements and associated recommendations of MP vary with the physiological status of the cows and the supplies of MP and energy, recommendations of MP are presented, assuming that energy is supplied to meet requirements. An efficiency of utilization of MP estimated to maximize export proteins has been defined as the “target” efficiency and is used to calculate recommendations.

Metabolizable Amino Acids

Of the 20 primary AAs that occur in proteins, 9 are usually classified as being “essential” AAs (EAAs) and 1 (arginine

[Arg]) as conditionally essential. AAs are termed “essential” if their carbon skeletons cannot be synthesized by animal cells or are synthesized at a rate insufficient to meet needs. The EAAs include histidine (His), isoleucine (Ile), Leu, Lys, Met, phenylalanine (Phe), threonine (Thr), tryptophan (Trp), and valine (Val). In contrast, AAs classified as “nonessential” (NEAAs) are those that can be synthesized *de novo* in adequate quantity given adequate supplies of N; these are alanine (Ala), asparagine (Asn), aspartate (Asp), cysteine (Cys), glutamate (Glu), glutamine (Gln), glycine (Gly), proline (Pro), serine (Ser) and tyrosine (Tyr). Although Arg is often classified as an NEAA in other animals (e.g., Hou et al., 2015), it has been traditionally classified as an EAA in dairy nutrition. Even if its *de novo* rate of synthesis is significant, it would be insufficient in high-producing dairy cows and is therefore usually considered part of the “10 EAAs.” However, jugular or abomasal infusion of Arg (Vincini et al., 1988) or deletion from a mixture of AA infused posttruminally (Schwab et al., 1976; Doepel and Lapierre, 2011) did not affect milk protein yield. *De novo* synthesis of arginine in dairy cows was estimated at approximately 30 g/d (Doepel et al., 2004; Martineau et al., 2014). In NRC (2001), the duodenal flows of EAA were estimated semiempirically by adjustment of the factorially determined flows using regression models based mainly on the RUP fraction. In the current edition, posttrimal flow of AA is estimated using a factorial method based on predictions of MCP, RUP, and endogenous duodenal CP. The flow of individual metabolizable AA, considered the net supply of AA, is calculated using the corresponding AA composition of feed factored to RUP and MCP and their respective digestibilities. Although endogenous duodenal proteins contribute to the duodenal protein flow, it is considered that these proteins are synthesized mainly from arterial supply and as such do not constitute a new net supply and therefore do not contribute to the net flow of metabolizable AA.

The committee investigated the use of variable ruminal degradabilities and intestinal digestibilities of individual EAAs among protein sources. Because White et al. (2017a) noted inconsistencies in methods used among feeds, the committee maintained the approach of factoring feed AA composition through RUP and dRUP. However, the committee recognizes that some, but not all, individual studies reported variable AA disappearance. For the AA composition of MCP, the committee adopted the approach derived from Sok et al. (2017). Although still retaining a constant AA profile, this approach modified the AA profile by accounting for protozoal contribution to MCP supply. It also addressed differences in AA profiles among fluid- and particulate-phase bacteria. The conversion of MCP to a TP basis was derived using AA recoveries accounting for hydrolysis losses rather than assuming 80 percent TP in the MCP, as done in NRC (2001). Recommendations for AA are based on a factorial approach, rather than the proportional approach used in NRC (2001); this means that the recommendations of each EAA

represent a quantitative assessment based on the sum of each EAA needed to fulfill each designated metabolic function. Recommendations are given for all EAAs, except Arg. The approach used for AA recommendations follows that used for the MP, and a target combined efficiency is used for each AA, assuming that energy requirements are adequately met. However, because of variable efficiency of AA use for several postabsorptive processes, MP and AA requirements are not constant and related to the energy supply and the physiological status of the cows. The Trp supply data are very limited, and thus predictions of Trp supply and recommendations need to be interpreted with caution.

METABOLIZABLE PROTEIN AND AMINO ACID SUPPLY

Rumen-Degraded and Undegraded

Protein Rationale for a Revised System

The NRC (2001) estimation of RDP and RUP resulted in mean (19 percent of the mean squared prediction error [MSPE]) and slope (22 percent of MSPE) biases when compared to observed NANMN flows (White et al., 2017b). Subsequent editing of the metadata by the committee to correct data entry errors reduced those biases to 7 percent of mean square error (MSE) each with a mean bias of 120 g/d and a slope bias of -0.31 g/g. An empirical prediction of ruminal digestibility based only on the A, B, and C fractions without use of k_d and k_p was considered by the committee; however, that failed to capture adequate variation among feeds. The addition of more feed categories for the B fraction degradation coefficient did not solve the sensitivity problem. For example, soybean meal, expeller-processed soybean meal, and nonenzymatically browned soybean meal had essentially equal calculated RUP values: 0.28, 0.30, and 0.28 g/g of CP. The committee assumed that a k_d would be required to match expectations of increased RUP with certain processing (Schwab and Broderick, 2017). Although k_d requires multiple time points and has procedural issues (Broderick et al., 2010), recommendations are discussed to improve its accuracy (see Chapter 18).

Considerations for Kinetics of Degradation

The current committee assessed the NRC (2001) technique for estimating kinetics of ruminal N degradation *in situ* for feedstuffs and recommends its continued use (see Chapter 18). The current database has a few notable considerations for users. The literature contains two types of fishmeal: ruminant and nonruminant grades (England et al., 1997), whereas users might be more likely to be using ruminant grade. Many of the earlier studies assessing fishmeal used a 24-hour *in situ* incubation; these values were included in NRC (2001) but removed from the current database because 24 hours is not long enough to properly estimate the C fraction (Liebe et al., 2018). In contrast, because of analytical issues, com

gluten meal has N disappearance in situ that seems to be underestimated (Murphy and Kennedy, 1987), which has been addressed (Stern et al., 1997).

As in the previous report, the current committee had to retain studies without correction for bacterial contamination to provide a robust database. Lack of correction for bacterial N contamination can underestimate the k_d of the B fraction (Wanderley et al., 1993). Bacterial N contamination represents up to 9 percent of the residual feed dry matter (DM) at 72 hours of incubation (Yang et al., 1999). Adding starch to the diet decreased the k_d of N in grasses (De Visser et al., 1998). In addition, the C fraction would be overestimated (and B fraction underestimated) by the degree to which bacterial N contamination is a proportion of residual N. The effect on predicted RDP would therefore depend on a culmination of factors, including errors associated with deriving k_d and A, B, and C fractions; that is, estimates of k_d are correlated with predicted A, B, and C fractions (Woods et al., 2003). Bacterial contamination can decrease the estimation of RDP by up to 5 percentage units (Alexandrov, 1998). However, this error is likely to partially (but not necessarily consistently among feeds) compensate for error from particle washout from bags.

Although in situ approaches have problems, various other approaches also have limitations or need further validation against in vivo measurements (Stern et al., 1997). Derivation of k_d presents a logistical challenge for feed analyses labs because more time points are needed. Based on omasal flow estimates, Reynal and Broderick (2005) suggested that NRC (2001) underestimated k_d , but this effect depended on the protein source (Brito et al., 2007b). In a meta-analysis of omasal N flows, similar conclusions were reached (Broderick et al., 2010). Despite these issues, the committee recommends further research to improve the accuracy of in situ kinetics used in calculation of RUP in protein sources. Numerous literature sources (Liebe et al., 2018) have become available since NRC (2001) to improve the feed library for nearly all common feeds. White et al. (2017b) and Liebe et al. (2018) explained exclusion criteria and corrections made when A + B + C did not sum to 100 percent.

Considerations for Ruminant Passage Rate

The committee considered using k_p data from the scientific literature or deriving static k_p empirically. Much of the k_p data used by NRC (2001) were derived from feed particles marked with rare earths, which can migrate to small particles and bacteria (Bernard and Doreau, 2000). Therefore, the B fraction may not pass at the same rate as the rare earths marking the particulate fraction. Predictions varied considerably from passage rates of undegradable NDF (Krizsan et al., 2010). Likewise, undegradable NDF does not necessarily pass at the same rate as potentially degradable NDF (Firkins et al., 1998). Soluble protein (fraction A) passes at rates much faster than particulate protein and can contribute significantly to RUP (Broderick et al., 2010; Huhtanen et al., 2014). Mechanisti-

cally, actual ruminal passage kinetics probably are even more complex than either of these approaches (Gregorini et al., 2015). The foundational principle that the marker should pass with the nutrient of interest (B protein in this case), not the particles or adherent bacteria, remains an elusive problem. Because k_d was required and measured k_p from markers were deemed problematic, the committee empirically derived a static k_p that allowed RUP to be predicted using a first-order model without bias as compared to NANMN flows (corrected for endogenous N) and converted to a CP basis.

An existing database containing postruminal N flows and diet descriptions from publications (Hanigan, 2006; Bateman et al., 2008) was updated (Roman-Garcia et al., 2016) and used to evaluate and update the model. For each feed in each diet, the mean CP, A, B, C, and k_d values were imported from the current feed library (see Chapter 19). Standard regression techniques yielded unrealistic parameter estimates for static k_p . The committee initially attempted to address this by increasing the number of feed categories; however, the static k_p values remained unrealistic and extremely variable for some feed categories. The NANMN calculation is derived by difference of nonammonia N (NAN) and microbial N flows and therefore aggregates error of both measurements. In addition to the limited number of observations for some feeds, fitting to NANMN is a challenge as it represents an aggregate of all feeds contributing RUP and is subject to potential variation introduced from endogenous N flows. The combination of large variation in observed NANMN flows, the aggregation of many ingredient contributions to NANMN, and the unbalanced nature of the data likely contributed to the unreasonable results with a mixed model. However, Bayesian hierarchical models allow study-specific parameters to follow a distribution for which the mean vector is the set of parameters common to all studies (Moraes et al., 2018b). Using this approach, the static k_p were estimated to be closer to expectations while accounting for random effects of studies.

Revised Estimates of Static k_p

The committee assumed that the k_p from wet and dry forages and concentrates underpinning the Seo et al. (2006) model provided information that could be leveraged to derive a set of static k_p values that would remove the bias in over-predicting RUP. The model used was of the following form:

$$Dt_RUP = \sum_{c=1}^{N_c} \left((1 - DCa_c) \times (Dt_CPAIn_c - Dt_CPAIn_{NPN,c}) + \frac{Kp_c}{Kp_c + Kd_c} \times Dt_CPBIn_c + Dt_CPCIn_c \right)$$

(Equation 6-1)

where

$Dt_$ = total diet concentration of the specified nutrient, subscript c represents the class of feed the ingredient belongs to,

DCa_c = ruminal degradability of fraction A in feed class c,
 $CPAIn$ = CP intake of fraction A within feed class c, $CPAIn_{NPNc}$
 = intake of ruminally degraded nonprotein N $\times$ 6.25 (CP
 equivalents) within feed class c,

$CPBin_c$ = intake of crude protein of fraction B within feed class
 c,

$CPCIn$ = CP intake of fraction C within feed class c,

Kp_c = static k_p for feed class c (e.g., forage or concentrate), and
 Kd_c is the weighted average of the in situ determined k_d for
 each feed in class c.

A Bayesian approach was used to derive estimates for DCa_c
 and Kp_c . The observations of Kp from Seo et al. (2006) were
 used as priors. DCa_c was included to allow for potential escape of
 the A fraction as reported by Broderick et al. (2010). DCa_c was
 initialized with a noninformative prior. The equation was fitted to
 observed postruminal flows of NANMN minus estimated
 endogenous N flows (see the subsequent section), and N was
 multiplied by 6.25 to convert to CP.

No gains in precision were realized for models with greater
 complexity than a single static k_p for all forages and another static
 k_p for concentrates. The solution using only a k_p for forage
 and another for concentrates yielded an estimate of 6.4 ± 4
 percent escape of protein in the A fraction, a static k_p for
 the B fraction of 5.28 ± 0.63 percent/h, and
 4.87 ± 0.33 percent/h for concentrates and forages, respec-
 tively. The concordance correlation coefficient (CCC) was 0.54, root
 mean square error (RMSE) was 40.9 percent of the mean, and
 mean and slope biases were 0.1 and 3.8 percent of MSE.
 These static k_p are not affected by dietary factors, whereas the k_p
 from NRC (2001) all increased modestly with increasing dry
 matter intake (DMI), and two k_p were affected by NDF and
 concentrate percentages. The committee concluded that the
 model with static k_p for forage and concentrates with some passage
 of the A fraction represented the best compromise to yield unbiased
 RUP values (versus NANMN $\times$ 6.25).

Metabolizable Protein Supply

Endogenous Protein Flow to the Duodenum

Because endogenous proteins arriving at the duodenum have
 been synthesized in gastric and pregastric compartments, the AA
 used for their synthesis is mainly, if not totally, from arterial origin.
 This means that these AAs have been previously absorbed and then
 used by gut tissues to synthesize secreted proteins. As such, they
 are not a new input of AA to the cow and need to be removed
 from the duodenal flow to estimate the net MP and AA supply
 (Lapierre et al., 2006). Due to the technical challenge of assessing
 endogenous N (EN) flow in ruminants, duodenal EN
 measurements are scarce. Nevertheless, duodenal EN flow has
 been determined in growing and mature cattle but by different
 methods. A limited database (see Table 6-1) was constructed to
 develop a regression between EN and DMI. Orskov et al. (1986)
 reported an EN flow from the rumen and abomasum when cattle
 received no feed and were only infused into the rumen with
 volatile fatty acids (VFAs). The data from steers fed at low intakes
 indicated a disproportionate ratio of duodenal EN relative to
 DMI when compared to cows fed at higher intakes. These data
 support the inclusion of an intercept in the regression, which was
 not included in the NRC (2001) estimation. The physiological
 status (growing versus mature) was not significant when included
 in the regression. Therefore, duodenal EN (g N/d) can be estimated
 as

$$Du_EndN (g/d) = 15.4 \pm 2.6 + 1.21 \pm 0.24 \times DMI$$

(Equation 6-2)

where DMI is kg/d.

Both the slope and intercept are significant ($P \leq 0.01$). Compared
 with this equation, the estimation of duodenal EN in NRC
 (2001) of 1.9 g N/kg DMI would underestimate EN in cattle
 with a DMI lower than 22.3 and overestimate when DMI is
 higher than 22.3 kg/d.

TABLE 6-1 Duodenal EN (g N/d) Flows from Cattle as Reported

Animal	DMI, kg/d ^a	BW, kg	EN	Method ^b	Reference
2 dairy cows	8.34	500	18.8	¹⁵ N	Brandt et al. (1980)
2 dairy cows	0 (3.4)	675	15.4	Rumen VFA inf	Orskov et al. (1986)
4 dairy cows	14.4	625	34.0	¹⁵ N-Leu dilution	Ouellet et al. (2002)
4 dairy cows	17.6	607	40.0	¹⁵ N-Leu dilution	Ouellet et al. (2010)
Dairy cows	18.0	600	40.9	Meta-analysis	Marini et al. (2008)
Dairy cows	17.9	597	29.5	Meta-analysis	Sauvant et al. (2013)
3 steers	2.86	300	23.0	RDP-free diet	Hart et al. (1990)
2 steers	0 (2.2)	278	11.6	Rumen VFA inf	Orskov et al. (1986)
4 steers	0 (1.8)	278	13.3	Rumen VFA inf	Orskov et al. (1986)
4 steers	3.14	424	24.4	RDP-free diet	Hannah et al. (1991)
4 steers	3.37	372	26.2	RDP-free diet	Lintzenich et al. (1995)
Growing cattle	6.82	359	24.2	Meta-analysis	Marini et al. (2008)

^a When VFA were infused, DMI = 0 and value in parentheses = kg of DM infused in the rumen.

^b RDP = rumen degraded protein; VFA inf = volatile fatty acid infusion.

Microbial Protein Supply

Adequate RDP is needed to maximize MCP production, NDF digestibility, and DMI (Schwab and Broderick, 2017). Microbial protein is the major supplier of MP, but its prediction also is important for estimating NANMN and therefore requirements of RUP. If the MCP supply is predicted to decrease, then RUP supply would need to increase, and vice versa, to maintain MP supply. Calculations of microbial N flow and ruminal protein balance (calculated as RDP-MCP-endogenous CP) are not independent. The committee did not use apparent ruminal N balance as a variable to assess RDP requirements for microbes because, although relevant biologically, its quantification propagates nonrandom error (i.e., bias) to an unknown degree. Rather than using RDP derived from flow data, Galyean and Tedeschi (2014) assessed the effect of total CP to influence EMPS in beef studies. Few studies intentionally limit CP for dairy cattle, and often total CP is maintained while exchanging RDP for RUP. Decreasing RDP in dairy studies is associated with decreasing MCP (Ipharraguerre and Clark, 2014).

Microbial N was predicted from RDP and predicted rumen-degraded carbohydrate (starch plus NDF) using an asymptotic integrated form of the Michaelis-Menten equation (Thornley and France, 2007). Previously, this equation form was used for microbial growth (Russell et al., 1992). The following equation was derived using the same data set as used for RDP estimates and derivations for rumen-degraded starch (RDS; kg/d) and rumen-degraded NDF (RDNDF, kg/d) as in White et al. (2017b). The same studies from White et al. (2017b) were used in the current derivation of microbial N flow with the major difference being that RDP (kg/d) was modified to reflect the current k_d/k_p approach and was fit using a Bayesian approach (Moraes et al., 2018a).

$$\text{Microbial N (g/d)} = [\beta_0 + (\beta_1 \times \text{RDP})] / [(1 + \beta_2/\text{RDNDF}) \times (1 + \beta_3/\text{RDS})] \quad (\text{Equation 6-3})$$

$$\begin{aligned} \text{Where RDNDF} = & [-31.9 + (0.721 \times \text{NDF}) - (0.247 \times \text{St}) + (6.63 \times \text{CP}) - (0.211 \times \text{CP}^2) - (38.7 \times \text{ADF}/\text{NDF}) \\ & - (0.121 \times \text{ForWet}) + (1.51 \times \text{DMI}) \times ((\text{NDF}/100) \times \text{DMI})/100 \end{aligned} \quad (\text{Equation 6-4a})$$

$$\begin{aligned} \text{RStDig} = & [(71.2 - (1.45 \times \text{DMI}) + (0.424 \times \text{fNDF}) \\ & + (1.39 \times \text{St}) - (0.0219 \times \text{St}^2) - (0.154 \times \text{ForWet})] \\ & \times (\text{St}/100) \times \text{DMI}/100 \end{aligned} \quad (\text{Equation 6-4b})$$

ForWet is concentration of wet forage (greater than 20 percent DM) in the diet (percentage of DM), DMI is in kg/d and NDF, and starch (St), CP, acid detergent fiber (ADF), and forage NDF (fNDF) are a percentage of diet DM. The RDNDF prediction had a negative slope for dietary

starch percentage, accounting for negative associative effects (White et al., 2016).

Predicted values should be multiplied by 6.25 to convert to CP. The numerator contains an intercept ($p_0 = 101 \pm 11$) and a coefficient for RDP intake ($(3, = 82.6 \pm 4.2)$, where RDP intake is kg/d. The denominator represents the main ruminally degraded carbohydrates, NDF ($P_2 = 0.094 \pm 0.028$) and starch ($P_3 = 0.027 \pm 0.010$), both of which are kg/d. The RMSE for the fit and root mean squared error of prediction from cross-validation were 29.7 and 29.8 percent of the mean (278 g/d), respectively. The CCC was 0.52 and 0.50 for the fit and cross-validation, respectively.

To limit unrealistic estimated requirements for RDP, observed microbial N flow was fitted against predicted RDP (percent DM) and ruminal NDF and starch degradabilities (percentage of respective intakes using the generalized additive mixed model from the mcgv R package) (Wood et al.,

2016). In the resulting smoothed response, microbial N flow was visualized to maximize at 12.0 percent RDP and to decline linearly as RDP decreased below 12.0, but the confidence interval widened, particularly as RDP declined below 10.0 percent. Because deficient RDP also depresses DMI (Firkins et al., 2006), which is the major driver of MCP production, the committee recommends a minimum of 10.0 percent RDP (derived with the current approach, which is generally higher than NRC, 2001) and no more than 12.0 percent RDP to optimize MCP supply.

A limited number of studies in dairy cows where diet RDP varied and RUP was held constant have been conducted. Kalscheur et al. (2006) reported a linear increase in milk protein yield and a curvilinear increase in DMI as diet RDP increased from 6.8 percent to 11.0 percent in diets containing 5.8 percent RUP. Substitution of urea N for 0, 1.2, 2.4, and 3.7 percent of dietary RDP resulted in a linear decrease in DMI, milk protein, and MCP in diets containing 5.6 percent RUP (Broderick and Reynal, 2009). Maximal DMI and milk protein yield occurred at 12.2 percent and 12.3 percent RDP, respectively, in diets that varied in 10.6 to 13.2 percent RDP (Reynal and Broderick, 2005). These results are consistent with the committee's general recommendations of 10 to a maximum of 12 percent RDP.

Equations 6-4a and 6-4b (see Figure 6-1) were deemed a more biological representation than the NRC (2001) approach to predict MCP, but there are some important caveats. The database is underrepresented with studies using small dairy breeds and with DMI greater than 30 kg/d. However, because the equations predicted percentage (not absolute) rumen degradabilities of NDF and starch (White et al., 2016), the predicted MCP flow should scale with DMI. From standard Michaelis-Menten form, the substrates for microbial protein synthesis in Equations 6-4a and 6-4b are rumen-degraded starch and NDF, implying an EMPS relationship (g microbial N/kg degraded NDF + starch) with maximal rates set by the RDP supply. Attempts to directly predict EMPS (g microbial N/kg organic matter [OM] truly degraded in the rumen) from

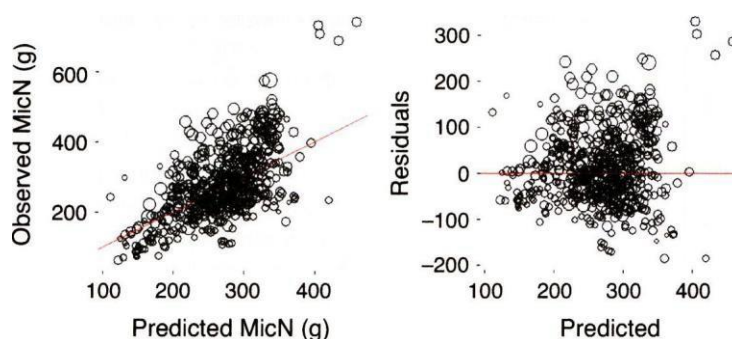


FIGURE 6-1 Observed, predicted, and residual (observed minus predicted) microbial N (MicN) flows (g/d). Symbols are sized according to $1/SE$ of the respective study. Number of treatment means = 580.

the source data had evidence for heteroscedasticity, and the model for ruminal OM digestibility had several coefficients that were $0.05 \leq P \leq 0.10$ (White et al., 2016). In addition, future refinements in rumen-degraded starch models (i.e., inclusion of adjustments for processing) lend themselves more directly to the equation than to equations predicting EMPS from OM degradation.

Bacteria can make AA from carbon skeletons derived from central metabolism, and the integration of metabolic pathways influences EMPS (Hackmann and Firkins, 2015b). Although some currently used models assume that preformed AAs only affect yield of amylolytic bacteria, this assumption ignores the complexity of the mixed community that includes fibrolytic and amylolytic bacteria (Firkins, 2010). NRC (2001) summarized reports for lactating cows fed diets with differing rumen-degraded starch and RDP, concluding that there was no consistent benefit in synchronization that could be distinguished from other potential explanations. Despite the clear indication of such synchrony from various *in vitro* studies, the current committee agrees with NRC (2001) *in vivo*-based conclusion. In theory, synchronization should be important and could be modeled in formulation software. However, synchrony of rumen-degraded carbohydrate and RDP is hard to distinguish in practice (Hall and Huntington, 2007). For example, total mixed rations allow disparate rates of RDP and carbohydrate degradation to cross over multiple meals. Certainly, BUN contributes to microbial N assimilation, but a distinctive qualification must be made for high-producing dairy cattle compared with beef cattle and sheep. With low CP diets often fed to beef and sheep, ruminal ammonia is buffered by BUN transfer to the rumen, particularly when ruminal VFA concentration is increased as would happen with higher-concentrate diets (Lu et al., 2014; Patra and Aschenbach, 2018). Urea synthesized by the liver and transferred back to the gut is an important short-term reservoir of N for growing and lactating cattle resulting from intermittent eating patterns (Reynolds and Kristensen, 2008). However, the response became incrementally more important as dietary CP decreased below 15 percent. Moreover, most of

these studies were short-term experiments from which labile body protein reserves are more plentiful. Calsamiglia et al. (2010) concluded that “urea-N salvaging mechanism, despite its obvious theoretical benefit to ruminant N efficiency, is poorly utilized under common applied dietary conditions of dairy cattle.” They concluded that increasing dietary CP percentage increased the urea excreted in urine rather than cycling to the gut. However, this is inconsistent with analyses of bovine studies utilizing double-labeled urea (meta-analysis by Li et al., 2019). In the latter work, urinary output and gut entry increased throughout the range of dietary CP intake, although the fractional proportion of urea released by the liver excreted in urine increased and that entering the gut decreased as dietary CP increased.

The committee recognizes the need for research to improve efficiency of N capture into milk protein; however, few data from lactating dairy cattle support relying on BUN to buffer against insufficient RDP intake. The microbial N derived from urinary urea-N (MNU; product/precursor ratio of enrichments of bacterial ^{15}N /urinary urea- ^{15}N) was 21 percent of the urea-N recycled to the gastrointestinal tract when dietary CP was 15 percent and declined with increasing CP (Batista et al., 2017). Those authors argued that the urea-N used for anabolism computation should not be used as a representation of microbial assimilation of urea-N. A direct evaluation of MNU avoids all misrepresentations in ^{15}N -urea kinetics plus errors in postruminal flows of microbial N. In four studies with dairy cattle (see Figure 6-2), the MNU averaged 14.6 percent when dietary CP ranged between 12.5 and 18.6 percent. Use of the product/precursor ratio in this case may result in biased estimates due to label dilution in the ruminal ammonia pool, that is, urea is not the direct precursor for microbial protein synthesis (Li et al., 2019), suggesting BUN could be a significant source of microbial N in the rumen under conditions when RDP is limiting.

Amino-N probably enhances EMPS when degraded carbohydrate is available (Hackmann and Firkins, 2015b), as would be expected in lactating dairy cattle. Dietary urea replacement of amino-N decreased MCP in dairy cattle (Brito

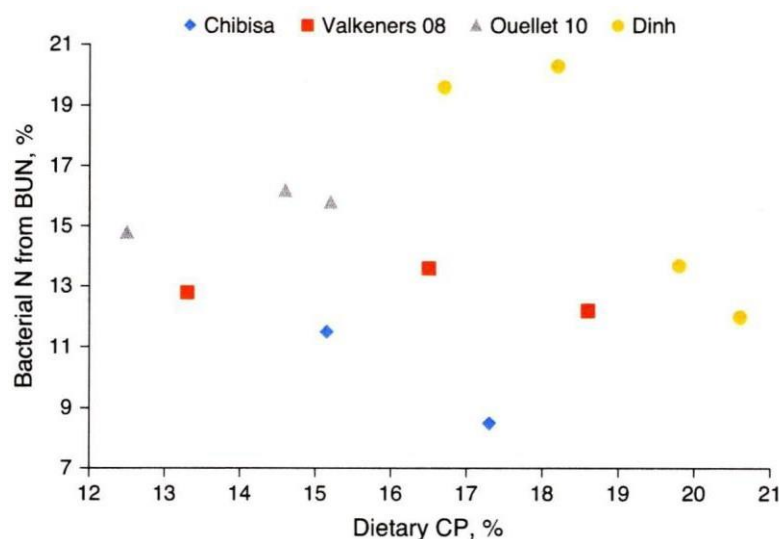


FIGURE 6-2 The product/precursor ratio of ^{15}N enrichment in ruminal bacteria divided by the ^{15}N enrichment in urinary urea-N (assumed in equilibrium with BUN) converted to a percentage and expressed against dietary CP percentage. Accounting for the random effect of study and excluding dietary CP >19.0 percent resulted in an average of 14.6 percent with a standard error of mean (SEM) = 2.1 percent. The plotted data from the studies (Dinh, 2007; Valkeners et al., 2008; Ouellet et al., 2010; Chibisa and Mutsvangwa, 2013) are unadjusted for the random effect of study.

et al., 2007; Broderick and Reynal, 2009). Hence, BUN recycling to the rumen was deemed to provide an important N source to buffer intermittent CP consumption, but these factors were considered when choosing not to include BUN transfer in the MCP model. Additionally, because the microbial equations were derived from in vivo observations, those factors were intrinsic to the derivation to the extent they can be captured in the carbohydrate and RDP terms. This contrasts with the approach taken by the beef NASEM (2016), which assumed 100 percent efficiency in converting RDP to MCP, implying that the balance of ammonia absorption from the rumen and BUN transfer into the rumen could compensate completely for RDP deficiencies.

Beet and citrus pulp (high in soluble fiber) and sugars are extensively degraded in the rumen. A moderate amount of beet pulp replacing high-moisture corn supported similar MCP production in dairy cattle (Voelker and Allen, 2003). Soluble fiber is highly degradable (Miron et al., 2002). The enzymatic assay reported as “starch” in nearly all of the source data used to derive the prediction equations included sugars. The responses to supplemental sugars have been mixed (Oba, 2011). Because of limited data, the equation might underpredict MCP when feeding high amounts of soluble fiber or sugars, but no evidence was available to make an adjustment.

Supply of MCP was not adjusted for supplemental fat for various reasons. One equation in the beef NASEM (2016) predicts rumen MCP supply as a function of fat-free TDN. Fat has not had consistent effects on MCP production (Oldick et al., 1999). Hanigan et al. (2013) observed a positive effect of dietary fat on microbial protein outflow using a data set with 227 treatment means. Schmidely et al. (2008) did not

observe an effect of fat on EMPS unless OM digestibility was depressed. In that case, EMPS is predicted to increase because the denominator decreased, not that the numerator (MCP outflow) increased. Those authors suggested that the improved EMPS from supplemental fat was from suppressed protozoa and their resultant predation of bacteria. Hanigan et al. (2013) suggested that uptake of preformed FAs could spare carbon for synthesis of other cellular components and for adenosine triphosphate (ATP) production by fermentation. However, FAs might be internalized as droplets rather than be directly used for membranes (Bauchart et al., 1990). The sparing effect was not well supported in one of the few studies that addressed bacterial growth directly (Maczulak et al., 1981). Improved efficiency of MCP synthesis from feeding fat has been attributed to k_p or other factors (Nagaraja et al., 1997). Moreover, high amounts of fat can be toxic to bacteria (Hackmann and Firkins, 2015a). Bacteria take up exogenous fatty acids to a lesser extent than do protozoa (Karnati et al., 2009). Thus, decreasing protozoal outflow from supplementing fats could increase usage of preformed FAs because of greater bacterial outflow resulting from less predation. Because of high DMI, bacterial predation should affect lactating dairy cows less than nonlactating ruminants (Firkins et al., 2007). For all of these reasons, the current committee chose to not account for fat.

Intestinal Digestibility of Rumen-Undegradable Protein

The current feed library has a much more robust inclusion of dRUP values compared with NRC (2001). The values are derived from a mix of in vitro and mobile bag studies. Proper analytical procedures must be followed to obtain accept-

able results (Liebe et al., 2018; see Chapter 18). Intestinal digestibility of RUP remains a critical research area, including standardization, prevention of analytical artifacts, and a better understanding of differential AA digestibility within RUP. Validation of all analytical methods against in vivo observations is essential to ensure they are representative across a broad range of feeds. An in vivo, isotope-based approach demonstrated the inaccuracy of the three-step procedures for assessing dRUP for several important ingredients and the extent of variation in digestibility of individual AA within the RUP fraction (Estes et al., 2018; Huang et al., 2019). That method is not adaptable to commercial settings, but it can be used to evaluate in vitro and in situ approaches.

Metabolizable Amino Acid Supply

Correction of Amino Acid Composition of Proteins Obtained from Hydrolysis

Because the committee moved toward a factorial AA approach, special attention was devoted to accurately defining the AA composition of each protein fraction. Correction factors for incomplete recovery of AA from a 24-hour acid hydrolysis have been published (Lapierre et al., 2019) and are detailed in Chapter 18. Those recovery factors were used to correct all reported AA composition obtained from protein hydrolysis and are thus intrinsic to all protein equations in the final model. The abbreviation AA_{corr} will be used when

referring to AA concentrations corrected for incomplete recovery from 24-hour hydrolysis. Users of the derived equations must keep in mind that the equations were built from corrected AA flows (i.e., estimations of the flows of metabolizable AA, secretions, and accretions as well as efficiency calculations were all done using true corrected AA flows). However, predictions of duodenal AA flows were converted to uncorrected flows for presentation purposes and to allow direct comparisons with published data.

Metabolizable Amino Acids from Microbial Crude Protein

Assessing metabolizable AA supply from MCP in ruminal outflow requires knowing (1) the TP/CP ratio of MCP, (2) the AA composition of microbial TP, and (3) the digestibility of microbial TP. Assumptions of 80 percent TP in MCP and 80 percent digestibility of TP in the small intestine (NRC, 2001) have been used since NRC (1985). Despite the importance of these constants in predicting metabolizable AA supply, few studies have evaluated them, particularly for dairy cattle (Patton et al., 2014). To evaluate the TP/CP ratio, the recovery of total AA after hydrolysis of bacteria was investigated. Formalin used as a preservative for bacteria decreased the recovery of some AA and should be avoided (Volden et al., 1999); therefore, data from these studies have been excluded. Accounting for the incomplete recovery of AA after a 24-hour hydrolysis, hydration upon hydrolysis (1 g of TP

would yield approximately 1.15 g of free AA), and that Gin and Asn are transformed into Glu and Asp, respectively, TP/CP averaged 82.4 percent in a broad survey of published AA composition of fluid- and particle-associated bacteria (Sok et al., 2017). The remainder of the CP that is not TP is mostly nucleic acids. The current edition assumes this proportion, 17.6 percent, which is slightly lower than the 20 percent used in NRC (2001). Bacterial RNA-N/N plus DNA-N/N is typically less than 15 percent (McAllan, 1982), which is consistent with other measurements (Czerkawski, 1976) or theoretical calculations (Hespeell and Bryant, 1979). RNA-N ranged from 9 to 13 percent of total N in bacteria (Susmel et al., 1993). In another survey, the total nucleic acid-N (including DNA) ranged from 11 to 24 percent of total N (mean of 14 percent) in bacteria and was lower in protozoa (Fujihara and Shem, 2011). Glucosamine is ignored in these studies but comprises about 2 percent of polysaccharide-free bacterial cells (Czerkawski, 1976); it is 7.8 percent N (about half of the 16 percent N in CP), so it should contribute about

1 percent CP. Increasing ruminal k_p and therefore growth rate of bacteria could increase the nucleic acid-N contribution to total CP (Bach et al., 2005), although the nucleic acid-N/N ratio is probably less variable in the ruminal bacteria of dairy cows (Firkins et al., 2006). Taking all this into consideration, the committee modified 80 percent TP to 82.4 percent TP in MCP. Undegraded nucleic acids in RUP should be negligible (Calsamiglia et al., 1996).

The AA composition differs between fluid- and particle-associated bacteria and protozoa, but insufficient data were available to determine whether AA profile of microbes is affected by feeding condition (Sok et al., 2017). For the model, a constant contribution of fluid- and particle-associated bacteria and protozoa to ruminal outflow was derived (33.4, 50.1, and 16.5 percent, respectively). After weighting for these proportions and accounting for differential recoveries of AA after hydrolysis, a AA_{corr} composition of the TP of microbial protein was derived (see the section “Secretion and Accretion”) and used with the factorial method to assess the contribution of MCP to metabolizable AA flow. The updated AA profiles reflect more current measurements and expectations but still retain limitations associated with dietary influence on bacteria and protozoa AA composition.

The inclusion of the protozoa contribution to MCP (16.5 percent) becomes important because protozoa contain much more Lys than bacteria (Jensen et al., 2006), with values up to twice as high (Reynal et al., 2003; Fessenden et al., 2016). Because of the potential for bacterial contamination of protozoal samples (Sylvester et al., 2005), the difference in Lys might even be greater. The high Lys concentration in protozoal protein is likely due to diaminopimelic acid metabolism (Williams and Coleman, 1992; Martin et al., 1996). Merchen and Titgemeyer (1992) did not note a difference in supply of Lys resulting from defaunation of growing ruminants. The greater Lys concentration of protozoa can make a

significant difference if protozoal outflow is enhanced by faster k_p (Jouany, 1996), as expected for dairy versus beef cattle. Unfortunately, protozoal Lys outflow has rarely been studied in dairy cattle. The Lie concentration was also higher for protozoa, whereas Met, Thr, and Val concentrations were lower than in bacteria; particulate-phase bacteria have increased Arg and Phe but decreased Thr compared with fluid-phase bacteria (Sok et al., 2017).

NRC (2001) assumed that the TP in MCP reaching the duodenum was 80 percent digestible. Results using isotopes or intragastric infusion of bacterial protein suggest digestibility greater than 80 percent, and the digestibility of both fungal and protozoal protein is probably higher than that of bacterial protein (Jouany, 1996). Tas et al. (1981) estimated true digestibility of microbial protein at 87 percent, compared to 81 percent from intragastric infusion (Storm et al., 1983). Apparent absorption of nucleic acid-free bacterial ^{15}N in steers was 74 percent (Salter and Smith, 1984), which was higher than apparent absorption of nonammonia ^{15}N in growing steers (Firkins et al., 1987). In that study, ^{15}N absorption between duodenum and ileum was similar to total N (including feed N). Low intakes by these nonlactating ruminants make large differences between apparent and true digestibility; for example, from 69 (apparent) to 86 (true) percent of N entering the duodenum (Tas et al., 1981). Fonseca et al. (2014) reported that fluid- and particle-associated bacteria had standardized digestibilities of 76.8 and 75.5 percent, respectively, for total AA in cecectomized roosters. These results are near the average of 76 percent from their literature search.

Limited data on intestinal digestibility of MCP in dairy cattle are available. The estimated mean digestibility of bacterial AA-N in dairy cattle ranged from 75 to 77 percent (Larsen et al., 2001). Those authors derived bacterial N and AA from duodenal and ileal samples for this calculation, although the digestibility could be underestimated because of bacterial cells produced in the ileum. The digestibility of AA in bacterial cell walls (especially diaminopimelic acid) is lower than the rest of the TP-AA, particularly Gram-positive bacteria. Thus, EAA digestibilities were all higher than the mean for digestibility of total AA-N. There is insufficient support to have differential digestibilities for individual AA in microbial protein. In fluid-associated bacteria, His and Met had higher digestibilities than total AA, and Val was lower; in particle-associated bacteria, Arg was higher, and Val was lower in cecectomized roosters (Fonseca et al., 2014). Based on this limited information, the committee extended the constant 80 percent true digestibility for all microbial EAA. Consequently, the conversion of MCP to MP is assumed to be 82.4 percent TP in CP at 80 percent digestibility = 65.9 percent (slightly higher than 64 percent in NRC, 2001).

Metabolizable Amino Acids from Rumen-Undegradable Protein

White et al. (2017a) evaluated the ruminal degradability and intestinal digestibility of EAA as affected by feed

category. Only 10 published studies (53 treatment means) reported ruminal AA disappearance in situ (at 12 or 16 hours) that met inclusion criteria. The committee deemed that a broader database would be needed before differential EAA ruminal degradabilities or intestinal digestibilities could be predicted with confidence. Therefore, the AA profile of the RUP fraction of feedstuffs is assumed to be the same as in the original feedstuff, which is consistent with NRC (2001), the French PDI (Rulquin et al., 1998), the Dutch DVE/OEB2010 system (van Duinkerken et al., 2011), and the Cornell Net Carbohydrate and Protein System (CNCPS) (Van Amburgh et al., 2015). Changes in AA profiles probably are greater for feeds with higher RDP (Boucher et al., 2009a). Ratios of AA profile after/before exposure to rumen fluid changed only modestly (Mupeta et al., 1997; Harstad and Prestleken, 2001) or greatly, depending on the protein source (Cozzi et al., 1995; Paz et al., 2014). Met, Leu, and Phe increased and His and Lys decreased after incubation of extruded peas and soybeans (Walhain et al., 1992). No statistics were done, but EAA ratios depended on protein source (Piepenbrink and Schingoethe, 1998). In canola meal, Thr, the branched-chain AAs, and aromatic AA concentrations increased after incubation (Boila and Ingalls, 1992). Excessive heating can convert L- to D-racemers or crosslink AA, which decreases digestibility (Vrese et al., 2000), and intentional heating confirmed lower Lys intestinal digestibility for some feed sources (Boucher et al., 2009b).

The intestinal digestibility of AA within the RUP fraction was assumed to be the same as the CP digestibility for all AAs. Recent in vivo work suggests that assumption is likely incorrect (Estes et al., 2018; Huang et al., 2019). However, the literature data were inadequate to derive individual AA digestibilities for the range of ingredients used in dairy diets. Future research is needed to improve accuracy and consistency of predicting digestible RUP and AA. In addition, researchers should acknowledge the bias caused by underestimating AA recovery during AA hydrolysis, particularly Met, if not appropriately protected prior to acid hydrolysis (Higgs et al., 2015), and report values for Trp. To develop the models of this revised version, the AA composition of feed ingredients as reported in Table 19-2 was corrected to account for incomplete recovery using correction factors (Lapierre et al., 2019). Ruminal degradability and intestinal digestibility of total AA, not just individual AA, need to be reported so total AA can be used for standardization among studies (White et al., 2017a).

Amino Acid Composition of Postruminal Endogenous Protein

The AA composition assigned to the postruminal endogenous duodenal flow, which was removed from the observed flow to assess the net supply, was derived from Orskov et al. (1986). The AA composition of rumen and abomasal fluids from cattle nourished with N-free intragastric infusion was

TABLE 6-2 AA Composition of the CP and TP Fractions Involved in the Estimation of AA Supply and Recommendations

AA	gAA _{corr} /100gCP			g AA _{corr} /100 g TP ^a		gAA _{corr} /100gTP ^b
	Duodenal Endogenous	Microbial ^c	Scurf	Whole Empty Body	Metabolic Fecal	Milk
Ala	4.69	7.38	9.17	8.59	6.32	3.59
Arg	4.61	5.47	9.60	8.20	5.90	3.74
Asx	4.75	13.39	8.39	9.61	7.56	8.14
Cys	2.58	2.09	2.70	1.74	3.31	0.93
Glx	11.31	14.98	14.69	15.76	15.67	22.55
Gly	5.11	6.26	21.08	14.46	8.45	2.04
His	2.90	2.21	1.75	3.04	3.54	2.92
Ile	4.09	6.99	2.96	3.69	5.39	6.18
Leu	7.67	9.23	6.93	8.27	9.19	10.56
Lys	6.23	9.44	5.64	7.90	7.61	8.82
Met	1.26	2.63	1.40	2.37	1.73	3.03
Phe	3.98	6.30	3.61	4.41	5.28	5.26
Pro	4.64	4.27	12.35	9.80	8.43	10.33
Ser	5.24	5.40	6.45	5.73	7.72	6.71
Thr	5.18	6.23	4.01	4.84	7.36	4.62
Trp	1.29	1.37	0.73	1.05	1.79	1.65
Tyr	3.62	5.94	2.62	3.08	4.65	5.83
Val	5.29	6.88	4.66	5.15	7.01	6.90

^a g AA_{corr}: AA composition corrected to account for the incomplete recovery of AA with 24-hour hydrolyses, expressed in hydrated form, and therefore sum to more than 100 for a given protein. Table modified from Lapierre et al. (2020).

^b g AA_{corr}: AA composition calculated from the primary structure of the reference protein of each family, expressed in hydrated form, and therefore sum to more than 100.

^c Adapted from Sok et al. (2017) using the correction factors proposed by Lapierre et al. (2019).

averaged; for Leu, only the rumen value was retained. The AA composition of endogenous protein contributing to the duodenal flow, once corrected for incomplete recovery with 24-hour hydrolysis, is presented in Table 6-2.

POSTABSORPTIVE USE OF
METABOLIZABLE PROTEIN
AND AMINO ACID

In NRC (2001), MP requirements included metabolic fecal, endogenous urinary, scurf, growth, gestation, and lactation. Those functions were retained in this version, although the approaches to estimate use of MP and AA changed. The greatest amount of available data was for milk protein yield, which allowed development of more comprehensive models for that function.

Estimation of Milk Protein Yield

NRC (2001) predicted the MP required to support a specific level of milk and milk protein yield (MPY). Additionally, a conceptual framework for the effects of individual AA on MPY was provided, and recommendations for Met and Lys were suggested. Shifting from an MP system to one based on AA implies that it is more accurate at predicting MPY. Therefore, the performance of existing MP-based equations was used as a benchmark. Data for all work on milk protein were collected from the literature by several teams (NRC, 2001; Hanigan et al., 2002; Bateman et al., 2008; Roman-Garcia et al., 2016) and additional data col-

lected from studies describing the effects of infused AA and AA fed in rumen-protected (RP) form. The complete data set contained 1,149 treatment means from 275 experiments. Of these, 898 treatments from 216 studies reporting milk production and milk protein content or output were used.

Evaluation of the National Research Council
2001 Model, Other Metabolizable Protein-
Based Models, and the First Limiting AA
Concept

The overall relationship between MP supply and MPY is quadratic, reflecting the decreased marginal return in MPY as protein supply increases (Hanigan et al., 1998; Huhtanen and Hristov, 2009; Lapierre et al., 2012a). However, in the NRC (2001) model as in many other models, MPY was linearly related to MP supply. The NRC (2001) MP system was evaluated using revised ingredient composition data (see Chapter 19) and revised energy supply equations (see Chapter 3). Estimation of MP allowable MPY by the 2001 model had an RMSE of 24.9 percent with significant slope bias. Overall, the NRC (2001) model underpredicted MPY by 28 g/d and overpredicted responses to changing MP supply, indicating that the partial efficiency of MP use (fixed at 0.67) in response to increased MP supply was too high. Net energy allowable MPY had an RMSE of 21.3 percent with significant slope bias. The model overpredicted MPY by 32 g/d on average (P<0.01) and overpredicted the response to varying net energy for lactation (NEL), indicating the partial efficiency of NEL use may be overpredicted. Protein

TABLE 6-3 Residuals Analyses for NRC (2001)-Based Predictions of Milk Protein Production as Compared to Predictions from the Revised Model^a

	NEL	NRC (2001) ^b		Equation 6-6
		MP	Minimum of NEL and MP	
N	922	926	922	926
Observed mean, g/d	922	921	922	921
Predicted mean, g/d	953	890	831	924
ccc	0.77	0.65	0.71	0.75
RMSE, g/d	194	228	210	133
RMSE, percent mean	21.0	24.9	22.8	14.4
Mean bias, percent MSE	3	2	19	0.0
Slope bias, percent MSE	38	32	21	3.1
Mean bias, g/d	-31	31	91	-2.7
Slope bias, g/g	-0.38	-0.44	-0.34	0.156
P _{Mean Bias}	0.0001	0.0002	0.0001	0.54
P _{Slope Bias}	0.0001	0.0001	0.0001	0.0001

^a Predictions were not adjusted for random study effects.

^b Predicted milk protein (g/d) = allowable milk (kg/d) × observed milk protein (%) × 10.

synthesis is an integrated process responding to both net energy and protein supply. Historically, the combination has been considered using a first limiting nutrient concept, which was modeled as the minimum of MPY available from MP or from NEL. Predictions based on that concept resulted in an RMSE of 22.9 percent with both mean and slope bias (see Table 6-3). Those biases and errors indicated the 2001 NRC models required revision.

Predicting Milk Protein from Essential Amino Acids and Energy Supply

Initial regression work indicated that digested energy intake (DEI) and AA supply were strong determinants of MPY. DEI was used to represent the effects of energy on milk protein synthesis as MEI cannot be calculated until the amount of catabolized protein is determined. Given that DEI includes DE from MP, there is an inherent correlation among the inputs that was addressed by removal of the DE associated with MP, resulting in a nonprotein DEI (DEInp). Additionally, this approach resolves the issue of MP having a greater DE value than carbohydrate, which can lead to optimizers based on DE (but not ME or NEL) to use MP rather than carbohydrates for energy. Thus, DEInp was used for the following work. Substitution of DE concentration for DEI or DEInp resulted in much poorer predictions. Because several EAA and energy-yielding substrates regulate rates of protein synthesis (Dos et al., 2004; Gan et al., 2011; Appuhamy et al., 2014) and all of the EAA are substrates for protein synthesis, one may expect several of these to be important drivers of production. This is a potential challenge as many of the dietary nutrients are correlated, in particular among the EAAs. Met supply had the lowest correlations with other EAAs, ranging from 0.74 to 0.85 due to the

large number of studies that utilized independent additions of Met through infusions and feeding RP-Met. Hanigan et al. (2000) demonstrated that representations of milk protein synthesis and milk yield as a function of the most limiting nutrient (an EAA or energy) were inadequate. An additive representation with independent effects of three or more EAAs and energy supply resulted in significantly better fits to the data and very large increases in variation explained by the models. This concept is well supported at the tissue level, where independent protein synthesis and cell signaling responses to at least Met, Leu, Ile, Thr, insulin, and acetate concentrations have been observed (Appuhamy et al., 2012, 2014; Arriola Apelo et al., 2014c), and responses in MPY have been verified (Schwab et al., 1976; Yoder et al., 2020). The relative supplies of each AA also interact to regulate mammary (Bequette et al., 2000; Hanigan et al., 2000) and liver AA transport activity (Myers et al., 2017), resulting in variable efficiency of transfer from the gut to the mammary cells, which explains the diminishing returns response to MP supply (Whitelaw et al., 1986; Lapierre et al., 2007a).

The large number of factors controlling synthesis results in a complicated response surface and hampers experimental and modeling progress. Because it is almost impossible to isolate and control concentrations of all of the factors, future experimentation must report measurements of all controlling factors and include treatments that independently manipulate those factors to ensure that derived responses are truly representative of each factor. Additionally, experiments using multilevel treatments are needed to better define the response surfaces that can be expected to be nonlinear. There is a critical need for larger central composite designs to assess and quantify potential interactions among the factors. In the absence of those types of experiments, the surfaces must be derived across experiments, which is subject to much greater

variation and covariation as many factors are not controlled or fully reported across experiments. Such work should include genomic characterization of the animals so that variation among animals can eventually be at least partially described based on genetic potential.

Given the biological underpinnings and despite the challenges with data sufficiency, the committee was able to develop a multiple regression equation to predict MPY using an “all-models” approach (Burnham and Anderson, 2002) to evaluate all possible combinations of the absorbed supply of individual EAA supply; DEInp; digestible starch (dSt), digestible NDF (dNDF), and digestible FAs (dFAs); and BW and parity. Dietary macronutrients other than these were initially screened and found to be unrelated to MPY. Three different global equation forms were evaluated. The first utilized metabolizable EAA and DEInp as the primary driving variables:

$$\begin{aligned} \text{Milk Protein, g/d} = & \text{Arg} + \text{His} + \text{Ile} + \text{Leu} + \text{Lys} + \text{Met} + \text{Phe} \\ & + \text{Thr} + \text{Trp} + \text{Val} + \text{NEAA} + \sum (\text{EAA}^2) \\ & + \text{DEInp} + \text{dFA} + \text{dNDF} + \text{BW} + \text{Parity} \\ & + \text{PubID} \end{aligned} \quad (\text{Equation 6-5})$$

where DEInp (Mcal/d) represented DE intake minus the energy contribution from MP assuming 5.65 kcal/g of MP; the individual AAs represented metabolizable supplies (g/d) of each; NEAA (g/d) represented the absorbed supply of NEAA; the $\sum (\text{EAA}^2)$ term represented the sum of squared individual EAA supplies; dFA and dNDF are expressed as percentage of dietary DM; BW represented the reported BW (kg); Parity was a continuous variable ranging from 0 to 1 based on the reported number of animals in each parity, with

0 and 1 representing first parity and multiparous animals, respectively; and PubID represented the random effect of study. The second and third equations contained the same terms as the first equation, but the individual EAAs were expressed as a percentage of the total EAA or as a ratio to DEInp, respectively.

The top 5,000 models based on the Akaike information criterion (AIC) were extracted from each set of solutions for further consideration and summarization. The equations based on EAA as a ratio to DEInp proved statistically inferior and were abandoned. The model using AA, represented as a ratio to EAA, had very similar performance; however, the distinct disadvantage of this form is the inconsistency in responses to a given EAA caused by the division of each EAA by the total EAA. With this form, the addition of one EAA to the diet results in a greater denominator, resulting in a reduction in the calculated ratios for all other EAA. This results in nonadditive responses to individual AAs when they are added to the total versus substituted for another AA, which is not biologically consistent with the known mechanisms (Arriola Apelo et al., 2014b,c,d). Thus, equations expressing AA as a ratio to MP or total EAA supply were excluded from

further consideration, and all efforts focused on the general form represented by Equation 6-5.

Many of the top equations derived from Equation 6-5 performed equally well, and thus the choice of an equation for use was partially dependent on the objectives, how well the equation of choice matched the biological underpinnings, and the stability of the derived coefficients and model predictions. Of the EAAs, Trp and Val were each present in half of the top 5,000 equations, but the slope estimates were generally negative and deemed nonbiological, and neither performed well in cross-evaluations having unstable solution estimates (slope estimates varied by more than 100 percent using a bootstrap approach with resampling; Simon, 2007). Phe and Thr were also present in 48 percent and 47 percent of the solutions, respectively, but did not perform well in cross-evaluations. Arg was present in 68 percent of the top solutions and was stable on cross-evaluation but was deemed suspect due to its conditional essentiality. The NEAA term was also present in 68 percent of the top solutions and was stable on cross-evaluations, indicating some apparent effects of total N supply. Parity and dFA were also present in about half of the solutions but unstable during cross-evaluations. The remaining terms were present in more than 70 percent of the models (excepting Ile and Leu at 59 percent and 57 percent, respectively), and all were stable during cross-evaluations. Based on these statistical and biological arguments, models containing individual linear terms for Arg, Phe, Thr, Trp, Val, dFA, and Parity were not considered. Filtering to remove equations containing those terms and conducting further testing of the combination of the excluded EAA and the NEAA (OthAA) resulted in the following solution that was selected for use in the model:

$$\begin{aligned} \text{Milk Protein (g/d)} = & -97.0 + 1.68 \times \text{His} + 0.885 \times \text{Ile} + 0.466 \\ & \times \text{Leu} + 1.15 \times \text{Lys} + 1.84 \times \text{Met} + 0.077 \\ & \times \text{OthAA} - 0.00215 \times \sum_{i=1}^{N_{\text{EAA}}} \text{EAA}_i^2 + 10.8 \\ & \times \text{DEInp} - 4.60 \times (\text{dNDF} - 17.06) - 0.420 \\ & \times (\text{BW} - 612) \end{aligned} \quad (\text{Equation 6-6})$$

where the individual EAA terms are expressed as g absorbed/d, DEInp is Mcal/d, dNDF is percent of DM, BW is in kg, and EAA_i^2 represents the squared supply of each of the EAAs present in the equation (His, Ile, Leu, Lys, and Met) denoted by the subscript *i*. OthAA represents the absorbed supply of the NEAA plus Arg, Phe, Thr, Trp, and Val. dNDF and BW are centered to the mean of the data for use in the equation, and thus their mean values (17.06 percent and 612 kg, respectively) were included as subtractions in the equation. The selected model also reflects refitting after the squared EAA term was reduced to a summation of the squares of His, Ile, Leu, Lys, and Met. All terms in Equation 6-6 were stable under cross-evaluation, and the equation had

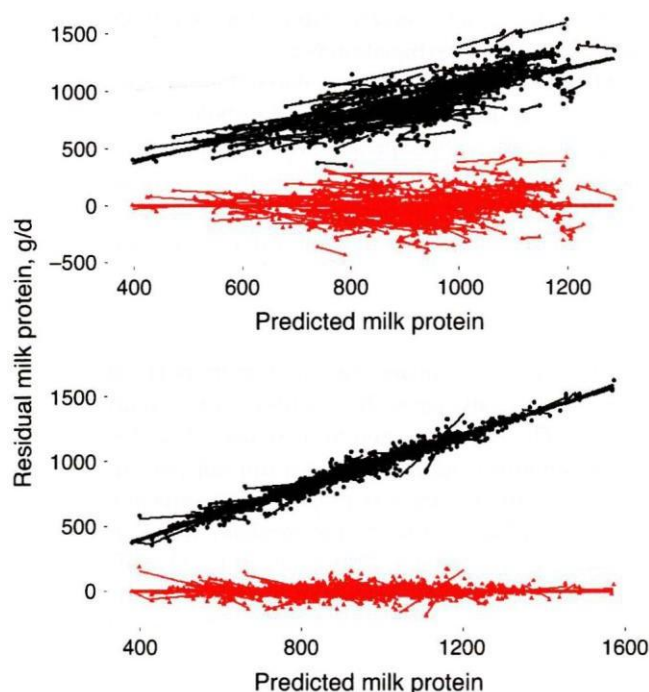


FIGURE 6-3 Residual errors for predictions of true milk protein by Equation 6-6. Data in the top panel were not corrected for the random effects of study, and data in the bottom panel were corrected for those effects. Data colored black represent observed versus predicted, and the data colored red represent residuals versus predicted. The lines connecting points (i.e., splines) are the linear regressions within study.

stable RMSE and CCC estimates across the evaluations. Protein yield predictions from this equation represent the predicted milk NP supported by a given diet (NP-milk).

The RMSE for Equation 6-6 was 14.4 percent with 0 percent mean bias and 3.1 percent of the MSE as slope bias, and the CCC was 0.75 and represents a substantial improvement in fit over the NRC (2001) equation (see Table 6-3). The residual errors for the predictions are displayed in Figure 6-3.

Correlations among parameter estimates should be low if the data are adequate to uniquely define those parameters, and this was the case for all of the linear coefficients in Equation 6-6 with the exception of a correlation of 0.67 between the coefficients for Leu and Lys, -0.51 for OthAA and lie coefficients, and -0.43 for Lys and His coefficients. The remainder were all below 0.32 in absolute terms.

Residual errors for MPY were plotted and regressed against the supplies of each individual metabolizable EAA and of total metabolizable EAA to appraise performance relative to each of the primary inputs. If the effects of these inputs are appropriately captured in the prediction equations, there should be no observable slope or mean deviation in the residual errors. Conversely, if the residuals are correlated with an input that is represented in the prediction equations,

that is an indication that the representation is inadequate. If the residuals are correlated with inputs not represented in the prediction equations, that is an indication that those inputs should be added to the scheme. As expected, no evidence of biased responses to any of the EAA supplies was observed, indicating that the effects of those inputs are appropriately captured in the model. The residual errors were centered on the line of unity, and there were no systematic deviations from that line throughout the range of metabolizable AA supplies. The responses to individual EAAs were further evaluated in a similar manner using only studies that utilized infused protein or AA sources. The results were the same as for evaluations using all of the data. When studies using RP-AA were isolated in the same manner, the responses were also predicted without bias. Thus, the committee concluded that the responses to individual EAAs were unbiased.

Residual errors from Equation 6-6 were also plotted and regressed against other dietary and animal descriptors (see Figure 6-4). As above, if the descriptor was represented in the model, then the effects should be captured and there should be no pattern to the residuals. If the effect was not in the model, there should also be no pattern to the residuals if the exclusion of that effect was warranted. If there is a pattern to the residuals, this suggests the factor does have an effect and thus should be added to the model. The committee observed no clear pattern to the residuals versus EAA divided by DEInp, indicating that the independent effects of these two terms were properly captured in the model. No obvious patterns to the residuals were observed across or within studies when plotted against dietary concentrations of dFA, dNDF, or dSt, supporting the selection of a model that included dNDF but excluded dFA and dSt. The residuals were not significantly related to any other variables except for days in milk (DIM) (-0.79 g/d per DIM) and milk fat content (-0.96 g/d per milk fat percent unit). The DIM effect appeared to be associated with three studies in which DIM changed by a few days within study and thus was deemed an unreliable estimate. The committee felt the addition of milk fat may be problematic for high milk fat breeds, which were very poorly represented in the data used for model derivation. The potential effects of DIM and milk fat content should be examined further in the future.

Although the revised equation cannot be directly compared to the NRC (2001) model as the latter only provided predictions of milk production, those predictions can be converted to milk protein predictions by multiplying the predicted milk production times the reported milk protein concentration for each treatment in the meta-data.

Because the EAA coefficients of Equation 6-6 represent a reference set of parameters reflecting a group of animals producing on average 920 g of milk protein per day, additional work was undertaken to address the limitation of using a quadratic equation with a fixed maximum. Indeed, many herds today have pens of cows that produce more milk protein than can be achieved at the apex predicted by Equation 6-6. The

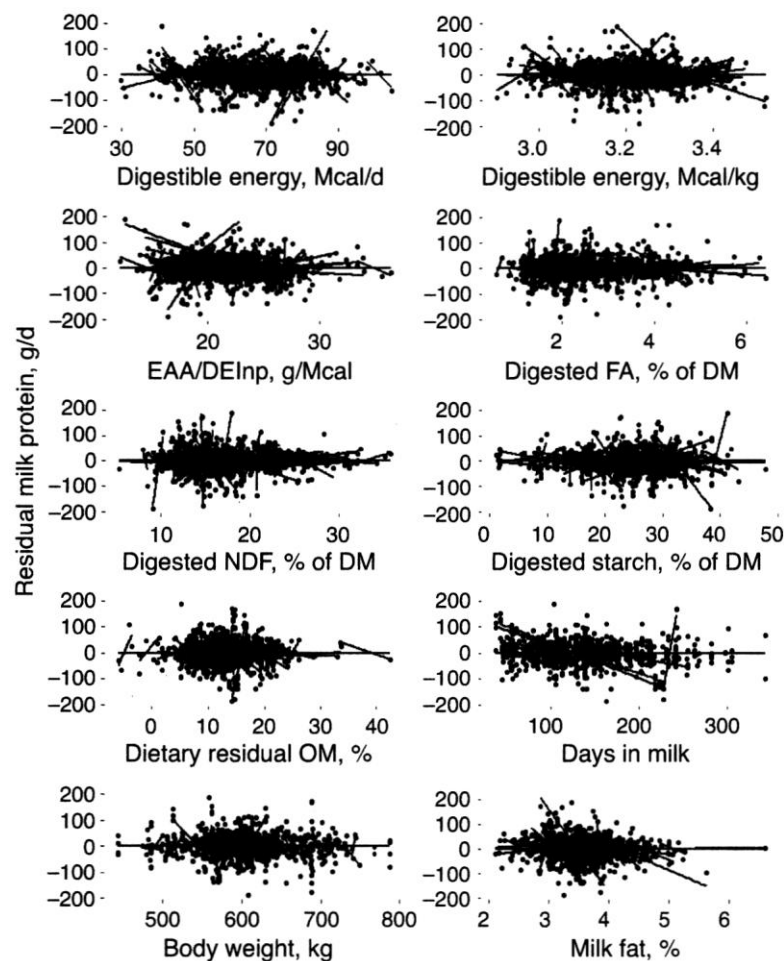


FIGURE 6-4 Residual errors from Equation 6-6 corrected for study effects versus various diet and animal factors.

form of the equation remained the same, but a set of algebraic equations was devised to allow scaling of the linear and quadratic coefficients for the EAA based on rolling herd average for milk protein (see Chapter 20).

Metabolizable Protein and Amino Acid Recommendations

The challenge in moving from a linear representation of the relationship between MP or AA supply and MPY to a multifactorial, nonlinear function with additive responses to independent variables is that one cannot define a set requirement for any of the driving variables. Indeed, an infinite combination of inputs would achieve a common level of production, and this prevents identification of a single set of requirements. Because of all of the possible combinations of EAA yielding a given level of production that can vary without bounds according to the above model, establishing reasonable biological bounds is critical. These bounds or suggested guidelines are provided for both MP and EAA based on net EAA use calculated in a factorial manner and respective target efficiencies. The factorial approach first requires the identification

and quantification of the functions that create a direct net demand on the EAA supply, either being protein secretion or protein accretion, often referred to as NP requirement. These functions are directly using EAAs that are removed, on a net basis, from the available pool of EAAs and therefore must be replenished by at least an equivalent amount on a timely basis. The second step is to assign an efficiency of utilization of the MP or EAA supply to support these different functions. The recommendations are calculated as the NP requirement divided by an efficiency. Estimation of target efficiency for MP and each EAA is detailed below.

Although the committee developed an equation to predict MPY, described in the section above, in parallel, the committee also defined MP and EAA recommendations to serve as general guidelines when MPY is entered rather than predicted, because an equation with multiple independent variables could still hide EAA imbalance. For example, a diet providing extremely low Lys but high Leu supplies (as in corn-based diets) could still predict high, but unachievable, MPY due to EAA imbalance. The first part of this section will describe the NP secretion and accretion. In the second

part, the basis of the calculation for the target efficiencies of utilization of MP and EAA is presented.

Secretion and

Accretion

The quantification of protein secretion and accretion, the basis for the estimation of the MP and EAA recommendation, is detailed below. To be consistent among the different functions, each function is quantified as TP (as opposed to CP), and the secretion or accretion of AA is calculated as TP secretion or accretion multiplied by its respective AA composition. Furthermore, secretion and accretion represent NP or net AA demand; hence, those values need to be divided by efficiency to obtain total MP (or metabolizable AA) requirement. Rather than using the word “maintenance” to refer to requirements related to scurf, endogenous urinary loss, and metabolic fecal protein (MFP), reference will be made to nonproductive functions (INRA, 2018) in this edition. Indeed, MFP represents the largest contribution to so-called maintenance, but its magnitude is driven by losses associated with the high DMI of the lactating dairy cow and, as such, cannot truly be considered a maintenance requirement necessary to sustain cow basal metabolic function. Once the net TP or AA export or accretion is determined for each function, recommendations are calculated as the sum of export and accretion, each divided by its respectively assigned (endogenous urinary loss and gestation) or target (scurf + MFP+MPY + growth + body reserves) efficiencies.

Scurf

f

NRC (2001) based the estimation of scurf requirements on Swanson (1977). After examination of more than 2,400 individual calorimetry observations on cattle collected in the U.S. Department of Agriculture-Beltsville from the mid-1960s to the mid-1990s, the equation for scurf protein from Swanson (1977) has been retained in the current revision, but the committee adjusted the equation from a CP basis to a TP basis:

$$\text{NP-scurf (g/d)} = 0.20 \times \text{BW}^{0.60} \times 0.85 = 0.17 \times \text{BW}^{0.60} \quad (\text{Equation 6-7a})$$

where 0.85 represents the TP/CP ratio of scurf, based on its AA composition and total N content.

To assess the net AA demand for scurf secretion, NP-scurf is multiplied by its AA composition. The AA composition of scurf was estimated using the head, hide, feet, and tail composition reported by Williams (1978) and van Amburgh et al. (2015). The mean of these studies, corrected to account for incomplete recovery of AA with 24-hour hydrolysis (Lapierre et al., 2019), is reported on a TP basis in Table 6-2.

$$\text{NetAA-scurf (g/d)} = \text{NP-scurf} \times [\text{AA}_{\text{corrScurf}}] / 100 \quad (\text{Equation 6-7b})$$

where $[\text{AA}_{\text{corr Scurf}}]$ is in g AA/100 g TP.

Endogenous Urinary Excretion

The estimate of endogenous urinary NP requirement used by most models is 2.75 g CP/kg BW^{0.50} based on Swanson (1977). However, urinary N excretion is derived from the catabolism of N metabolites and is not a protein secretion per se as for the other nonproductive functions. To better define the AA composition of this secretion, a literature review was conducted to quantify the composition of endogenous urinary-N (Lapierre et al., 2020). Briefly, the major N-metabolites in urine contributing to endogenous urinary losses are endogenous urea, endogenous purine derivatives (PD), creatinine and creatine, hippuric acid, and 3-methyl-His. From this review, daily excretion (per kg of BW) of endogenous urea has been quantified as 10 mg N, creatinine excretion as 9.46 mg N (25.5 mg creatinine), and creatine excretion estimated as 0.37 that of creatinine (i.e., 3.5 mg N). Daily urinary excretion of endogenous PD was estimated to average 27.1 mg N/BW^{0.75}. Urinary excretion of His (mg/d) as 3-methyl-His equals

$7.82 + 0.55 \times \text{BW}$. Using a database from Spek et al. (2013), the remainder of the “measured” endogenous urinary-N excretion, representing 46 percent of this fraction, was assumed to be hippuric acid, formed in the liver to detoxify benzoic acid originating from rumen fermentation of dietary phenolic compounds. Although hippuric acid cannot be purely defined as “endogenous,” it has probably been included in previous estimates of endogenous urinary excretion. Beside hippuric acid, most of the estimations of urinary excretion were based on BW, and therefore the endogenous urinary-N excretion was expressed relative to BW and averaged 53 mg N/kg BW. Using a different approach, the INRA (2018) estimated a daily endogenous urinary-N loss averaging 50 mg N/kg BW. Compounds constituting endogenous urinary-N excretion are not protein per se but N-metabolites that have AA as their origin; therefore, the standard N to CP conversion is assumed, meaning that the TP/CP ratio was set as 1.0.

$$\text{NP-endogenous urinary (g/d)} = 53 \times 6.25 \times \text{BW} \times 0.001 \quad (\text{Equation 6-8a})$$

The initial reason to revisit the endogenous urinary excretion was to identify which AAs were upstream of urinary endogenous losses. After examination of the synthesis process of the N-compounds detailed above, only the endogenous urea and 3-methyl-His excretions create a direct demand on EAA if Arg is considered as a conditionally essential AA. Endogenous PD are synthesized from Asp, Gin, and Gly; creatine and creatinine from Arg and Gly; and hippuric acid from Gly. Therefore, for all EAAs except His, endogenous urinary urea excretion (0.010 g N/BW per day) is used to estimate the AA endogenous urinary loss, assuming that this loss has the AA composition of the whole empty body reviewed in Lapierre et al. (2020; see Table 6-2). Therefore, for all EAAs except His:

$$\text{NetAA-endogenous urinary (g/d)} = 0.010 \times 6.25 \times \text{BW} \times [\text{AA}_{\text{corr-WholeEmptyBody}}] / 100$$

(Equation 6-8b)

where BW is in kg, and $[\text{AA}_{\text{corr-WholeEmptyBody}}]$ is in g AA / 100 g TP.

For His, the loss of urinary 3-methyl His must be added:

$$\begin{aligned} \text{NetHis-endogenous urinary (g/d) His, g/d} &= (0.010 \times 6.25 \times \text{BW} \times [\text{His}_{\text{corr-WholeEmptyBody}}] / 100) + (7.82 \\ &+ 0.55 \times \text{BW}) / 1,000 \end{aligned} \quad (\text{Equation 6-8c})$$

where $[\text{His}_{\text{corr-WholeEmptyBody}}]$ is in g His/100 g TP.

Metabolic Fecal Protein

The MFP secretion should represent the endogenous proteins secreted or sloughed in the gut lumen and not digested in the small intestine. They create a net demand on the digestible flow of AA: indeed, except for Phe, more than 80 percent of the EAAs used by the gut are derived from circulating (i.e., from previously absorbed) AA (MacRae et al., 1997), whereas the remainder would be from intestinal supply but is still derived from estimated metabolizable AA. NRC (2001) estimated MFP from DMI. However, rumen microbial synthesis from urea does not create a demand on metabolizable AA and should not be included in MFP. Accordingly, because this MFP value consists of “bacteria and bacterial debris synthesized in the cecum and large intestine, keratinized cells, and a host of other compounds” (Swanson, 1982), NRC (2001) corrected the initial calculation for the amount of undigested ruminal bacterial CP appearing in the feces of dairy cattle, and MFP (g MP/d) was estimated as $[(\text{DMI (kg)} \times 30) - 0.50((\text{bacterial MP}/0.80) - \text{bacterial MP})]$. This estimate was used as is for MP requirements, with no conversion of CP secretion to TP and no efficiency factor assigned to MFP excretion.

Using an isotopic dilution approach, Ouellet et al. (2002, 2010) developed a model to estimate MFP, allowing quantification of undigested endogenous proteins. Also, in a meta-analysis using 65 growing-finishing cattle studies (291 treatment means) and 43 dairy cow studies (164 treatment means), Marini et al. (2008) developed an equation regressing intake of digestible CP on CP intake. The equation had an intercept of 30 g CP/kg DMI, which represented MFP plus undigested bacteria synthesized from urea. This equation, including the NDF concentration of the rations, was adapted to fit the measured fecal MFP originating only from undigested endogenous proteins and adjusted to represent the ileal endogenous flow observed by Ouellet et al. (2002, 2007, 2010). In addition, endogenous secretions occurring across the hindgut were included based on observations in sheep (Sandek et al., 2001) and using the ileal flow of small intestinal endogenous protein secretion of 5.1 g CP/kg DMI

(Ouellet et al., 2007). However, due to the scarcity of data on the exact origin of this N, half of this input was assumed to originate from endogenous proteins and the other half from urea. Therefore, the daily loss of CP as MFP was estimated from Lapierre et al. (2020) as

$$\text{CP-MFP (g/d)} = (11.62 + 0.134 \times \text{NDF}_{\% \text{DM}}) \times \text{DMI} \quad (\text{Equation 6-9a})$$

Note that the term MFP is kept, although the forestomach and small intestinal loss was truly measured at the ileum. Assuming 73 percent of TP in MFP, based on its AA composition and total N content,

$$\text{NP-MFP (g/d)} = \text{CP-MFP} \times 0.73 \quad (\text{Equation 6-9b})$$

The AA composition of the MFP is based on the AA composition of ruminal and abomasal isolates from Ørskov et al. (1986), except for Leu, for which only the rumen isolates were used, and the endogenous flow at the ileum in pigs (Jansman et al., 2002). The committee assumed that 70 percent of the MFP is from undigested duodenal flow and the remaining 30 percent from the intestine (Ouellet et al., 2002, 2010). The averaged composition is detailed in Table 6-2. Therefore, individual AA secretion in MFP would be

$$\text{NetAA-MFP (g/d)} = \text{NP-MFP} \times [\text{AA}_{\text{corr-MFP}}] / 100 \quad (\text{Equation 6-9c})$$

where $[\text{AA}_{\text{corr-MFP}}]$ is in g AA/100 g TP.

Milk

Milk TP secretion is the easiest export protein to measure. The factor to convert the N concentration into CP in milk should be 6.34, related to the AA composition of milk (Karmann and van Boekel, 1986), rather than 6.38. As for the other secreted proteins, it is expressed as TP. If the TP/CP ratio is not known, the NPN content of the milk CP is assumed to be 4.9 percent (DePeters and Cant, 1992), and a fixed ratio of TP/CP of 0.951 will be used by the model.

Because early studies reported less than a 3 percent difference in the EAA composition of milk protein produced from cows fed solely NPN sources and milk protein from control cows (Syvaoja and Virtanen, 1965) and no effect of forage/grain ratio on the AA composition of milk (Featherston et al., 1964), the AA composition of milk protein has been assumed to be constant. This may prove to be untrue under more severe dietary manipulations such as experienced during AA deletion studies. However, additional data are required to further test that hypothesis. Milk AA composition has therefore been calculated based on the primary structure of the reference protein of each family, as detailed by

Farrell et al. (2004) using the approach of Swaisgood (1995). Milk TP fractions reported in 15 manuscripts published between 1980 and 2012 (Lapierre et al., 2020) averaged 82.4 percent casein (percentage of total protein: 35.2 percent as 1-casein, 7.6 percent as2-casein, 30.9 percent (3-casein, and 8.7 percent K-casein) and 17.6 percent whey (percentage of total protein: 3.7 percent α -lactalbumin, 10.5 percent (3-lactoglobulin, 1.04 percent albumin, 1.64 percent IgG1, 0.18 percent IgG2, 0.04 percent IgA, 0.33 percent IgM, and 0.21 percent lactoferrin). The AA composition of milk TP calculated using this procedure is presented in Table 6-2. With this approach, there is no need to correct for incomplete recovery from 24-hour hydrolyses.

$$\text{NetAA-Milk (g/d)} = \text{NP-Milk (g/d)} \times [\text{AA}_{\text{calc-Milk}}] / 100$$

(Equation 6-10)

where NP-milk = milk TP yield and $[\text{AA}_{\text{calc-Milk}}]$ is in g AA/100 g TP.

Pregnancy

Knowledge of the rates of nutrient accretion in conceptus tissues (fetus, placenta, fetal fluids, and uterus) is limited for dairy cattle. Pregnancy requirements were calculated as described for energy in Chapter 3. The size of the gravid uterus at a given gestation day was estimated based on calf birth weight (see Chapter 3). From that function, the daily gain in mass (kg/d) of the gravid uterus ($\text{Gain}_{\text{GrUter}}$) was calculated (see Chapter 3), and that gain was assumed to contain 125 g of protein/kg of wet weight (Bell et al., 1995).

$$\text{NP-Gestation (g/d)} = \text{Gain}_{\text{GrUter}} \times 125$$

(Equation 6-11a)

Use of a constant fraction of protein for gravid uterine gain based on that derived at parturition will introduce some bias at time points prior to parturition, but the errors are likely small and will cancel out by parturition. The model includes involution of the uterus postpartum. The rate of involution of uterine tissue postpartum and the fate of the AA from the involuting tissue are unknown. Because of the lack of data, the committee assumed complete involution required 4 weeks (see Chapter 3), and related AA would contribute to NP supply.

The AA composition of protein accretion associated with pregnancy is based on the AA composition of the whole empty body (see Table 6-2) because direct data are unavailable. Individual AA accretion for pregnancy is

$$\text{NetAA-Gestation (g/d)} = \text{NP-gestation} \times [\text{AA}_{\text{corr-WholeEmptyBody}}] / 100$$

(Equation 6-11b)

where $[\text{AA}_{\text{corr-WholeEmptyBody}}]$ is in g AA/100 g TP.

Based on the equations above and assuming typical DMI of dry cows, diet DM would need to contain 829 g of MP (9.6 percent CP) at 60 days prepartum and 956 g of MP (13.5 percent CP) at 5 days prepartum, if the efficiency of converting MP to NP-Gestation is 0.33 as used by NRC (2001) and 60 percent of dietary CP is retained in MP. Data are very limited on AA metabolism in gestating, nonlactating dairy cows. In cows at this physiological status, liver removal of group 1 AA (His, Met, Phe + Tyr, and Trp) relative to net portal absorption is approximately twice as large as the ratio observed in lactating dairy cows (Wray-Cahen et al., 1997; Larsen et al., 2015). Although it is not clear if high liver removal of group 1 AA prepartum is related to excess protein feeding or difference in physiological status, data are not sufficient to change the efficiency used in NRC (2001).

Using milk yield in the subsequent lactation as the response variable, dry cows' diets with as little as 11 percent CP appear adequate (see Chapter 12). However, few individual studies included dry cows fed diets with < 10 percent CP. Low-protein diets may reduce DMI and fiber digestion, resulting in less NEL than predicted. Furthermore, based on a meta-analysis, diets with at least 14.5 percent CP yielded positive responses when fed to late-gestation nulliparous animals (see Chapter 12). To provide adequate RDP (-10 percent of DM, discussed above) for DMI and digestibility, dry cows' diets would need to contain about 12 percent CP (greater concentrations would likely benefit late-gestation heifers), and the committee recommends maintaining that concentration of CP. Another justification for recommending higher concentrations of CP is because the protein required to produce colostrum is not included in requirement calculations. Holstein cows may secrete more than 1 kg of protein in first milking colostrum. Although colostrum synthesis only occurs over a few days, it still represents a significant demand for AA. Estimated MP requirements for gestation by a dry cow producing a calf with a birth weight of 44 kg (see Table 3-3, Chapter 3) are about 25 percent less at 60 days prepartum and 37 percent greater at 5 days prepartum compared to NRC (2001).

Growth

Target frame growth rates for an average Holstein cow (mature birth weight of 700 kg) during first and second lactations are 0.19 and 0.15 kg/d, respectively, assuming that postpartum BW at first and second calvings is 82 percent and 92 percent of mature birth weight. For Jersey cows (with mature birth weight of 520 kg), targets are 0.14 and 0.11 kg/d for first and second lactations. These target rates assume that frame growth occurs consistently over the lactation, regardless of changes in intake, milk production, and body condition. The protein concentration of empty gain is a function of BW relative to mature weight, and empty BW gain associated with frame growth is considered 85 percent of live BW gain in cows. Using equations from Chapter 11 for cows at 82 percent and 92 percent of mature BW, the protein content of BW gain

associated with frame growth would be 11.2 and 10.6 percent for first- and second-lactation cows. Because values are so similar, growth during lactation was considered to contain 11.0 percent protein. Therefore, during lactation:

$$\text{NP-growth (g/d)} = \text{Frame weight gain (g/d)} \times 0.11 \times 0.86 \quad (\text{Equation 6-12a})$$

Additional details are in Chapter 11. However, if the change in BW is not frame growth but rather a change in body reserves, the protein content is assumed to be 8.0 percent protein. In the text, NP-growth is the sum of the NP for frame growth plus NP for body reserves.

The default values for frame gain during lactation can be altered by users. The 0.86 is the ratio of TP/CP derived from the AA_{corr} composition of the CP. The AA composition of growth is based on the AA composition of the whole empty body (see Table 6-2). Individual AA accretion for growth is

$$\text{NetAA-growth (g/d)} = \text{NP-growth} \times [\text{AA}_{\text{corr-WholeEmptyBody}}] / 100 \quad (\text{Equation 6-12b})$$

where $[\text{AA}_{\text{corr-WholeEmptyBody}}]$ is in g AA/100 g TP.

Under most conditions, the amount of TP and AA required for growth during lactation will be extremely low (at target rates of growth, TP will equal 13 to 18 g/d).

Efficiency of Utilization of Metabolizable Protein and Amino Acids

The approach used to estimate MPY acknowledges that the efficiency of utilization of MP or AA is variable. European models (e.g., NorFor, 2011; Van Duinkerken et al., 2011; INRA, 2018) have adopted a variable efficiency of utilization of MP. Because the efficiency of utilization of MP and AA is variable, the concept of a “single,” fixed MP or AA requirement is no longer tenable. One must consider the problem from a marginal return basis where the system is optimized when the marginal return from the last unit of input nutrient becomes 0. However, only response function for milk protein could be derived; therefore, the overall NP response to nutrient inputs is not reflected in the model system. Despite this limitation, if one assumes that nonproductive NP requirements are met before milk protein output is maximized, a target efficiency for use of AA and MP within the model system can likely be identified as an approximation of the point where the marginal return becomes 0.

The use of a combined efficiency for use of AA and MP for the secretion functions of scurf, MFP, and milk has been suggested (Lapierre et al., 2007a). The efficiency of use for gestation is not known, and different efficiencies (e.g., 0.50, 0.33, and 0.65) have been proposed by different committees (NRC, 1989, 2001; NASEM, 2016) with little or no supporting data. Because of lack of data, the committee retained the

efficiency (0.33) used by the previous committee (NRC, 2001)

for nonlactating animals. The efficiency of AA for growth is also lower (0.40; NASEM, 2016). However, the NP needed to support growth and gestation by lactating cows is very small relative to other NP uses. For simplicity, growth was assumed to have the same efficiency as other functions for lactating cows but not for growing heifers or dry cows. An efficiency of 1.0 was used for endogenous urinary losses, because, contrary to the other secretions that are proteins, these losses are end products of N-metabolite metabolism, as in INRA (2018). For lactating cows that are not pregnant, efficiency of utilization of MP (Eff_MP) and of individual AA (Eff_AA) for the other functions is assumed variable and is calculated as

$$\text{Eff_MP} = (\text{NP-scurf} + \text{NP-MFP} + \text{NP-milk} + \text{NP-growth}) / (\text{MP supply} - \text{NP-endogenous urinary}) \quad (\text{Equation 6-13a})$$

and

$$\text{Eff_AA} = (\text{NetAA-scurf} + \text{NetAA-MFP} + \text{NetAA-Milk} + \text{Net AA growth}) / (\text{metabolizable AA} - \text{NetAA-endogenous urinary}) \quad (\text{Equation 6-13b})$$

Using a combined efficiency for scurf, MFP, and lactation, and 100 percent efficiency for endogenous urinary resulted in improved predictions of the efficiency of MP use compared to fixed efficiencies for the nonproductive functions and a variable efficiency solely for MPY (Sauvant et al., 2015). The CNCPS-Version 6.5 (van Amburgh et al., 2015) also opted for a combined efficiency of utilization of AA, which included scurf, MFP, lactation, and endogenous urinary loss.

The efficiency of utilization of MP and individual AAs was calculated as detailed above for each treatment included in the database described in the section “Estimation of MPY.” The committee then, a priori, removed some studies for uncertainties regarding the true availability of AA supply (e.g., dietary Met analogs) and studies involving intravenous infusions to end up with a final database of 921 treatment means. As expected, the calculated efficiencies were highly variable (see Table 6-4).

Target Efficiencies of Utilization of Metabolizable Protein and Amino Acids

Although the efficiency of utilization of MP and AA to support protein export plus accretion is variable, guidelines (not requirements) for adequate supplies of MP and individual EAAs are presented, based on the assumption that energy requirements are met. Target efficiencies were estimated as follows. First, studies were coded to look specifically at the increment of MP supply. Because sufficient details were not included to estimate protein accretion, an average growth of 0.19 g/d (equivalent to 18 g TP/d as $0.19 \times 0.11 \times 0.86$, where

TABLE 6-4 Proposed Target Efficiencies of Converting MP and EAA to Export Proteins and Body Gain and the Associated Descriptive Statistics for the Database

MP or AA	Target	Mean	SD	Minimum	Maximum
MP	0.69	0.66	0.102	0.35	1.00
His	0.75	0.78	0.141	0.34	1.21
Ile	0.71	0.61	0.088	0.36	0.93
Leu	0.73	0.67	0.120	0.24	1.01
Lys	0.72	0.67	0.106	0.35	1.05
Met	0.73	0.71	0.120	0.36	1.13
Phe	0.60	0.54	0.087	0.23	0.82
Thr	0.64	0.58	0.078	0.35	0.86
Trp	0.86	0.77	0.125	0.42	1.20
Val	0.74	0.66	0.099	0.37	1.01

0.11 = protein/empty BW gain and 0.86 = empty BW/BW) was assumed for first-lactation Holstein cows and 0.14 kg/d (13 g TP/d) for first-lactation Jersey cows, as established in the previous section, whereas gestation requirements were set to 0. Equations relating the sum of export plus accretion of NP or individual NetAA to their respective efficiency of utilization (dependent variable) were developed using the `rma.mv` function from the `metafor` package in R (Viechtbauer, 2010; R Core Team, 2013, version 3.4.1, 2017-06-30). The hierarchy of the studies was considered, and data were weighted by

$\sqrt{n}$. Unbiased estimates of fixed effects and valid estimates of SE were obtained using the `robust` function in the `metafor` package. The linear and quadratic terms were significant ($P < 0.02$) for MP and all EAAs. The target efficiencies (i.e., the efficiency when the sum of NP or EAA export plus accretion was maximal) were obtained using the first derivative of these equations relating the sum of export plus accreted NP or individual EAA to the efficiency of utilization of MP or individual EAA. Combining these results with careful examination of efficiencies observed in deletion studies, the target efficiency of Phe was increased from 0.57 to 0.60 to better reflect observations reported when only Phe supply was modified. These target efficiencies, as well as statistics of observed efficiencies, are in Table 6-4. The target efficiency of Phe and Thr may be slightly underestimated, but due to the lack of data, an underestimate of efficiency was preferred to a resultant underestimate of recommended supply.

Target efficiencies can be used to assess diets and identify AAs that may be limited or in excess. Efficiencies greater than the target efficiencies are an indication that for those EAAs, supply may be short and negatively impact MPY. On the other hand, EAAs with efficiencies less than the target are more likely to be in excess, which may not be harmful to the animal (unless there is a severe imbalance) but has economic and environmental costs. Under certain situations, efficiencies greater than target can be achieved without necessarily adversely affecting MPY, as evidenced by the number of observations above the target values. Unfortunately, despite their crucial biological roles, involvement of EAAs in functions like loss or regain of body tissue, reproduction, and

immune response is less well described in the literature than the functions used in the current version to estimate the efficiency of EAAs, especially when it comes to quantification; hence, they could not be included in the current calculations. However, even if not accounted in the current estimations of the efficiencies, a ranking could be done and the EAA with the largest difference between the observed and the target efficiencies would be the EAA with the shortest supply. So even though the true efficiency of each EAA might not be exactly right, the ranking among EAAs might be a useful indication of their relative supply. Also, the current framework based on biological functions could be adapted as more knowledge is gained on these different functions, and they could be included in the equations of estimations of the efficiency of utilization of EAAs. Additional work is required to more clearly identify the potential optimal efficiencies that could be achieved with rations balanced for all of the EAAs as well as defining confidence intervals surrounding these efficiencies.

Recommendations of Metabolizable Protein and Essential Amino Acids

After characterization of the daily NP and NetAA secretion or accretion and defining target efficiencies, recommendations for adequate MP and AA supply can be determined by dividing NP or NetAA by the target efficiencies. Efficiencies of using MP for growth (approximately 0.40) and gestation (0.33) are substantially lower than the efficiency for other uses. In lactating cows, growth is a very minor use of MP, and because of the assumption needed, when deriving recommendations for MP and EAA, it was given the same target efficiency as milk. The NP accreted for gestation in late-lactation, lower milk-yielding cows can be substantial; hence, its efficiency was kept at 0.33. The NP supply equations discussed above are used to estimate NetAA requirements for the various functions and thus are based on protein content calculated from AA composition rather than CP (i.e., nitrogen $\times$ 6.25). The base data for those equations, however, were CP; therefore, MP requirements (but not AA requirements) are calculated as

TABLE 6-5 Example of Adequate EAA Supplies for a Mature Nonpregnant Cow (650 kg BW) Consuming 26 kg/d of a Diet with 34 Percent NDF with Graded MPY, Based on Target Efficiencies in Table 6-4

MPY, g/d	His	Ile	Leu	Lys	Met	Phe	Thr	Trp	Val	MP
1,000	55	112	187	159	50	117	110	26	125	2,122
1,200	63	130	216	183	58	134	124	30	144	2,411
1,400	71	147	245	208	67	152	139	34	162	2,701
						% MP				
1,000	2.60	5.29	8.83	7.48	2.35	5.51	5.19	1.23	5.89	—
1,200	2.61	5.37	8.97	7.60	2.42	5.57	5.16	1.24	5.96	—
1,400	2.62	5.44	9.08	7.69	2.46	5.63	5.14	1.25	6.01	—

$$\begin{aligned} \text{Recommended MP supply} = & [(\text{NP-scurf} + \text{NP-MFP} \\ & + \text{NP-milk} + \text{NP-growth}) / \text{Target_Eff_MP}] \\ & + (\text{NP-gestation} / 0.33) + \text{NP-endogenous urinary} \end{aligned}$$

(Equation 6-14a)

$$\begin{aligned} \text{Recommendation for individual EAA} \\ \text{digestible flow} = & [(\text{NetAA-scurf} + \text{NetAA-MFP} \\ & + \text{NetAA-Milk} + \text{NetAA-growth}) / \text{Target_Eff_AA}] \\ & + (\text{NetAA-gestation} / 0.33) + \text{AA-endogenous urinary} \end{aligned}$$

(Equation 6-14b)

For late-gestation cows and heifers, equations are

$$\begin{aligned} \text{Recommendation for MP supply} = & (\text{NP-scurf} + \text{NP-MFP}) \\ & / \text{Target_Eff_MP} + (\text{NP-gestation} / 0.33) + (\text{NP-growth} \\ & / 0.40) + \text{NP-endogenous urinary} \end{aligned}$$

(Equation 6-14c)

$$\begin{aligned} \text{Recommendation for individual EAA} \\ \text{digestible flow} = & [(\text{NetAA-scurf} + \text{NetAA-MFP}) \\ & / \text{Target_Eff_AA}] + (\text{NetAA-gestation} / 0.33) \\ & + (\text{NetAA-growth} / 0.40) + \text{AA-endogenous urinary} \end{aligned}$$

(Equation 6-14d)

Emphasis should be placed on meeting the EAA rather than MP recommendations. Indeed, in studies where EAA but not MP recommendations were met, cows fed low MP diets and infused with mixtures of EAA (meeting EAA but below MP recommendation) had similar MPY than cows receiving an infusion of EAA + NEAA (at EAA and MP recommendations) (e.g., Schwab et al., 1976; Metcalf et al., 1996; Doepel and Lapierre, 2010). Recommendations for MP are given mainly for general comparison and are, on average, lower than the NRC (2001) recommendations. The current recommendations do not include any requirement for endogenous duodenal flow, as it is not included in the supply. Also, in the current edition; (1) the relative proportion of EAA to MP is slightly higher than in the previous edition because the digestible flows of AA are corrected for incom-

plete recovery of AA in 24-hour hydrolyses and because MP recommendations are lower than in the previous edition, and (2) a recommendation for a single ratio of AA to MP is not given because this ratio changes as the proportion of the net output of milk increases relative to total output (see Table 6-5). However, when calculated in g/d for an average cow, the recommendations of Lys and Met are in a similar range compared with the estimations obtained with the proportional approach in NRC (2001).

MEETING THE RECOMMENDATIONS FOR METABOLIZABLE PROTEIN AND AMINO ACIDS

Balancing for Metabolizable Protein

The relationship between MPY and protein supply is better when supply was expressed as MP compared with CP, with a slight improvement of the relationship when the quadratic term of MP was included (e.g., Huhtanen and Hristov, 2009; Lapierre et al., 2009). Because MP supply is the sum of MP from MCP and from RUP, economics have generally favored achieving maximum or near-maximum MCP because of the relatively low cost of RDP and the good AA profile of MCP. Santos et al. (1998) published a comprehensive review of the effects of replacing soybean meal with various sources of RUP on protein metabolism and production, and in 76 percent of the metabolism studies, higher RUP (i.e., lower RDP) decreased MCP flow to the small intestine. Similarly, Ipharraguerre and Clark (2005) reported that the mean milk production responses to replacing soybean meal with RUP supplements varied from

-2.5 to +2.75 percent. In addition, inadequate RDP supply can reduce digestibility and energy supply (Lee et al., 2012b; Luo et al., 2018). Therefore, RDP and fermentable energy provisions should be complementary to support efficient MCP synthesis. Although balancing diets for MP can be used as a general guide, balancing for individual EAAs derived from the MP, assuming that RDP supply is sufficient to maintain an efficient rumen fermentation,

should allow achieving similar animal performance but at a lower total MP and CP supply. Improved balance or supply of certain metabolizable EAAs may also increase DMI (see later section).

Balancing for Individual Amino Acids

AAAs are the nutrients used as building blocks for the synthesis of all proteins. Metabolizable AAAs are also vital to support a multitude of metabolic pathways, but to a lesser quantitative extent. For example, Met contributes to multiple pathways, including provision of a methyl group for many transmethylation reactions (Manjarin et al., 2014). Also, AAAs except Leu and Lys can serve as precursors for gluconeogenesis; all AAAs can be converted to FAs or serve as immediate sources of energy when oxidized to CO₂ (e.g., Bequette et al., 1996; Lapierre et al., 2002). Whether these needs are fully met when needs for protein synthesis are met is not clear. Some of these reactions and molecules can influence longer-term animal health, and thus the effect of a deficiency may not manifest within the time frame used for many production studies. In addition, some AAAs play an active role in intracellular signaling, especially involving the mechanistic target of mTOR (Arriola Apelo et al., 2014b). The effects of AA, energy supply, insulin, and other factors on these signaling pathways appear to explain why the concept of the barrel and stave with a single limiting AA does not fully explain observed lactational responses, whereas the additive model represented by Equation 6-6 captures more of the observed variation. Current knowledge, however, limits the estimation of the EAA recommendations to their effect on protein secretion and accretion and the associated efficiency of utilization. Further research is required to explore interactions among EAAs and energy (Arriola Apelo et al., 2014a). Defining recommendations based only on such productive functions, however, does not mean that the roles of AAAs in functions other than direct incorporation into protein are not important but rather reflects the lack of knowledge on the adequate assessment of these roles and the quantification of these demands.

Recommendations for MP are mainly given as a general guideline and to maintain historical perspective. As pointed out by NRC (2001), diets that provide sufficient amounts of MP may still be deficient in one or more EAAs. Conversely, diets that are apparently deficient in MP may also be completely sufficient in EAAs, and thus the emphasis should be put on balancing for individual EAAs. Indeed, from experiments where the supply of a single AA or a group of AAAs has been changed, total MP supply per se is not the best predictor of MPY. For example, deletion of His, Lys, or Met from a total AA mixture infused postruminally did not change total MP supply significantly but decreased MPY by 20 percent (Weekes et al., 2006). Deletion of His or Phe from an EAA mixture decreased MPY by 23 percent (Doelman et al., 2015), whereas deletion of Phe from a complete AA

mixture decreased MPY by 16 percent (Doepel et al., 2016) with a minimal impact on MP supply. Infusion of NEAA did not affect MPY, despite a significant increase in MP supply (Doepel and Lapierre, 2010). Admittedly, the variation of the proportion of NEAA to MP supply is often beyond the expected physiological range in infusion studies, but variations due to dietary alterations can also be substantial. For example, from the database used by Doepel et al. (2004), with digestive flows of EAA estimated with NRC (2001), the proportion of total EAA relative to MP supply in control treatments varied from 42 to 48 percent (mean 45.4 percent); so, a ration providing 3,000 g/d of MP could supply between 1,260 and 1,440 g of EAA. In Lee et al. (2012a) and Giallongo et al. (2016), inclusion of RP-His, -Lys, and -Met allowed MP to decrease by 450 g/d while maintaining MPY. In Haque et al. (2015), substituting EAA to maintain the EAA supply while decreasing MP supply by 375 g/d maintained MPY. Together, these results indicate that the supply of individual EAAs and not total MP drives MPY. These studies show that one AA per se cannot be declared “limiting.” The limitation occurs when the supply is sufficiently low to affect protein synthesis, which can be encountered for all EAAs, as shown with deletion studies (e.g., Weekes et al., 2006; Doelman et al., 2015; Doepel and Lapierre, 2016) and as reflected in Equation 6-6.

Earlier work suggested that His might be limiting in grass silage-based diets (Vanhatalo et al., 1999; Korhonen et al., 2000; Huhtanen et al., 2002). However, diets fed in these studies were also low in CP (between 13.2 and 14.7 percent CP). Similarly, in corn silage-based diets with low protein concentration (14 percent CP), RP-Met plus RP-Lys were unsuccessful in restoring MPY to the level observed with adequate MP diets (Lee et al., 2012b,c). The addition of RP-His to RP-Met and RP-Lys to a deficient MP diet restored MPY to the level of the adequate MP diet. Inclusion of single AA as RP-His or RP-Lys or RP-Met to a deficient MP diet was insufficient to restore MPY to that of the adequate MP diet, but the combination of the three RP-AAAs was effective in maintaining MPY similar to the adequate MP diet (Giallongo et al., 2016). The overall conclusions were as follows: (1) at low CP concentrations, the proportion of MP originating from microbial protein increases; (2) although MCP is considered to have a good AA profile, it is low in His (see Table 6-2) relative to the composition of proteins involved in the determination of requirements (milk, MFP, and endogenous urinary); and (3) when the proportion of MCP relative to total MP supply increases, His supply decreases more rapidly than the supply of the other AAAs and may create a deficiency. Low-protein diets may have multiple EAA deficiencies, which will be apparent when metabolizable EAA supply is evaluated.

With sufficient knowledge, rations can be balanced for individual EAAs, and once the recommendation for each EAA is met, there should be no need to balance for MP. The modeling work undertaken herein and by others represents a significant step on that path. The relationships among His,

Lys, and Met and MPY are well defined by the literature, but other EAAs are not as well studied. Whether those EAAs are generally sufficient in most diets is uncertain, and the lack of clean comparisons presumably contributed to greater uncertainties associated with their coefficients in Equation 6-6. Clearly, all EAAs are important, and it is possible that some NEAAs are important (Luo et al., 2018). More research is needed to better characterize these additional AAs. The current work represents a first step and provides a framework that is analogous to the approach taken in swine and poultry nutrition, where diets are prepared with large proportions of individual EAA ingredients. This practice has even led to a shortage of total N for de novo synthesis of NEAA, which can be alleviated through supply of NPN in the diet (e.g., Mansilla et al., 2017). Such a shortage of NPN is less likely to happen in dairy rations because of the significant recycling of BUN to the rumen and the need for adequate RDP to support maximal microbial growth.

Amino Acid Metabolism

Essential Amino Acids

Essential AAs were initially classified into two functional groups, groups 1 and 2, based on their pattern of utilization by the mammary gland of dairy cows (Mephram, 1982): AAs from group 1 have a stoichiometric transfer of mammary uptake to milk protein, and group 2 AAs have a mammary uptake greater than secretion in milk protein. Further measurements of AA net flux across the portal-drained viscera (PDV), the liver, and the mammary gland confirmed these classes. This grouping does not mean that the EAAs in the same group share common metabolic pathways. Group 1

AAs, including His, Met, Phe + Tyr, and Trp are, on a net basis, substantially extracted by the liver, barely removed by peripheral tissues, and extracted by the mammary gland on a net basis nearly equal to that secreted in milk protein (Lapierre et al., 2012a). In contrast, group 2 AAs, including Ile, Leu, Lys, and Val, are catabolized by the PDV, barely removed by the liver, catabolized by peripheral tissues, and extracted by the mammary gland at a higher rate than secreted into milk protein. The excess uptake of group 2 AAs relative to milk protein output increases with increased AA supply. Although their metabolism pattern differs from the branched-chain AAs and Lys, Arg and Thr could also be categorized in group 2. Mammary uptake of Arg and perhaps Thr is larger than secretion in milk protein; it did not vary with their supply (Lapierre et al., 2012a). The excess mammary uptake of group 2 AAs, relative to milk supply, can be used for mammary synthesis of NEAAs. For example, the carbons from Leu were incorporated into Glu (Wohlt et al., 1977). The N of excess Leu in goat (Rubert-Aleman et al., 1999) and of Lys in cows (Lapierre et al., 2009) was transferred to many NEAAs, either used for milk protein synthesis or released into the mammary vein. Similarly, the large excess of Arg uptake

is used to support mainly Pro but also Ala, Asp, Glu, Gly, and Ser (Clark et al., 1975; Roets et al., 1979). Some AAs such as Leu and Lys can provide ketogenic products that are used for fat synthesis and ATP or NADPH production. A balance model summarizing these interchanges, including metabolism of nonnitrogenous compounds, was found to largely be in balance for carbon and nitrogen, suggesting that knowledge regarding inputs and outputs for the mammary gland is mostly complete (Hanigan et al., 2001).

Nonessential Amino Acids

The classification of AA as being EAA or NEAA originates from research with nonruminant animals (Hou et al., 2015). Research with ruminants, especially dairy cattle, is extremely limited but indicates the classification is similar to nonruminants (Black et al., 1952, 1955, 1957; Black and Kleiber, 1958). In these former studies, however, Arg was included in the NEAA group. NRC (2001) termed Arg as essential because even if Arg can be synthesized by animal tissues, it would be at rates insufficient to meet requirements, particularly for high levels of production. However, deletion of Arg from an AA mixture did not affect MPY (Doepel and Lapierre, 2011), but Arg was significant in the model selection process above. It was removed from the model due to the conditional-essentiality designation rather than for statistical reasons. Given its presence in the best models, further work to define its role in MPY is warranted.

Other studies supported the general distribution of AAs into EAAs and NEAAs in a more indirect way. Postruminal infusions of a mixture of the 9 EAAs plus Arg was sufficient to increase MPY, and addition of the NEAA to the infused mixture did not further increase MPY (Schwab et al., 1976; Metcalf et al., 1996; Doepel and Lapierre, 2010). These observations indicate that under normal feeding conditions, individual NEAAs absorbed in amounts less than required for metabolic need can be synthesized at a sufficient rate such that MPY is not affected. Although there is no evidence that NEAAs as a group can limit MPY when dairy cattle are fed conventional diets, research is too limited to totally rule out the potential importance of specific NEAAs. Indeed, regulation of the rate of protein synthesis may be dominant over direct substrate effects (Luo et al., 2018).

There might also be an impact of low NEAA supply because EAAs are required to synthesize several of the NEAAs. For example, in nonruminants such as swine and poultry, the EAAs, Met and Phe, are precursors of the NEAAs, Cys and Tyr, respectively. Jorgensen and Larson (1968) reported that liver and mammary tissue from cows was able to synthesize Tyr from Phe. In lactating goats, 5.3 percent of the net Phe taken up by mammary glands was converted to Tyr (Hanigan et al., 2009). In dairy cows, the conversion of Phe to Tyr within the mammary gland represented

7 percent of the Tyr secreted into milk protein (Lemosquet et al., 2010). If the digestive flows of Cys or Tyr are limited,

their *de novo* synthesis from their EAA-precursor might be insufficient. No clear studies involving dairy cattle have measured the extent that these two NEAAs, Cys and Tyr, can “spare” Met and Phe. Pruekvimolphan and Grummer (2001) concluded from an experiment with lactating dairy cows fed a Met-deficient diet that Cys in feather meal probably cannot substitute for Met in MP. In fact, Cys concentration is very low in milk protein, which is the largest net demand on metabolizable AA. Therefore, the ratio of Cys to Cys + Met is much lower in milk (0.23) compared with that in microbial protein (0.46) or in most of the feed ingredients, especially the plant proteins, usually averaging greater than 0.40 (see Table 19-2). Therefore, the likelihood that digestive flow of Cys would be low enough to decrease Met availability due to its *de novo* synthesis is low in lactating cows fed typical diets. However, the ratio of Tyr to Tyr + Phe in milk (0.53) is higher than in MCP (0.48) and in most feed ingredients (usually less than 0.45). Increased MPY has been observed when formaldehyde-treated canola meal was supplemented with dietary Tyr (Rae and Ingalls, 1984), but substantial amounts of Tyr were destroyed or rendered unavailable by formaldehyde treatment (Rae et al., 1983). The MPY response in the former study may have occurred because of deficiency in Tyr or in Phe if used to synthesize Tyr; however, because no differences were observed between treatments on Phe and Tyr plasma concentrations, determining which mechanism was responsible for the lower MPY is not possible. Deletion of Phe from a total AA mixture infused posttruminally in cows fed a protein-deficient diet decreased MPY and plasma Phe concentrations but had no effect on plasma Tyr concentrations (Doepel et al., 2016), whereas when Phe was deleted from an EAA mixture, plasma Tyr concentrations decreased (Doelman et al., 2015). Overall, no clear evidence suggests that for lactating cows, Cys and Tyr are of concern. However, more research is needed to clearly determine whether under some circumstances, *de novo* synthesis of these two NEAAs would occur to an extent that would decrease the availability of their EAA-precursor and directly limit protein synthesis.

Two other NEAAs have received some attention. Pro and Gin (including its intermediate precursor Glu) are similar in that (1) concentrations of both are considerably higher in milk protein (10.3 and 22.5 percent of TP, respectively) than in MCP (4.3 and 15.0 percent of TP, respectively) or in most feedstuffs, (2) extraction by the lactating dairy mammary gland is considerably less than the quantities secreted in milk protein (Clark et al., 1978; Illg et al., 1987), and (3) both can be synthesized in the mammary gland from Arg, either through transfer of the C skeleton for Pro or through transamination of the excess N from Arg for Gin. Duodenal infusions of Pro were only effective in increasing MPY in mid-lactation cows but not in early lactation (Bruckental et al., 1991) and had no effect in another study (Alumot et al., 1983); however, mammary uptake of Arg decreased and milk fat concentration increased in all three studies. Gin has been

hypothesized to limit milk protein synthesis in cows during early lactation (Meijer et al., 1995). The reasons for the hypothesis were low plasma Gin concentrations after calving and slower recovery than for the other AAs (Meijer et al., 1995; Doepel et al., 2002), as well as the multiple functions involving Gin, including the immune system, purine and pyrimidine synthesis, and as an energy source (Doepel et al., 2006), all of which are very demanding in early lactation. However, posttrimal infusion of 300 g/d of Gin during the first 3 weeks postcalving had only a limited effect on milk production, metabolic parameters, and immune function (Doepel et al., 2006).

The above milk protein meta-analysis demonstrated that collectively the NEAAs plus Arg, Phe, Thr, Trp, and Val had a small but highly significant impact on milk protein synthesis (Equation 6-6). Consideration of only the NEAA was also significant, and thus the effect is not driven solely by Arg, Phe, Thr, Trp, and Val. Because this effect is small and not clearly demonstrated in controlled trials, the committee recommends that the focus for balancing AAs should be placed on the EAAs, excluding Arg.

Rumen-Protected Amino Acids

Because ingested free AAs are readily catabolized by the microbes in the rumen, AAs need to be fed in a protected form to avoid or limit degradation by rumen microbes; however, this protection should not significantly interfere with absorption from the intestine. Without direct AA supplementation, minimal concentrations of free AAs are present in rumen fluid (Lewis and Emery, 1962; Velle et al., 1997; Volden et al., 2001) because AAs arising from hydrolysis of RDP are rapidly used for microbial protein synthesis or deaminated (Lewis and Emery, 1962). Even with diets averaging 66 percent of the CP as RDP, the contribution of free AAs to the total duodenal flux of AAs was less than 2 percent (Volden et al., 2001).

When unprotected forms of Lys, Met, and Thr were given in amounts similar to those recommended for protected forms (9 to 20 g), apparent ruminal degradation over an 8-hour period averaged 88 and 90 percent (Velle et al., 1997; Volden et al., 1998). Ruminal escape of 10 to 12 percent would be unlikely to elicit a detectable response at such low feeding levels. However, at higher doses (48 to 120 g), ruminal degradation decreased to 73 to 78 percent (Velle et al., 1997; Volden et al., 1998), suggesting that the need for rumen protection could be eliminated with high oral doses of free AAs. Such high doses, however, can be costly and may have detrimental or toxic effects on the cow. For example, with the high dose of unprotected Met, it was noted that “after administration of Met at the 120 g dosage, an unpleasant odor emanated from the cows. Mucous membranes were discolored, and feed intake was transiently depressed” (Velle et al., 1997), suggesting that the animals experienced some level of sulfur toxicity.

In addition, rumen populations may adapt to these high levels and become more efficient at degrading the AAs over a longer period but may be overwhelmed in the short term.

Types of Protection

Because Lys and Met often limit MPY, considerable effort has been made to develop technologies that would allow them to escape ruminal degradation without compromising their absorption in the small intestine. The bioavailability of a RP-AA is the combination of its rumen escape and intestinal digestibility. At the time of this writing, only these two AAs are commercially available in RP form. For research projects, however, other RP-AAs have been produced and have been effective (e.g., His [Lee et al., 2012a]; Leu [Arriola Apelo et al., 2014a]; Ile and Val [Leal-Yepes et al., 2019]). Commercial products differ in the technology used to protect the AAs from ruminal degradation. Details are not given for specific commercial products, but briefly, the matrix used for encapsulation is a combination of pH-sensitive polymer and lipid, lipid, a combination of fiber and lipid, or calcium salts of long-chain FAs. The pH-sensitive coating depends on the differences in pH between the rumen (encapsulated at rumen pH) and abomasum (released at acid pH). Physical protection systems (e.g., lipid coating) must provide a reasonable degree of protection against ruminal degradation while providing a reasonable degree of intestinal release. Physical handling of the RP-AA prior to ingestion might alter the bioavailability as some coatings are susceptible to damage during feed manufacturing (e.g., pelleting) and diet preparation (Wu and Papas, 1997). Methods to assess bioavailability of RP-AA are detailed in Chapter 18.

D- and L-isomers of an Amino Acid

The D- and L-isomers of an AA are chemically identical, but one is the mirror image of the other, with the amino group being on one side or the other of the carbon chain. Despite this small difference, mammals can only incorporate the L-isomer of AAs into proteins. Small amounts of D-AAs exist in bacterial cell walls and in free form in some plants. The AAs produced industrially in pure form by fermentation (e.g., Lys, Thr, and Trp) are L-isomers. In contrast, AAs produced from chemical synthesis (e.g., Met) are a DL-racemic mixture. Animals need to convert the D-form into the L-form before it can be utilized for protein synthesis. Such conversion in mammals involves D-AA oxidase. The D form of Met is deaminated to yield the keto acid, 2-oxo-4-methylthiobutanoate. This can then be reaminated to the L-form of Met (Friedman and Gumbmann, 1989). The efficiency of conversion of D-Met to L-Met is species dependent: rats, chicks, pigs, rabbits, and dogs all demonstrate conversion to L-Met when D-Met is administered by either oral or intravenous routes, although this is not the case with

primates (Lewis and Baker, 1995). In growing cattle, abomasal infusions of D- and L-Met produced similar increases in N retention (Campbell et al., 1996), although it tended to be less with DL-Met infusion compared with an equimolar dose of L-Met (Titgemeyer and Merchen, 1990). In dairy cows, a minimum of 75 percent of a bolus dose D-[^{13}C]Met was transformed into L-[^{13}C]Met (Lapierre et al., 2012b). The behavior of the two isomers was, however, totally different: the half-life of D-Met was much greater than the half-life of L-Met (52 versus 8 minutes). The mammary gland did not extract any D-Met, and the fractional hepatic removal of D-Met was numerically lower than the fractional extraction of L-Met, leading to an accumulation of the D-isomer in plasma (Lapierre et al., 2012b). Therefore, L-Met synthesized from the D-isomer elsewhere in the body, not in the liver or the mammary gland, could be used to support milk protein synthesis. No detrimental effects of the D-isomer over the L-isomer of Met are evident. The longer half-life of D-Met could offer the opportunity to delay clearance of the absorbed Met and act as a potential reservoir for L-Met synthesis (Lapierre et al., 2012b).

Met Analogs

Feeding an analog of AA may be an alternative to coated or encapsulated forms of AA because it would not be affected by handling such as pelleting and may be less expensive. Many analogs have been tested (Schwab, 1995), but the most studied is an analog of Met, DL-2-hydroxy-4-methylthiobutanoate (HMTBA), often referred to as HMB. The analog HMTBA has long been proposed as a means to provide Met and increase milk and protein yields of dairy cows fed rations limited in Met (Polan et al., 1970). The Ca salt of HMTBA has also been studied as a supplement for increasing milk and milk fat production (e.g., Loerch and Oke, 1989). Liquid HMTBA is available and extensively used in the poultry and swine industry as a substitute for Met.

There is no consensus on the “Met bioavailability” of HMTBA (ruminal escape $\times$ intestinal absorption $\times$ conversion to Met), but HMTBA is more resistant to rumen microbial degradation than free Met (Belasco, 1972, 1980; Patterson and Kung, 1988). It can be absorbed across the ruminal and omasal epithelium (McCollum et al., 2000), and ruminants possess the enzymes to convert HMTBA to Met (Belasco, 1972, 1980). Blood concentrations of Met increased linearly to HMTBA dose (Feng et al., 2018). Estimations of post-ruminal availability of HMTBA vary greatly. In dairy cows, up to 50 percent (Koenig et al., 1999, 2002) of the ingested dose was reported to flow from the rumen. Also, in dairy cows, net portal absorption of HMTBA averaged 13 percent of the ingested dose (Lapierre et al., 2007b), similar to the 12.5 percent observed in sheep (Lobley et al., 2006). Total bioavailability needs to account for HMTBA conversion to Met and metabolism in the gut tissues (McCollum et al.,

2000), and such consideration resulted in an estimate of 18 percent of the dose ingested being available for postabsorptive use (Lobley et al., 2006).

The isopropyl ester of HMTBA (HMBi) has also been studied. Rumen escape is greater than HMTBA because it is rapidly absorbed across the rumen wall (Graulet et al., 2004). This would explain why a positive response in milk protein concentration has been observed with HMBi supplementation despite only 2.3 percent of the HMBi recovered as HMTBA in omasal digesta (Noftsker et al., 2005). Absorbed HMBi is converted to HMTBA after or during absorption and subsequently converted to the keto acid of Met and then transaminated to L-Met (Baker, 1994). The HMTBA supplement is a racemic mixture of D- and L-isomers that can both be converted to 2-oxo-4-methylthiobutanoate followed by amination to L-Met. The oxidase for L-HMTBA exists predominantly in the peroxisomes within liver and kidney, whereas the dehydrogenase for D-HMTBA is a mitochondrial enzyme found in most tissues (Dibner and Knight, 1984; McCollum et al., 2000; Dibner, 2003). In vitro (McCollum et al., 2000) and ovine data (Lobley et al., 2006) indicate that HMTBA conversion to L-Met occurs in many tissues. In dairy cattle, intravenous infused HMTBA provided 15 percent of the Met used for milk protein secretion, of which 85 percent was produced in other tissues and transported to the mammary gland through blood. The remaining 15 percent was synthesized directly in the mammary gland. The liver removed 38 percent of the infused dose of HMTBA, but that removed was not apparently converted to Met as net Met hepatic release declined (Lapierre et al., 2010). Altogether, this would explain why measuring Met plasma concentration is not an appropriate way of estimating HMTBA availability and its potential value in supplying Met.

Effect of Rumen-Protected Amino Acid Supplementation

Since the publication of the last NRC (2001), numerous studies have been published on effects of balancing dairy rations for individual AAs, especially Lys and Met, with most of them based on the proportional approach (expressing the supply of each AA as a percentage of MP). Based on a meta-analysis of data from experiments involving postprandial AA supplementation, Vyas and Erdman (2009) reported a response in MPY to AA supplementation with a marginal efficiency decreasing from 39 to 25 percent for Lys and from 44 to 22 percent for Met over the range of the predicted AA supply. A review by Robinson (2010) concluded that RP-Met increased milk energy yield and the efficiency of N utilization for milk (N milk/N intake), RP-Lys decreased DMI but increased the milk/DMI ratio, whereas the combination of RP-Met and RP-Lys increased milk and milk energy yield, milk protein percentage, the efficiency of N utilization, and milk/DMI. However, these improvements were small and unpredictable based on dietary characteristics. A meta-analysis (Patton, 2010) performed on two commercial RP-Met sources

concluded that overall RP-Met increased milk protein percentage (+0.07 percent) and yield (+27 g/d). However, milk protein response to RP-Met could not be related to dietary characteristics. A meta-analysis performed on the effect of supplementing RP-Met plus RP-Lys to diets with <15 percent CP reported a small improvement in MPY when RP-AAs were fed (Sinclair et al., 2014). The authors concluded that cows fed low CP diets could respond to RP-Met and RP-Lys supplementation if DMI and other AAs were not limiting. Finally, a meta-analysis compared responses to Met supplementation via postprandial DL-Met infusion, HMTBA, or an RP-Met product (Zanton et al., 2014). In that analysis, MPY increased 2.23 g of protein/g of metabolizable Met, irrespective of the Met source, until reaching a breakpoint. Milk protein concentration increased for all modes of supplementation except for HMTBA, in which only milk yield tended to increase in response to the Met analog. The observed MPY responses observed by Zanton et al. (2014) are similar to the slope coefficient for Met in Equation 6-6. The above authors also observed a positive milk fat yield response that did not differ among DL-Met supplements (1.9 g fat/g Met).

Studies have been conducted to evaluate the use of RP-Met during the transition period because of its multiple roles during this critical period. Feeding RP-Met to cows starting a few weeks prepartum and continuing a few weeks postpartum has increased early lactation milk and milk component yields (Osorio et al., 2013; Batistel et al., 2017) and improved neutrophil function (Osorio et al., 2013; Batistel et al., 2018). Feeding RP-Met during transition has also reduced various measures of oxidative stress (Batistel et al., 2018), which may partially be responsible for improved immune function. Production responses may in part be caused by Met's role as a methyl donor and its role in transmethylation reactions via activation of S-adenosylmethionine, which is involved in more than a hundred metabolic reactions (Finkelstein, 1990). Supplementation of RP-Met to transition cows may also have positive epigenetic effects on the immune system of the calf (Alharthi et al., 2019).

PROTEIN AND DRY MATTER INTAKE

NRC (2001) evaluated (without including study effects) the effects of RDP and RUP on milk protein and DMI. When study was accounted for, the effect of RDP on DMI persisted but to a lesser extent (Firkins et al., 2006). Numerous studies used ingredients that were high in RDP to isonitrogenously substitute for either a different protein source or the same source processed to be higher in RUP. Based on a review of the literature, a limitation of RDP decreases MCP flow to the duodenum and DMI (Santos et al., 1998). Many of those studies are in the database used by the current committee. However, many additional studies with greater range and diversity were added for the current work. In essentially all cases, RDP has been estimated using feed library values or by subtraction of observed microbial N and an estimate

of endogenous N from observed total nonammonia flows. Increasing RDP (if limited) increases MCP production and often DMI, and both MCP and DMI increase MP supply even though RDP itself does not contribute directly to MP. In studies designed to test the specific role of RDP directly, RDP had a large influence on MCP flow to the omasum (Reynal and Broderick, 2005). Replacing TP sources of RDP with urea also progressively depressed MCP flow (Brito et al., 2007; Broderick and Reynal, 2009). An RDP limitation depressed DMI linearly (Cyriac et al., 2008).

Dietary CP percentage is positively associated with DMI (Allen, 2000). This association was reinforced, regardless of whether evaluated in crossover or continuous lactation experiments (Zanton, 2016). Although CP typically was measured, RDP typically was not measured. In another meta-analysis using model outputs (Daniel et al., 2016), DMI was related nonlinearly to MP but still not maximized at the centered point of 9.7 percent MP (about 17.2 percent CP). Yet, mechanisms for DMI depression with decreasing CP percent, long related to improvements in fiber digestibility and alleviation of fill, still await further characterization (Sinclair et al., 2014). Some potential mechanisms are discussed in Chapter 2. Increasing RDP from 9.4 to 10.3 percent of DM increased DMI, but DMI decreased when RDP was increased from 11.0 to 12.1 percent of DM (Mutsaers et al., 2016). Martineau et al. (2016) suggested that provision of MP from casein via postprandial infusions (not increasing RDP) increased DMI when dietary MP was deficient but induced a satiety effect when dietary MP exceeded recommendations. In the latter case, more ammonia from RUP would need to be cleared by the liver. Mobilization of body protein reserves could contribute to hypophagia in early lactation (see Chapter 2). Carder and Weiss (2017) noted that careful attention to providing metabolizable AAs increased production of milk and milk components in early lactation, apparently sparing body protein mobilization, and even showing potential for a longer-term response after peak milk. Clearly, more research is needed on the role of metabolizable AA supply for cows in the first 30 days of lactation.

Numerous studies have been conducted since NRC (2001) to evaluate lowering dietary CP (typically avoiding limited RDP) while maintaining supply of limiting EAA. Decreasing CP from 17.1 to 15.8 percent decreased DMI, but supplementing RP-Met increased DMI (Broderick et al., 2009). Despite lowering NDF digestibility, feeding MP-deficient diets did not depress DMI, whereas supplementing RP-His to the MP-deficient diet increased DMI and MPY (Giallongo et al., 2015, 2017). When MP was deficient, dairy cattle decreased DMI; however, DMI was numerically but not always significantly recovered by supplementing RP-Met, RP-Lys, and RP-His; when His alone was added, DMI was almost the same as supplementation with all three RP-AAs, which fully recovered MPY (Giallongo et al., 2016). Of the RP-AAs typically studied, the greatest response in DMI appears to be from RP-His (Patton et al., 2015; Giallongo et al., 2017).

When MP was deficient, Met and Lys extraction into tissues was estimated to increase relative to those AAs that were not deficient (Lee et al., 2015). Metabolizable His appears to become increasingly limiting as microbial protein makes up an increasing proportion of the MP (Giallongo et al., 2016), and increased supply of metabolizable His (when it is limited in MP) seems to be directed preferentially toward the mammary gland (Sadri et al., 2016). Thus, increasing MPY could increase mammary extraction of blood EAA in addition to His and pull DMI. Patton et al. (2015) explained the relative constancy of plasma EAA concentrations over the lactation cycle and the challenge to relate concentrations to physiological functions such as control of voluntary feed intake. Clearly, the role of EAA in short- and long-term regulation of milk protein synthesis awaits further understanding (Arriola Apelo et al., 2014b; Cant et al., 2018), and future efforts should help delineate a role for metabolizable AA supply affecting DMI, particularly in longer-term studies (Hristov et al., 2019).

EFFECTS OF PROTEIN ON REPRODUCTION

Based largely on the seminal efforts described by Butler et al. (1998), NRC (2001) discussed studies regarding the negative correlation between excess dietary CP and fertility in dairy cattle. Since then, numerous studies have had specific objectives to address a mechanism for such a response, with likely responses related to urea or ammonia toxicity on oocyte, embryo, and uterine environments (Thatcher et al., 2011; Berry et al., 2016). One meta-analysis supported a role for excess BUN depressing fertility (Lean et al., 2012). In contrast, another meta-analysis (Sinclair et al., 2014) lessened such a role, but the authors noted that many of the studies in the prior meta-analysis used low-producing cows. Berry et al. (2016) noted a clear distinction among production systems in which studies with pasture-fed cattle lacked any negative association even though grazing cattle typically have much higher BUN than those fed total mixed rations. Based on herd records, lowering milk urea N (MUN; a proxy for BUN), within the range of 9.0 to 18.0 mg/dL, was suggested to have only a modest opportunity to improve fertility through dietary adjustments (Guo et al., 2004). Those authors noted that MUN was more related to individual cows within herds than among herds such that dietary formulation per se played a minimal role compared with other factors in the negative association of MUN with fertility. An MP deficiency could promote problems in early lactation and therefore lower fertility (Drackley and Cardoso, 2014), whereas diets with excess MP relative to ME should be avoided (Diskin et al., 2016).

Dairy cows suffer a myriad of issues in the transition period (see Chapter 12) that could influence tissue energy balance and other factors related to reproduction. For example, services per conception leading to pregnancy increased by 0.46 when mastitis was experienced preservice and increased to 0.72 services per conception when both pre- and postservice mastitis was included (Dolecheck et al.,

2018). Energetic efficiency was expected to decrease by up to 15 percent when cows experienced inflammatory responses (Bertoni et al., 2015). However, Bertoni et al. (2015) and Enger (2019) could not attribute quantitative requirements for MP or AAs despite increased synthesis of acute-phase proteins, antibodies, and other proteins during the inflammatory response associated with clinical mastitis. Similarly, the role of mobilization of non-esterified FA in inflammatory responses that depress fertility is well recognized (Le Blanc, 2014; Sordillo et al., 2016), whereas there is a more limited understanding of how inflammatory responses are related to MP supply. Enhancing the postprandial supply of Met and His in the transition cow likely increases the ability to reduce fat infiltration in the liver (Coleman et al., 2020), which likely enhances reproductive success. Those authors also noted that DMI could increase and that inflammatory markers could decrease in the transition period by increasing metabolizable Met supply. Arg and several other AAs (including nonessentials) were inferred to have a role in moderating inflammation, but studies are lacking in ruminants and especially with respect to reproduction.

Appropriate protein feeding in the transition cow is critical to provide metabolizable AA to limit the mobilization of body protein reserves, enhance peak milk production, and maintain higher milk production. The postpartum cow has a supply of labile protein reserves, but these are less than her supply of energy from adipose tissue (Drackley and Cardoso, 2014). As the postpartum cow adapts to high milk production, tissue-mobilized AA must compete for oxidation and gluconeogenesis by the liver and will not necessarily be diverted toward mammary protein synthesis (Larsen et al., 2015). Proper supply of starch should help optimize the efficiency of AA metabolism by the mammary gland (Cantalapiedra-Hijar et al., 2015). Increasing starch supply (without causing ruminal acidosis) should help limit mobilization of AA from muscle or direct more AA toward replenishing peripheral tissues (Nichols et al., 2016) and FAs from adipose tissue (Drackley and Cardoso, 2014) at the same time that increasing insulin secretion helps the reproductive tissues compete for nutrients (Lucy et al., 2014). Thus, MP must be adequate to support the increasing milk production and to stimulate increasing DMI in the postpartum cow (Amanlou et al., 2017). Optimization of metabolizable AA and ME will help to limit fertility issues that have been associated with high BUN.

Concentrations of MUN can be useful at the farm level to evaluate supply of CP. For example, Patton et al. (2014) described opportunities to use MUN in managerial approaches such as to safely lower dietary CP while meeting metabolizable AA targets. However, the usefulness of these values is diminished by variation among cows and herds due to genetic and management factors (Wood et al., 2003; Aguilar et al., 2012). For example, dietary concentrations of Na and K are important determinants of plasma and MUN concentrations

(Spek et al., 2013). These factors likely explain some of the variation among studies (Hristov et al., 2018). Greater attention to herd and other dietary characteristics might help explain variability in MUN and its potential to improve reproductive management. For example, a drop in MUN from before versus after artificial insemination was associated with increased risk for conception failure, which was more important than the MUN concentration per se (Albaaj et al., 2017). Given the current proclivity to decrease CP to minimize environmental pressure and its integration into an improved AA supply-requirement system, the current committee does not support a role of excess protein (i.e., to be beyond a reasonable safety factor) in the impairment of fertility unless the high BUN results from insufficient energy and excessive body tissue mobilization (Patton et al., 2014). Instead, the committee agrees that more attention is needed to assess fertility in light of lower-protein diets. Within a reasonable target for MP supply relative to requirements, fertility should not be impaired.

UREA RECYCLING AND ENERGY COST

Ammonia derived from microbial protein degradation in the gut is absorbed into the portal vein and detoxified by conversion to urea in the liver, whereas excess absorbed AAs that are catabolized in nonhepatic tissues are shuttled to the liver in Ala and Glu and the N removed and converted to urea. In dairy cows, the equivalent of 60 to 85 percent of digested N is converted to urea in the liver (e.g., Lapierre and Lobley, 2001; Ruiz et al., 2002; Recktenwald et al., 2014). To sustain anabolism when protein intake is low, ruminants have evolved to salvage part of the synthesized urea by recycling into the lumen of the gut, either through saliva or by direct transfer from the blood across the epithelial wall (Reynolds and Kristensen, 2008; Batista et al., 2017). Recycled urea is hydrolyzed to ammonia by bacterial ureases (Patra and Aschenbach, 2018), and ammonia can be either reabsorbed into blood as ammonia or used to support microbial protein synthesis. Part of microbial protein is digested and used to support anabolism, and part will be excreted in the feces (undigested rumen microbial protein and most of the hindgut-produced microbial protein). The potential futile cycle of BUN reconversion to ammonia is apparently regulated by expression of urea transporters (Calsamiglia et al., 2010). As explained previously, between 15 percent and 30 percent of rumen microbial N should be derived from BUN in dairy cattle (Li et al., 2019), which agrees with data from other ruminants (Batista et al., 2017).

Although adequate RDP is needed to support MCP production and DMI, excess RDP beyond a reasonable safety factor should be minimized to reduce urea excretion in urine and the environmental impact of dairy production systems (see Chapter 14). Excess dietary protein, particularly RDP, was positively associated with increased heat production

(Reed et al., 2017), although the latter results are inconsistent with earlier work (Tyrrell et al., 1970), which showed a minimal effect on heat production. Although hepatic urea synthesis requires ATP, this cost is minor relative to other energetic costs resulting from deamination of excess AAs and other metabolic processes (Reynolds, 2005). The committee does account for the energy associated with urinary N loss in derivation of metabolizable energy (see Chapter 3) but not any increase in heat production associated with increased urinary N loss. Urinary urea excretion is greater in dairy cattle experiencing heat stress, resulting from metabolic stress responses described by Ruts (2019). Therefore, based on limited data, feeding to meet but not exceed RDP and AA requirements is recommended when cows are experiencing heat stress.

REFERENCES

- Aguilar, M., M. D. Hanigan, H. A. Tucker, B. L. Jones, S. K. Garbade, M. L. McGilliard, C. C. Stallings, K. F. Knowlton, and R. E. James. 2012. Cow and herd variation in milk urea nitrogen concentrations in lactating dairy cattle. *J. Dairy Sci.* 95(12):7261-7268.
- Albaaj, A., G. Foucras, and D. Raboisson. 2017. Changes in milk urea around insemination are negatively associated with conception success in dairy cows. *J. Dairy Sci.* 100(4):3257-3265.
- Alexandrov, A. N. 1998. Effect of ruminal exposure and subsequent microbial contamination on dry matter and protein degradability of various feedstuffs. *Anim. Feed Sci. Technol.* 71(1-2):99-107.
- Alharthi, A. S., V. Lopreiato, H. Dai, R. Bucktrout, M. Abdelmegeid, F. Batistel, C. Parys, X. Shen, M. A. Ballou, E. Trevisi, I. A. Alhidary, M. M. Abdelahman, and J. J. Loor. 2019. Short communication: Supply of methionine during late pregnancy enhances whole-body innate immune response of Holstein calves partly through changes in mRNA abundance in polymorphonuclear leukocytes. *J. Dairy Sci.* 102:10599-10605.
- Allen, M. S. 2000. Effects of diet on short-term regulation of feed intake by lactating dairy cattle. *J. Dairy Sci.* 83:1598-1624.
- Alumot, E., I. Bruckental, A. Tadmor, C. Kennit, and P. Holstein. 1983. Effect of proline on arginine uptake and nitrogen metabolism of lactating goats. *J. Dairy Sci.* 66(6): 1243-1247.
- Amanlou, H., T. A. Farahani, and N. E. Farsuni. 2017. Effects of rumen undegradable protein supplementation on productive performance and indicators of protein and energy metabolism in Holstein fresh cows. *J. Dairy Sci.* 100(5):3628-3640.
- Appuhamy, J. A. D. R. N., N. A. Knoebel, W. A. D. Nayananjali, J. Escobar, and M. D. Hanigan. 2012. Isoleucine and leucine independently regulate mTOR signaling and protein synthesis in MAC-T cells and bovine mammary tissue slices. *J. Nutr.* 142(3):484-491.
- Appuhamy, J. A. D. R. N., W. A. Nayananjali, E. M. England, D. E. Gerard, R. M. Akers, and M. D. Hanigan. 2014. Effects of AMP-activated protein kinase (AMPK) signaling and essential amino acids on mammalian target of rapamycin (mTOR) signaling and protein synthesis rates in mammary cells. *J. Dairy Sci.* 97(1):419-429.
- Arriola Apelo, S. I., A. L. Bell, K. Estes, J. Ropelewski, M. J. de Veth, and M. D. Hanigan. 2014a. Effects of reduced dietary protein and supplemental rumen-protected essential amino acids on the nitrogen efficiency of dairy cows. *J. Dairy Sci.* 97(9):5688-5699.
- Arriola Apelo, S. I., J. R. Knapp, and M. D. Hanigan. 2014b. Current representation and future trends of predicting amino acid utilization in the lactating dairy cow. *J. Dairy Sci.* 97(7):4000-4017.
- Arriola Apelo, S. I., L. M. Singer, X. Y. Lin, M. L. McGilliard, N. R. St-Pierre, and M. D. Hanigan. 2014c. Isoleucine, leucine, methionine, and threonine effects on mammalian target of rapamycin signaling in mammary tissue. *J. Dairy Sci.* 97(2):1047-1056.
- Arriola Apelo, S. I., L. M. Singer, W. K. Ray, R. F. Helm, X. Y. Lin, M. L. McGilliard, N. R. St-Pierre, and M. D. Hanigan. 2014d. Casein synthesis is independently and additively related to individual essential amino acid supply. *J. Dairy Sci.* 97(5):2998-3005.
- Bach, A., and M. D. Stem. 2000. Measuring resistance to ruminal degradation and bioavailability of ruminally protected methionine. *Anim. Feed Sci. Tech.* 84:23-32.
- Bach, A., G. B. Huntington, S. Calsamiglia, and M. D. Stem. 2000. Nitrogen metabolism of early lactation cows fed diets with two different levels of protein and different amino acid profiles. *J. Dairy Sci.* 83(11):2585-2595.
- Bach, A., S. Calsamiglia, and M. D. Stem. 2005. Nitrogen metabolism in the rumen. *J. Dairy Sci.* 88(E. Suppl.):E9-E21.
- Baker, D. H. 1994. Utilization of precursors for L-amino acids. Pp. 37-61 in *Amino Acids in Farm Animal Nutrition*, J. P. D'Mello, ed. Wallingford, UK: CAB International.
- Bateman, H. G., II, J. H. Clark, and M. R. Murphy. 2005. Development of a system to predict feed protein flow to the small intestine of cattle. *J. Dairy Sci.* 88(1):282-295.
- Bateman, H. G., II, M. D. Hanigan, and R. A. Kohn. 2008. Sensitivity of two metabolic models of dairy cattle digestion and metabolism to changes in nutrient content of diets. *Anim. Feed Sci. Technol.* 140(3-4):272-292.
- Batista, E. D., E. Detmann, S. C. Valadares Filho, E. C. Titgemeyer, and R. F. D. Valadares. 2017. The effect of CP concentration in the diet on urea kinetics and microbial usage of recycled urea in cattle: A meta-analysis. *Animal* 11(8): 1303-1311.
- Batistel, F., J. M. Arroyo, A. Bellingeri, L. Wang, B. Saremi, C. Parys, E. Trevisi, F. C. Cardoso, and J. J. Loor. 2017. Ethyl-cellulose rumen-protected methionine enhances performance during the periparturient period and early lactation in Holstein dairy cows. *J. Dairy Sci.* 100:7455-7467.
- Batistel, F., J. M. Arroyo, C. I. M. Garces, E. Trevisi, C. Parys, M. A. Ballou, F. C. Cardoso, and J. J. Loor. 2018. Ethyl-cellulose rumen-protected methionine alleviates inflammation and oxidative stress and improves neutrophil function during the periparturient period and early lactation in Holstein dairy cows. *J. Dairy Sci.* 101:480-490.
- Bauchart, D., F. Legay-Carmier, M. Doreau, and B. Gaillard. 1990. Lipid metabolism of liquid-associated and solid-adherent bacteria in rumen contents of dairy cows offered lipid-supplemented diets. *Br. J. Nutr.* 63:563-578.
- Belasco, I. J. 1972. Stability of methionine hydroxy analog in rumen fluid and its conversion in vitro to methionine by calf liver and kidney. *J. Dairy Sci.* 55(3):353-357.
- Belasco, I. J. 1980. Fate of carbon-14 labeled methionine hydroxy analog and methionine in the lactating dairy cow. *J. Dairy Sci.* 63:775-784.
- Bell, A. W., R. Slepatis, and R. A. Ehrhardt. 1995. Growth and accretion of energy and protein in the gravid uterus during late pregnancy in Holstein cows. *J. Dairy Sci.* 78(9): 1954-1961.
- Bequette, B. J., J. A. Metcalf, D. Wray-Cahen, F. R. C. Backwell, J. D. Sutton, M. A. Lomax, J. C. Macrae, and G. E. Lobley. 1996. Leucine and protein metabolism in the lactating dairy cow mammary gland: Responses to supplemental dietary crude protein intake. *J. Dairy Res.* 63(2):209-222.
- Bequette, B., M. Hanigan, A. Calder, C. Reynolds, G. Lobley, and J. MacRae. 2000. Amino acid exchange by the mammary gland of lactating goats when histidine limits milk production. *J. Dairy Sci.* 83(4):765-775.
- Bernard, L., and M. Doreau. 2000. Use of rare earth elements as external markers for mean retention time measurements in ruminants. *Reprod. Nutr. Dev.* 40:89-101.
- Berry, D. P., N. C. Friggens, M. Lucy, and J. R. Roche. 2016. Milk production and fertility in cattle. *Annu. Rev. Anim. Biosci.* 4(1):269-290.
- Berthiaume, R., H. Lapierre, M. Stevenson, N. Cote, and B. W. McBride. 2000. Comparison of the in situ and in vivo intestinal disappearance of ruminally protected methionine. *J. Dairy Sci.* 83(9):2049-2056.
- Bertoni, G., A. Minuti, and E. Trevisi. 2015. Immune system, inflammation and nutrition in dairy cattle. *Anim. Prod. Sci.* 55(7):943-948.

- Black, A. L., and M. Kleiber. 1958. The transfer of carbon from propionate to amino acids in intact cows. *J. Biol. Chem.* 232(1):203-209.
- Black, A. L., M. Kleiber, and A. H. Smith. 1952. Carbonate and fatty acids as precursors of amino acids in casein. *J. Biol. Chem.* 197(1):365-370.
- Black, A. L., M. Kleiber, and C. F. Baxter. 1955. Glucose as a precursor of amino acids in the intact dairy cow. *Biochem. Biophys. Acta.* 17(3):346-353.
- Black, A. L., M. Kleiber, A. H. Smith, and D. N. Stewart. 1957. Acetate as a precursor of amino acids of casein in the intact dairy cow. *Biochem. Biophys. Acta.* 23(1):54-59.
- Boila, R. J., and J. R. Ingalls. 1992. In situ rumen digestion and escape of dry matter, nitrogen and amino acids in canola meal. *Can. J. Anim. Sci.* 72(4):891-901.
- Borucki Castro, S. I., H. Lapiere, L. E. Phillip, P. W. Jardon, and R. Berthiaume. 2008. Towards non-invasive methods to determine the effect of treatment of soya-bean meal on lysine availability in dairy cows. *Animal* 2(2):224-234.
- Boucher, S. E., S. Calsamiglia, C. M. Parsons, H. H. Stein, M. D. Stern, P. S. Erickson, P. L. Utterback, and C. G. Schwab. 2009a. Intestinal digestibility of amino acids in rumen-undegraded protein estimated using a precision-fed cecectomized rooster bioassay: II. Distillers dried grains with solubles and fish meal. *J. Dairy Sci.* 92(12):6056-6067.
- Boucher, S. E., S. Calsamiglia, C. M. Parsons, M. D. Stern, M. Ruiz Moreno, M. Vazquez-Anón, and C. G. Schwab. 2009b. In vitro digestibility of individual amino acids in rumen-undegraded protein: The modified three-step procedure and the immobilized digestive enzyme assay. *J. Dairy Sci.* 92(8):3939-3950.
- Brandt, M. W., K. Rohr, and P. Lebzien. 1980. Determination of endogenous protein-N in duodenal chyme of dairy cows using of N-15. *J. Anim. Physiol. Anim. Nutr.* 44:26-27.
- Brito, A. F., G. A. Broderick, and S. M. Reynal. 2007. Effects of different protein supplements on omasal nutrient flow and microbial protein synthesis in lactating dairy cows. *J. Dairy Sci.* 90(4): 1828-1841.
- Broderick, G. A., and S. M. Reynal. 2009. Effect of source of rumen-degraded protein on production and ruminal metabolism in lactating dairy cows. *J. Dairy Sci.* 92:2822-2834.
- Broderick, G. A., M. J. Stevenson, and R. A. Patton. 2009. Effect of dietary protein concentration and degradability on response to rumen-protected methionine in lactating dairy cows. *J. Dairy Sci.* 92(6):2719-2728.
- Broderick, G. A., P. Huhtanen, S. Ahvenjarvi, S. M. Reynal, and K. J. Shingfield. 2010. Quantifying ruminal nitrogen metabolism using the omasal sampling technique in cattle—a meta-analysis. *J. Dairy Sci.* 93:3216-3230.
- Bruckental, I., I. Ascarelli, B. Yosif, and E. Alumot. 1991. Effect of duodenal proline infusion on milk production and composition in dairy cows. *Anim. Sci.* 53(3):299-303.
- Burnham, K. P., and D. R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. New York: Springer-Verlag.
- Butler, W. R. 1998. Review: Effect of protein nutrition on ovarian and uterine physiology in dairy cattle. *J. Dairy Sci.* 81:2533-2539.
- Calsamiglia, S., and M. D. Stern. 1995. A three-step in vitro procedure for estimating intestinal digestion of protein in ruminants. *J. Anim. Sci.* 73(5): 1459-1465.
- Calsamiglia, S., M. D. Stern, and J. L. Firkins. 1996. Comparison of nitrogen-15 and purines as microbial markers in continuous culture. *J. Anim. Sci.* 74:1375-1381.
- Calsamiglia, S., A. Ferret, C. K. Reynolds, N. B. Kristensen, and A. M. van Vuuren. 2010. Strategies for optimizing nitrogen use by ruminants. *Animal* 4:1184-1196.
- Campbell, C. G., E. C. Titgemeyer, and G. St-Jean. 1996. Efficiency of D- vs L-methionine utilization by growing steers. *J. Anim. Sci.* 74(10):2482-2487.
- Cant, J. P., J. M. Kim, S. R. L. Cieslar, and J. Doelman. 2018. Symposium review: Amino acid uptake by the mammary glands: Where does the control lie? *J. Dairy Sci.* 101(6):5655-5666.
- Cantalapiedra-Hijar, G., I. Ortigues-Marty, and S. Lemosquet. 2015. Diets rich in starch improve the efficiency of amino acids use by the mammary gland in lactating Jersey cows. *J. Dairy Sci.* 98(10):6939-6953.
- Carder, E. G., and W. P. Weiss. 2017. Short- and longer-term effects of feeding increased metabolizable protein with or without an altered amino acid profile to dairy cows immediately postpartum. *J. Dairy Sci.* 100(6):4528-4538.
- Chibisa, G. E., and T. Mutsvangwa. 2013. Effects of feeding wheat or corn- wheat dried distillers grains with solubles in low- or high-crude protein diets on ruminal function, omasal nutrient flows, urea-N recycling, and performance in cows. *J. Dairy Sci.* 96(10):6550-6563.
- Clark, J. H., H. R. Spiers, and C. L. Davis. 1978. Uptake and metabolism of nitrogenous components by the lactating mammary gland. *Fed. Proc.* 37(5): 1233-1238.
- Clark, J. H., R. G. Derrig, C. L. Davis, and H. R. Spiers. 1975. Metabolism of arginine and ornithine in the cow and rabbit mammary tissue. *J. Dairy Sci.* 58:1808-1813.
- Coleman, D. N., V. Lopreiato, A. Alharthi, and J. J. Looor. 2020. Amino acids and the regulation of oxidative stress and immune function in dairy cattle. *J. Anim. Sci.* 98:S175-S193.
- Cozzi, G., I. Andrighetto, P. Berzaghi, and C. E. Polan. 1995. In situ ruminal disappearance of essential amino acids in protein feedstuffs. *J. Dairy Sci.* 78(1): 161-171.
- Cyriac, J., A. G. Rius, M. L. McGilliard, R. E. Pearson, B. J. Bequette, and M. D. Hanigan. 2008. Lactation performance of midlactation dairy cows fed ruminal degradable protein at concentrations lower than National Research Council recommendations. *J. Dairy Sci.* 91:4704-1713.
- Czerkawski, J. W. 1976. Chemical composition of microbial matter in the rumen. *J. Sci. Food Agric.* 27:621-632.
- Daniel, J. B., N. C. Friggens, P. Chapoutot, H. Van Laar, and D. Sauvant. 2016. Milk yield and milk composition responses to change in predicted net energy and metabolizable protein: A meta-analysis. *Animal* 10(12): 1975-1985.
- De Visser, H., A. Klop, C. J. Van der Koelen, and A. M. Van Vuuren. 1998. Starch supplementation of grass harvested at two stages of maturity prior to ensiling: Intake, digestion, and degradability by dairy cows. *J. Dairy Sci.* 81(8):2221-2227.
- DePeters, E. J., and J. P. Cant. 1992. Nutritional factors influencing the nitrogen composition of bovine milk: A review. *J. Dairy Sci.* 75:2043-2070.
- Dibner, J. J. 2003. Review of the metabolism of 2-hydroxy-4-(methylthio) butanoic acid. *World Poultry Sci. J.* 59:99-110.
- Dibner, J. J., and C. D. Knight. 1984. Conversion of 2-hydroxy-4-(methylthio) butanoic acid to L-methionine in the chick: A stereospecific pathway. *J. Nutr.* 114:1716-1723.
- Dinh, S. K. 2007. Urea-N Recycling in Lactating Dairy Cows. in Department of Animal and Avian Sciences. PhD diss., University of Maryland, College Park, MD.
- Diskin, M. G., S. M. Waters, M. H. Parr, and D. A. Kenny. 2016. Pregnancy losses in cattle: Potential for improvement. *Reprod. Fertil. Dev.* 28(2):83-93.
- Doelman, J., R. V. Curtis, M. Carson, J. J. M. Kim, J. A. Metcalf, and J. P. Cant. 2015. Essential amino acid infusions stimulate mammary expression of eukaryotic initiation factor 2B but milk protein yield is not increased during an imbalance. *J. Dairy Sci.* 98:4499-4508.
- Doepel, L., and H. Lapiere. 2010. Changes in production and mammary metabolism of dairy cows in response to essential and nonessential amino acid infusions. *J. Dairy Sci.* 93:3264-3274.
- Doepel, L., and H. Lapiere. 2011. Deletion of arginine from an abomasal infusion of amino acids does not decrease milk protein yield in Holstein cows. *J. Dairy Sci.* 94(2):864-873.
- Doepel, L., H. Lapiere, and J. J. Kennelly. 2002. Peripartum performance and metabolism of dairy cows in response to prepartum energy and protein intake. *J. Dairy Sci.* 85(9):2315-2334.
- Doepel, L., D. Pacheco, J. J. Kennelly, M. D. Hanigan, I. F. Lopez, and H. Lapiere. 2004. Milk protein synthesis as a function of amino acid supply. *J. Dairy Sci.* 87(5): 1279-1297.

- Doepel, L., M. Lessard, N. Gagnon, G. E. Lobley, J. F. Bernier, P. Dubreuil, and H. Lapiere. 2006. Effect of postprandial glutamine supplementation on immune response and milk production in dairy cows. *J. Dairy Sci.* 89(8):3107-3121.
- Doepel, L., I. I. Hewage, and H. Lapiere. 2016. Milk protein yield and mammary metabolism are affected by phenylalanine deficiency but not by threonine or tryptophan deficiency. *J. Dairy Sci.* 99(4):3144-3156.
- Dolecheck, K. A., A. Garcla-Guerra, and L. E. Moraes. 2019. Quantifying the effects of mastitis on the reproductive performance of dairy cows: A meta-analysis. *J. Dairy Sci.* 102:8454-8477.
- Dos, D. S., S. M. Ali, D.-H. Kim, D. A. Guertin, R. R. Latek, H. Erdjument-Bromage, P. Tempst, and D. M. Sabatini. 2004. Rictor, a novel binding partner of mTOR, defines a Rapamycin-insensitive and Raptor-independent pathway that regulates the cytoskeleton. *Curr. Biol.* 14(14): 1296-1302.
- Drackley, J. K., and F. C. Cardoso. 2014. Prepartum and postpartum nutritional management to optimize fertility in high-yielding dairy cows in confined TMR systems. *Animal* 8(Suppl. 1):5-14.
- Enger, B. D. 2019. Invited review: Reevaluating how mastitis reduces milk yield: Discussion of competitive substrate utilization. *Appl. Anim. Sci.* 35:408-415.
- England, M. L., G. A. Broderick, R. D. Shaver, and D. K. Combs. 1997. Comparison of in situ and in vitro techniques for measuring ruminal degradation of animal by-product proteins. *J. Dairy Sci.* 80(11):2925-2931.
- Estes, K. A., R. R. White, P. S. Yoder, T. Pilonero, H. Schramm, H. Lapiere, and M. D. Hanigan. 2018. An in vivo stable isotope-based approach for assessment of absorbed amino acids from individual feed ingredients within complete diets. *J. Dairy Sci.* 101(8):7040-7060.
- Farrell, H. M. J., R. Jimenez-Flores, G. T. Bleck, E. M. Brown, J. E. Butler, L. K. Creamer, C. L. Hicks, C. M. Hollar, K. F. Ng-Kwai-Hang, and H. E. Swaisgood. 2004. Nomenclature of the proteins of cows' milk—sixth revision. *J. Dairy Sci.* 87:1641-1674.
- Featherston, W. R., D. R. Frazee, D. L. Hill, C. H. Noller, and C. E. Parmelee. 1964. Constancy of amino acid composition of cow's milk protein under changing ration. *J. Dairy Sci.* 47:1417-1418.
- Feng, X., R. R. White, H. A. Tucker, and M. D. Hanigan. 2018. Meta-analysis of 2-hydroxy-4-methylthio-butyric acid supplementation on ruminal fermentation, milk production, and nutrient digestibility. *J. Dairy Sci.* 101(8):7182-7189.
- Fessenden, S. W., T. J. Hackmann, D. A. Ross, A. Foskolos, and M. E. Van Amburgh. 2017. Ruminal bacteria and protozoa composition, digestibility, and amino acid profile determined by multiple hydrolysis times. *J. Dairy Sci.* 100:7211-7226.
- Finkelstein, J. D. 1990. Methionine metabolism in mammals. *J. Nutr. Biochem.* 1:228-237.
- Firkins, J. L. 2010. Reconsidering rumen microbial consortia to enhance feed efficiency and reduce environmental impact of ruminant livestock production systems. *Rev. Bras. Zootec.* 39:445-457.
- Firkins, J. L., S. M. Lewis, L. Montgomery, L. L. Berger, N. R. Merchen, and G. C. Fahey. 1987. Effects of feed intake and dietary urea concentration on ruminal dilution rate and efficiency of bacterial growth in steers. *J. Dairy Sci.* 70:2312-2321.
- Firkins, J. L., M. S. Allen, B. S. Oldick, and N. R. St-Pierre. 1998. Modeling ruminal digestibility of carbohydrates and microbial protein flow to the duodenum. *J. Dairy Sci.* 81(12):3350-3369.
- Firkins, J. L., A. N. Hristov, M. B. Hall, G. A. Varga, and N. R. St-Pierre. 2006. Integration of ruminal metabolism in dairy cattle. *J. Dairy Sci.* 89(E. Suppl.):E31-E51.
- Firkins, J. L., Z. Yu, and M. Morrison. 2007. Ruminal nitrogen metabolism: Perspectives for integration of microbiology and nutrition for dairy. *J. Dairy Sci.* 90(E. Suppl.):E1-E16.
- Fleming, A. J., K. A. Estes, H. Choi, B. A. Barton, C. A. Zimmerman, and M. D. Hanigan. 2019. Assessing bioavailability of ruminally protected methionine and lysine prototypes. *J. Dairy Sci.* 102(5):4014-4024.
- Fonseca, A. C., S. M. Fredin, L. F. Ferraretto, C. M. Parsons, P. L. Utterback, and R. D. Shaver. 2014. Short communication: Intestinal digestibility of amino acids in fluid- and particle-associated rumen bacteria determined using a precision-fed cecotomized rooster bioassay. *J. Dairy Sci.* 97(6):3855-3859.
- Friedman, M., and M. R. Gumbmann. 1989. Dietary significance of D-amino acids. Pp. 173-190 in *Absorption and Utilization of Amino Acids*, Vol. II, M. Friedman, ed. Boca Raton, FL: CRC.
- Fujihara, T., and M. N. Shem. 2011. Metabolism of microbial nitrogen in ruminants with special reference to nucleic acids. *Anim. Sci. J.* 82(2): 198-208.
- Galyean, M. L., and L. O. Tedeschi. 2014. Predicting microbial protein synthesis in beef cattle: Relationship to intakes of total digestible nutrients and crude protein. *J. Anim. Sci.* 92(11):5099-5711.
- Gan, X., J. Wang, B. Su, and D. Wu. 2011. Evidence for direct activation of mTORC2 kinase activity by phosphatidylinositol 3,4,5-trisphosphate. *J. Biol. Chem.* 286(13): 10998-11002.
- Giallongo, F., A. N. Hristov, J. Oh, T. Frederick, H. Weeks, J. Wemer, H. Lapiere, R. A. Patton, A. Gehman, and C. Parys. 2015. Effects of slow-release urea and rumen-protected methionine and histidine on performance of dairy cows. *J. Dairy Sci.* 98(5):3292-3308.
- Giallongo, F., M. T. Harper, J. Oh, J. C. Lopes, H. Lapiere, R. A. Patton, C. Parys, I. Shinzato, and A. N. Hristov. 2016. Effects of rumen-protected methionine, lysine, and histidine on lactation performance of dairy cows. *J. Dairy Sci.* 99(6):4437-4452.
- Giallongo, F., M. T. Harper, J. Oh, C. Parys, I. Shinzato, and A. N. Hristov. 2017. Histidine deficiency has a negative effect on lactational performance of dairy cows. *J. Dairy Sci.* 100(4):2784-2800.
- Graulet, B., C. Richard, and J. C. Robert. 2004. The isopropyl ester of methionine hydroxy-analogue is absorbed through the rumen wall in the cow. *J. Anim. Feed Sci.* 13(Suppl. 1):269-272.
- Graulet, B., C. Richard, and J. C. Robert. 2005. Methionine availability in plasma of dairy cows supplemented with methionine hydroxy analog isopropyl ester. *J. Dairy Sci.* 88(10):3640-3649.
- Gregorini, P., P. Beukes, G. Waghorn, D. Pacheco, and M. Hanigan. 2015. Development of an improved representation of rumen digesta outflow in a mechanistic and dynamic model of a dairy cow, Molly. *Ecol. Model.* 313:293-306.
- Guo, K., E. Russek-Cohen, M. A. Varner, and R. A. Kohn. 2004. Effects of milk urea nitrogen and other factors on probability of conception of dairy cows. *J. Dairy Sci.* 87(6): 1878-1885.
- Hackmann, T., and J. Firkins. 2015a. Electron transport phosphorylation in rumen butyrate: Unprecedented ATP yield for glucose fermentation to butyrate. *Front. Microbiol.* 6:622.
- Hackmann, T. J., and J. L. Firkins. 2015b. Maximizing efficiency of rumen microbial protein production. *Front. Microbiol.* 6:465.
- Hall, M. B., and G. B. Huntington. 2007. Nutrient synchrony: Sound in theory, elusive in practice. *J. Anim. Sci.* 86:E287-E292.
- Hanigan, M. D. 2006. An ingredient based input scheme for Molly. Pp. 328-349 in *Nutrient Digestion and Utilization in Farm Animals: Modelling Approaches*, eds. Kebreab, J. Dijkstra, A. Bannink, W. J. J. Gerrits, and J. France, ed. Wallingford, UK: CABI Publishing.
- Hanigan, M., J. Cant, D. Weakley, and J. Beckett. 1998. An evaluation of postabsorptive protein and amino acid metabolism in the lactating dairy cow. *J. Dairy Sci.* 81(12):3385-3401.
- Hanigan, M. D., J. France, L. A. Crompton, B. J. Bequette, J. P. McNamara, and D. E. Beever. 2000. Evaluation of a representation of the limiting amino acid theory for milk protein synthesis. Pp. 127-144 in *Modelling Nutrient Utilization in Farm Animals*, J. P. McNamara, J. France, D. E. Beever, eds. Wallingford, UK: CABI Publishing.
- Hanigan, M., B. Bequette, L. Crompton, and J. France. 2001. Modeling mammary amino acid metabolism. *Live. Prod. Sci.* 70(1-2):63-78.
- Hanigan, M., L. Crompton, B. Bequette, J. Mills, and J. France. 2002. Modelling mammary metabolism in the dairy cow to predict milk constituent yield, with emphasis on amino acid metabolism and milk protein production: Model evaluation. *J. Theor. Biol.* 217(3):311-330.
- Hanigan, M. D., J. France, S. J. Mabbesh, W. C. McNabb, and B. J. Bequette. 2009. High rates of mammary tissue protein turnover in lactating goats are energetically costly. *J. Nutr.* 139(6): 1118-1127.

- Hanigan, M. D., J. A. D. R. N. Appuhamy, and P. Gregorini. 2013. Revised digestive parameter estimates for the Molly cow model. *J. Dairy Sci.* 96(6):3867–3885.
- Hannah, S. M., R. C. Cochran, E. S. Vanzant, and D. L. Harmon. 1991. Influence of protein supplementation on site and extent of digestion, forage intake, and nutrient flow characteristics in steers consuming dominant bluestem-range forage. *J. Anim. Sci.* 69:2624–2633.
- Haque, M. N., J. Guinard-Flament, P. Lamberton, C. Mustiere, and S. Lemosquet. 2015. Changes in mammary metabolism in response to the provision of an ideal amino acid profile at 2 levels of metabolizable protein supply in dairy cows: Consequences on efficiency. *J. Dairy Sci.* 98(6):3951–3968.
- Harstad, O. M., and E. Prestipkken. 2001. Rumen degradability and intestinal indigestibility of individual amino acids in corn gluten meal, canola meal and fish meal determined *in situ*. *Anim. Feed Sci. Technol.* 94(3–4): 127–135.
- Hart, F. J., and J. Leibholz. 1990. A note on the flow of endogenous protein to the omasum and abomasum of steers. *Anim. Prod.* 51:217–219.
- Hespehl, R. B., and M. P. Bryant. 1979. Efficiency of rumen microbial growth: Influence of some theoretical and experimental factors on Y_{ATP} . *J. Anim. Sci.* 49(6): 1640–1659.
- Higgs, R. J., L. E. Chase, D. A. Ross, and M. E. Van Amburgh. 2015. Updating the Cornell Net Carbohydrate and Protein System feed library and analyzing model sensitivity to feed inputs. *J. Dairy Sci.* 98(9):6340–6360.
- Hou, Y., Y. Yin, and G. Wu. 2015. Dietary essentiality of “nutritionally non-essential amino acids” for animals and humans. *Exp. Biol. Med.* 240(8):997–1007.
- House, W. A., and A. W. Bell. 1993. Mineral accretion in the fetus and adnexa during late gestation in Holstein cows. *J. Dairy Sci.* 76(10):2999–3010.
- Hristov, A. N., M. Harper, J. Oh, F. Giallongo, J. C. Lopes, G. Cudoc, J. Clay, R. Ward, and L. E. Chase. 2018. Short communication: Variability in milk urea nitrogen and dairy total mixed ration composition in the northeastern United States. *J. Dairy Sci.* 101(2): 1579–1584.
- Hristov, A. N., A. Bannink, L. A. Crompton, P. Huhtanen, M. Kreuzer, M. McGee, P. Noziere, C. K. Reynolds, A. R. Bayat, D. R. Yanez-Ruiz, J. Dijkstra, E. Kebreab, A. Schwarm, K. J. Shingfield, and Z. Yu. 2019. Invited review: Nitrogen in ruminant nutrition: A review of measurement techniques. *J. Dairy Sci.* 102(7):5811–5852.
- Huang, X., K. A. Estes, P. S. Yoder, C. Wang, N. Jiang, T. Pilonero, and M. D. Hanigan. 2019. Assessing availability of amino acids from various feedstuffs in dairy cattle using a stable isotope-based approach. *J. Dairy Sci.* 102:10983–10996.
- Huhtanen, P., A. Vanhatalo, and T. Varvikko. 2002. Effects of abomasal infusions of histidine, glucose, and leucine on milk production and plasma metabolites of dairy cows fed grass silage diets. *J. Dairy Sci.* 85(1):204–216.
- Huhtanen, P., and A. N. Hristov. 2009. A meta-analysis of the effects of dietary protein concentration and degradability on milk protein yield and milk N efficiency in dairy cows. *J. Dairy Sci.* 92(7):3222–3232.
- Huhtanen, P., A. Bayat, S. J. Krizan, and A. Vanhatalo. 2014. Compartmental flux and *in situ* methods underestimate total feed nitrogen as judged by the omasal sampling method due to ignoring soluble feed nitrogen flow. *Br. J. Nutr.* 111(3):535–546.
- Illg, D. J., J. L. Sommerfeldt, and D. J. Schingoethe. 1987. Lactational and systemic responses to the supplementation of protected methionine in soybean meal diets. *J. Dairy Sci.* 70(3):620–629.
- INRA (Institut National de la Recherche Agronomique). 2018. Feeding System for Ruminants. Wageningen, The Netherlands: Wageningen Academic Publishers.
- Ipharraguerre, I. R., and J. H. Clark. 2005. Impacts of the source and amount of crude protein on the intestinal supply of nitrogen fractions and performance of dairy cows. *J. Dairy Sci.* 88:E22–E37.
- Ipharraguerre, I. R., and J. H. Clark. 2014. A meta-analysis of ruminal outflow of nitrogen fractions in dairy cows. *Adv. Dairy Res.* 2:122.
- Jansman, A. J. M., W. Smink, P. van Leeuwen, and M. Rademacher. 2002. Evaluation through literature data of the amount and amino acid composition of basal endogenous crude protein at the terminal ileum of pigs. *Anim. Feed Sci. Technol.* 98:49–60.
- Jarosz, L., T. Hvelplund, M. R. Weisbjerg, and B. B. Jensen. 1994. True digestibility of protein in the small intestine and the hind gut of cows measured with the mobile bag technique using 15-N-labelled roughage. *Acta Agric. Scand. Sect. A Anim.* 44(3): 146–151.
- Jensen, C., M. R. Weisbjerg, and T. Hvelplund. 2006. Evaluation of methods for estimating the amino acid supply to the duodenum of microbial, endogenous, and undegraded feed protein on maize silage diets fed to dairy cows. *Anim. Feed Sci. Technol.* 131:1–24.
- Ji, P., H. A. Tucker, R. E. Clark, M. Miura, and C. S. Ballard. 2016. Short communication: Effect of on-farm feeding practices on rumen protected lysine products. *J. Dairy Sci.* 99(2): 1242–1246.
- Jorgensen, G. N., and B. L. Larson. 1968. Conversion of phenylalanine to tyrosine in the bovine mammary secretory cell. *Biochim. Biophys. Acta Gen. Subjects* 165(1): 121–126.
- Jouany, J.-P. 1996. Effect of rumen protozoa on nitrogen utilization by ruminants. *J. Nutr.* 126:1335S–1346S.
- Kalscheur, K. F., R. L. Baldwin VI, B. P. Glenn, and R. A. Kohn. 2006. Milk production of dairy cows fed differing concentrations of rumen-degraded protein. *J. Dairy Sci.* 89:249–259.
- Karman, A. H., and A. J. S. van Boekel. 1986. Evaluation of the Kjeldahl factor for conversion of the nitrogen content of milk and milk products to protein content. *Net. Milk. Dairy J.* 40:315–336.
- Kamati, S. K. R., J. T. Sylvester, C. V. D. M. Ribeiro, L. E. Gilligan, and J. L. Firkins. 2009. Investigating unsaturated fat, monensin, or bromoethanesulfonate in continuous cultures retaining ruminal protozoa: I. Fermentation, biohydrogenation, and microbial protein synthesis. *J. Dairy Sci.* 92:3849–3860.
- King, K. J., W. G. Bergen, C. J. Sniffen, A. L. Grant, D. B. Grieve, V. L. King, and N. K. Ames. 1991. An assessment of absorbable lysine requirements in lactating cows. *J. Dairy Sci.* 74(8):2530–2539.
- Koenig, K. M., and L. M. Rode. 2001. Ruminant degradability, intestinal disappearance, and plasma methionine response of rumen-protected methionine in dairy cows. *J. Dairy Sci.* 84(6): 1480–1487.
- Koenig, K. M., L. M. Rode, C. D. Knight, and P. R. McCullough. 1999. Ruminant escape, gastrointestinal absorption, and response of serum methionine to supplementation of liquid methionine hydroxy analog in dairy cows. *J. Dairy Sci.* 82(2):355–361.
- Koong, L. J., W. N. Garrett, and P. V. Rattray. 1975. A description of the dynamics of fetal growth in sheep. *J. Anim. Sci.* 41(4): 1065–1068.
- Korhonen, M., A. Vanhatalo, T. Varvikko, and P. Huhtanen. 2000. Responses to graded post-ruminal doses of histidine in dairy cows fed grass silage diets. *J. Dairy Sci.* 83(11):2596–2608.
- Krizsan, S. J., S. Ahvenjarvi, and P. Huhtanen. 2010. A meta-analysis of passage rate estimated by rumen evacuation with cattle and evaluation of passage rate prediction models. *J. Dairy Sci.* 93(12):5890–5901.
- Lapierre, H., and G. E. Lobley. 2001. Nitrogen recycling in the ruminant: a review. *J. Dairy Sci.* 84 (E. Suppl.):E223–E236.
- Lapierre, H., J. P. Blouin, J. F. Bernier, C. K. Reynolds, P. Dubreuil, and G. E. Lobley. 2002. Effect of supply of metabolizable protein on whole body and splanchnic leucine metabolism in lactating dairy cows. *J. Dairy Sci.* 85(10):2631–2641.
- Lapierre, H., D. Pacheco, R. Berthiaume, D. R. Ouellet, C. G. Schwab, P. Dubreuil, G. Holtrop, and G. E. Lobley. 2006. What is the true supply of amino acids for a dairy cow? *J. Dairy Sci.* 89(Suppl. 1):E1–E14.
- Lapierre, H., G. E. Lobley, D. R. Ouellet, L. Doepel, and D. Pacheco. 2007a. Amino acid requirements for lactating dairy cows: Reconciling predictive models and biology. Pp. 39–59 in *Proceedings of the Cornell Nutrition Conference for Feed Manufacturers*. Ithaca, NY: Department of Animal Science, Cornell University.
- Lapierre, H., M. Vazquez-Anon, D. Parker, P. Dubreuil, and G. E. Lobley. 2007b. Short communication: Absorption of 2-hydroxy-4-methylthiobutanoate in dairy cows. *J. Dairy Sci.* 90(6):2937–2940.

- Lapierre, H., L. Doepel, E. Milne, and G. E. Lobley. 2009. Responses in mammary and splanchnic metabolism to altered lysine supply in dairy cows. *Animal* 3:360-371.
- Lapierre, H., M. Vazquez-Anon, D. Parker, P. Dubreuil, G. Holtrop, and G. E. Lobley. 2011. Metabolism of 2-hydroxy-4-(methylthio)butanoate (HMTBA) in lactating dairy cows. *J. Dairy Sci.* 94:1526-1535.
- Lapierre, H., G. E. Lobley, L. Doepel, G. Raggio, H. Rulquin, and S. Lemosquet. 2012a. Mammary metabolism of amino acids in dairy. *J. Anim. Sci.* 90:1708-1721.
- Lapierre, H., G. Holtrop, G. Calder, J. Renaud, and G. E. Lobley. 2012b. Is D-methionine bio-available to the dairy cow? *J. Dairy Sci.* 95:353-362.
- Lapierre, H., S. Binggeli, M. Sok, D. Pellerin, and D. R. Ouellet. 2019. Estimation of correction factors to determine the true amino acid concentration of protein after 24-h hydrolysis. *J. Dairy Sci.* 102:1205-1212.
- Lapierre, H., R. Martineau, M. D. Hanigan, J. W. Spek, and D. R. Ouellet. 2020. Review: Impact of protein and energy supply on the fate of amino acids from absorption to milk protein in dairy cows. *Animal* 14(Suppl. 1):S87—S102.
- Larsen, M., T. G. Madsen, M. R. Weisbjerg, T. Hvelplund, and J. Madsen. 2001. Small intestinal digestibility of microbial and endogenous amino acids in dairy cows. *J. Anim. Physiol. Anim. Nutr.* 85(1—2):9—21.
- Larsen, M., C. Galindo, D. R. Ouellet, G. Maxin, N. B. Kristensen, and H. Lapierre. 2015. Abomasal amino acid infusion in postpartum dairy cows: Effect on whole-body, splanchnic, and mammary amino acid metabolism. *J. Dairy Sci.* 98(11):7944-7961.
- Le Blanc, S. J. 2014. Reproductive tract inflammatory disease in postpartum dairy cows. *Animal* 8:54-63.
- Leal Yepes, F. A., S. Mann, T. R. Overton, C. M. Ryan, L. S. Bristol, G. E. Granados, D. V. Nydam, and J. J. Wakshlag. 2019. Effect of rumen-protected branched-chain amino acid supplementation on production- and energy-related metabolites during the first 35 days in milk in Holstein dairy cows. *J. Dairy Sci.* 102(6):5657-5672.
- Lean, I. J., R. Celi, H. Raadsma, J. McNamara, and A. R. Rabiee. 2012. Effects of dietary crude protein on fertility: Meta-analysis and meta-regression. *Anim. Feed Sci. Technol.* 171(1):31—42.
- Lee, C., A. N. Hristov, T. W. Cassidy, K. S. Heyler, H. Lapierre, G. A. Varga, M. J. de Veth, R. A. Patton, and C. Parys. 2012a. Rumen-protected lysine, methionine, and histidine increase milk protein yield in dairy cows fed a metabolizable protein-deficient diet. *J. Dairy Sci.* 95(10):6042-6056.
- Lee, C., A. N. Hristov, C. J. Dell, G. W. Feyereisen, J. Kaye, and D. Beegle. 2012b. Effect of dietary protein concentration on ammonia and greenhouse gas emitting potential of dairy manure. *J. Dairy Sci.* 95(41):1930-1941.
- Lee, C., A. N. Hristov, K. S. Heyler, T. W. Cassidy, H. Lapierre, G. A. Varga, and C. Parys. 2012c. Effects of metabolizable protein supply and amino acid supplementation on nitrogen utilization, milk production, and ammonia emissions from manure in dairy cows. *J. Dairy Sci.* 95(9):5253-5268.
- Lee, C., F. Giallongo, A. N. Hristov, H. Lapierre, T. W. Cassidy, K. S. Heyler, G. A. Varga, and C. Parys. 2015. Effect of dietary protein level and rumen-protected amino acid supplementation on amino acid utilization for milk protein in lactating dairy cows. *J. Dairy Sci.* 98(31):1885-1902.
- Lemosquet, S., G. E. Lobley, R. Koopman, L. J. C. van Loon, A. K. Kies, and H. Lapierre. 2010. A large supply of phenylalanine is not oxidised by the mammary gland of dairy cows. Pp. 139-140 in *Energy and Protein Metabolism and Nutrition*, C. M. Crovetto, ed. EAAP Pub. No. 127. Wageningen, The Netherlands: Wageningen Academic Publishers.
- Lewis, A. I., and D. H. Baker. 1995. Bioavailability of D-amino acids and DL-hydroxy-methionine. Pp. 67-81 in *Bioavailability of Nutrients for Animal: Amino Acids, Minerals, and Vitamins*, 1st ed., C. Ammerman, D. Baker, and A. Lewis, eds. San Diego, CA: Academic Press.
- Lewis, T. R., and R. S. Emery. 1962. Relative deamination rates of amino acids by rumen microorganisms. *Mich. Agric. Exp. Station* 2945:765-768.
- Li, M. M., E. C. Titgemeyer, and M. D. Hanigan. 2019. A revised representation of urea and ammonia nitrogen recycling and use in the Molly cow model. *J. Dairy Sci.* 102(6):5109-5129.
- Liebe, D. M., J. L. Firkins, H. Tran, P. J. Kononoff, and R. R. White. 2018. Technical note: Methodological and feed factors affecting measurement of protein A, B, and C fractions, degradation rate, and intestinal digestibility of rumen-undegraded protein. *J. Dairy Sci.* 101(9):8046-8053.
- Lintzenich, B. A., E. S. Vanzant, R. C. Cochran, J. L. Beaty, R. T. J. Brandt, and G. St-Jean. 1995. Influence of processing supplemental alfalfa on intake and digestion of dormant bluestem-range forage by steers. *J. Anim. Sci.* 83:1187-1195.
- Lobley, G. E., T. J. Wester, G. Holtrop, J. J. Dibner, D. S. Parker, and M. Vazquez-Anon. 2006. Absorption and digestive tract metabolism of 2-hydroxy-4-methylthiobutanoic acid in lambs. *J. Dairy Sci.* 89(9):3508-3521.
- Loerch, S. G., and B. O. Oke. 1989. Rumen protected amino acids in ruminant nutrition. Pp. 187-200 in *Absorption and Utilization of Amino Acids*, Vol. III, M. Friedman, ed. Boca Raton, FL: CRC Press.
- Lu, Z., F. Stumpff, C. Deiner, J. Rosendahl, H. Braun, K. Abdoun, J. R. Aschenbach, and H. Martens. 2014. Modulation of sheep ruminal urea transport by ammonia and pH. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 307(5):R558-R570.
- Lucy, M. C., S. T. Butler, and H. A. Garverick. 2014. Endocrine and metabolic mechanisms linking postpartum glucose with early embryonic and foetal development in dairy cows. *Animal* 8(Suppl. 1):82—90.
- Luo, C., S. Zhao, M. Zhang, Y. Gao, S. Wang, M. D. Hanigan, and N. Zheng. 2018. SESN2 negatively regulates cell proliferation and casein synthesis by inhibition the amino acid-mediated mTORC1 pathway in cow mammary epithelial cells. *Sci. Rep.* 8(1):3912.
- MacRae, J. C., L. A. Bruce, D. S. Brown, and A. G. Calder. 1997. Amino acid use by the gastrointestinal tract of sheep given lucerne forage. *Am. J. Physiol. Gastr. Liver Physiol.* 273:G1200-G1207.
- Maczulak, A. E., B. A. Dehority, and D. L. Palmquist. 1981. Effects of long-chain fatty acids on growth of rumen bacteria. *Appl. Environ. Microbiol.* 42:856-862.
- Manjarin, R., B. J. Bequette, G. Wu, and N. L. Trottier. 2014. Linking our understanding of mammary gland metabolism to amino acid nutrition. *Amino Acids* 46:2447-2462.
- Mansilla, W. D., J. K. Htoo, and C. F. M. de Lange. 2017. Nitrogen from ammonia is as efficient as that from free amino acids or protein for improving growth performance of pigs fed diets deficient in nonessential amino acid nitrogen. *J. Anim. Sci.* 95:3093-3102.
- Marini, J. C., D. G. Fox, and M. R. Murphy. 2008. Nitrogen transactions along the gastrointestinal tract of cattle: A meta-analytical approach. *J. Anim. Sci.* 86:660-679.
- Martin, C., L. Bernard, and B. Michalet-Doreau. 1996. Influence of sampling time and diet on amino acid composition of protozoal and bacterial fractions from bovine ruminal contents. *J. Anim. Sci.* 74:1157-1163.
- Martineau, R., C. Cortes, I. Ortigues-Marty, D. R. Ouellet, and H. Lapierre. 2014. Evaluation of equations predicting the net portal appearance of amino acid nitrogen in ruminants. *J. Dairy Sci.* 97(3): 1670-1681.
- Martineau, R., D. R. Ouellet, E. Kebreab, and H. Lapierre. 2016. Casein infusion rate influences feed intake differently depending on metabolizable protein balance in dairy cows: A multilevel meta-analysis. *J. Dairy Sci.* 99(4):2748-2761.
- Maxin, G., D. R. Ouellet, and H. Lapierre. 2013. Effect of substitution of soybean meal by canola meal or distillers grains in dairy rations on amino acid and glucose availability. *J. Dairy Sci.* 96:7806-7817.
- McAuan, A. B. 1982. The fate of nucleic acid in ruminants. *Proc. Nutr. Soc.* 41:309-317.

- McCollum, M. Q., M. Vazquez-Anon, J. J. Dibner, and K. E. Webb, Jr. 2000. Absorption of 2-hydroxy-4-(methylthio)butanoic acid by isolated sheep ruminal and omasal epithelia. *J. Anim. Sci.* 78(4): 1078-1083.
- Meijer, G. A. L., J. Van Der Meulen, J. G. M. Bakker, C. J. Van Der Koelen, and A. M. Van Vuuren. 1995. Free amino acids in plasma and muscle of high yielding dairy cows in early lactation. *J. Dairy Sci.* 78(5): 1131-1141.
- Mephram, T. B. 1982. Amino acid utilization by lactating mammary gland. *J. Dairy Sci.* 65:287-298.
- Merchen, N. R., and E. C. Titgemeyer. 1992. Manipulation of amino acid supply to the growing ruminant. *J. Anim. Sci.* 70:3238-3247.
- Metcalf, J. A., L. A. Crompton, D. Wray-Cahen, M. A. Lomax, J. D. Sutton, D. E. Beever, J. C. MacRae, B. J. Bequette, F. R. C. Backwell, and G. E. Lobley. 1996. Responses in milk constituents to intravascular administration of two mixtures of amino acids to dairy cows. *J. Dairy Sci.* 79(8): 1425-1429.
- Miron, J., E. Yosef, D. Ben-Ghedalia, L. E. Chase, D. E. Bauman, and R. Solomon. 2002. Digestibility by dairy cows of monosaccharide constituents in total mixed rations containing citrus pulp. *J. Dairy Sci.* 85(1):89-94.
- Moraes, L. E., E. Kebreab, J. L. Firkins, R. R. White, R. Martineau, and H. Lapierre. 2018a. Predicting milk protein responses and the requirement of metabolizable protein by lactating dairy cows. *J. Dairy Sci.* 101(0):310-327.
- Moraes, L. E., R. R. White, and J. L. Firkins. 2018b. A new model to predict microbial protein synthesis in the rumen. *J. Dairy Sci.* 100(Suppl. 2):410 (Abstr.).
- Mupeta, B., M. R. Weisbjerg, T. Hvelplund, and J. Madsen. 1997. Digestibility of amino acids in protein rich tropical feeds for ruminants estimated with the mobile bag technique. *Anim. Feed Sci. Technol.* 69(1-3):271-280.
- Murphy, J. J., and J. J. Kennelly. 1987. Effect of protein concentration and protein source on the degradability of dry matter and protein in situ. *J. Dairy Sci.* 70(9): 1841-1849.
- Mutsaers, T., K. L. Davies, J. J. McKinnon, and D. A. Christensen. 2016. Effects of dietary crude protein and rumen-degradable protein concentrations on urea recycling, nitrogen balance, omasal nutrient flow, and milk production in dairy cows. *J. Dairy Sci.* 99(8):6298-6310.
- Nagaraja, T. G., C. J. Newbold, C. J. Van Nevel, and D. I. Demeyer. 1997. Manipulation of rumen fermentation. Pp. 523-632 in *The Rumen Microbial Ecosystem*, P. N. Hobson and C. S. Stewart, eds. London, UK: Chapman and Hall.
- NASEM (National Academies of Sciences, Engineering, and Medicine). 2016. *Nutrient Requirements of Beef Cattle*. 8th rev. ed. Washington, DC: The National Academies Press.
- Nichols, K., J. J. M. Kim, M. Carson, J. A. Metcalf, J. P. Cant, and J. Doelman. 2016. Glucose supplementation stimulates peripheral branched-chain amino acid catabolism in lactating dairy cows during essential amino acid infusions. *J. Dairy Sci.* 99(2): 1145-1160.
- Noftsker, S., N. R. St-Pierre, and J. T. Sylvester. 2005. Determination of rumen degradability and ruminal effects of three sources of methionine in lactating cows. *J. Dairy Sci.* 88(1):223-237.
- NorFor. 2011. *The Nordic Feed Evaluation System*. EAAP Publication No. 130. Wageningen, The Netherlands: Wageningen Academic Publishers.
- NRC (National Research Council). 1985. *Ruminant Nitrogen Usage*. Washington, DC: National Academy Press.
- NRC. 1989. *Nutrient Requirements of Dairy Cattle*. 6th rev. ed. Washington, DC: National Academy Press.
- NRC. 2001. *Nutrient Requirements of Dairy Cattle*. 7th rev. ed. Washington, DC: National Academy Press.
- NRC. 2003. *Air Emissions from Animal Feeding Operations: Current Knowledge, Future Needs*. Washington, DC: The National Academies Press.
- Oba, M. 2011. Review: Effects of feeding sugars on productivity of lactating dairy cows. *Can. J. Anim. Sci.* 91:37-16.
- Oldick, B. S., J. L. Firkins, and N. R. St-Pierre. 1999. Estimation of microbial nitrogen flow to the duodenum of cattle based on dry matter intake and diet composition. *J. Dairy Sci.* 82:1497-1511.
- Orskov, E. R., N. A. McLeod, and D. J. Kyle. 1986. Flow of nitrogen from the rumen and abomasum in cattle and sheep given protein-free nutrients by intragastric infusion. *Br. J. Nutr.* 56:241-248.
- Osorio, J. S., P. Ji, J. K. Drackley, D. Luchini, and J. J. Loo. 2013. Supplemental Smartamine M or Metasmar during the transition period benefits postpartal cow performance and blood neutrophil function. *J. Dairy Sci.* 96:6248-6263.
- Ouellet, D. R., M. Demers, G. Zuur, G. E. Lobley, J. R. Seoane, J. V. Nolan, and H. Lapierre. 2002. Effect of dietary fiber on endogenous nitrogen flows in lactating dairy cows. *J. Dairy Sci.* 85:3013-3025.
- Ouellet, D. R., D. Valkeners, G. Holtrop, G. E. Lobley, and H. Lapierre. 2007. Contribution of endogenous secretions and urea recycling to nitrogen metabolism. Pp. 1-24 in *Proceedings of the Cornell Nutrition Conference for Feed Manufacturers*. Ithaca, NY: Department of Animal Science, Cornell University.
- Ouellet, D. R., R. Berthiaume, G. Holtrop, G. E. Lobley, R. Martineau, and H. Lapierre. 2010. Effect of method of conservation of timothy on endogenous nitrogen flows in lactating dairy cows. *J. Dairy Sci.* 93(9):4252-4261.
- Overton, T. R., D. W. LaCount, T. M. Cicela, and J. H. Clark. 1996. Evaluation of a ruminally protected methionine product for lactating dairy cows. *J. Dairy Sci.* 79(4):631-638.
- Pacheco, D., R. A. Patton, C. Parys, and H. Lapierre. 2012. Ability of commercially available dairy ration programs to predict duodenal flows of protein and essential amino acids in dairy cows. *J. Dairy Sci.* 95(2):937-963.
- Patra, A. K., and J. R. Aschenbach. 2018. Ureases in the gastrointestinal tracts of ruminant and monogastric animals and their implication in urea-N/ammonia metabolism: A review. *J. Adv. Res.* 13:39-50.
- Patterson, J. A., and L. J. Kung. 1988. Metabolism of DL-methionine and methionine analogs by rumen microorganisms. *J. Dairy Sci.* 71:3292-3301.
- Patton, R. A. 2010. Effect of rumen-protected methionine on feed intake, milk production, true milk protein concentration, and true milk protein yield, and the factors that influence these effects: A meta-analysis. *J. Dairy Sci.* 93(5):2105-2118.
- Patton, R. A., A. N. Hristov, and H. Lapierre. 2014. Protein feeding and balancing for amino acids in lactating dairy cattle. *Vet. Clin. North Am. Food Anim. Pract.* 30(3):599-621.
- Patton, R. A., A. N. Hristov, C. Parys, and H. Lapierre. 2015. Relationships between circulating plasma concentrations and duodenal flows of essential amino acids in lactating dairy cows. *J. Dairy Sci.* 98(7):4707-4734.
- Paz, H. A., and P. J. Kononoff. 2014. Lactation responses and amino acid utilization of dairy cows fed low-fat distillers dried grains with solubles with or without rumen-protected lysine supplementation. *J. Dairy Sci.* 97(10):6519-6530.
- Paz, H. A., T. J. Klopfenstein, D. Hostetler, S. C. Fernando, E. Castillo-Lopez, and P. J. Kononoff. 2014. Ruminal degradation and intestinal digestibility of protein and amino acids in high-protein feedstuffs commonly used in dairy diets. *J. Dairy Sci.* 97(10):6485-6498.
- Piepenbrink, M. S., and D. J. Schingoethe. 1998. Ruminal degradation, amino acid composition, and estimated intestinal digestibilities of four protein supplements. *J. Dairy Sci.* 81(2):454-461.
- Polan, C. E., P. T. Chandler, and C. N. Miller. 1970. Methionine hydroxy analog. Varying levels for lactating cows. *J. Dairy Sci.* 53:607-610.
- Pruekvimolphan, S., and R. R. Grummer. 2001. Lactation responses to sulfur-containing amino acids from feather meal or rumen-protected methionine. *J. Dairy Sci.* 84(11):2515-2522.
- R Core Team. 2013. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing.
- Rae, R. C., and J. R. Ingalls. 1984. Lactational response of dairy cows to oral administration of L-tyrosine. *J. Dairy Sci.* 67(7): 1430-1438.
- Rae, R. C., J. R. Ingall, and J. A. McKirdy. 1983. Effect of dietary crude protein level and formaldehyde treated canola meal on milk produc-

- tion and nitrogen utilization during early lactation. *Can. J. Anim. Sci.* 63:905-915.
- Recktenwald, E. B., D. A. Ross, S. W. Fessenden, C. J. Wall, and M. E. Van Amburgh. 2014. Urea-N recycling in lactating dairy cows fed diets with 2 different levels of dietary crude protein and starch with or without monensin. *J. Dairy Sci.* 97:1611-1622.
- Reed, K. F., H. C. Bonfa, J. Dijkstra, D. P. Casper, and E. Kebreab. 2017. Estimating the energetic cost of feeding excess dietary nitrogen to dairy cows. *J. Dairy Sci.* 100(9):7116-7126.
- Reynal, S. M., and G. A. Broderick. 2003. Effects of feeding dairy cows protein supplements of varying ruminal degradability. *J. Dairy Sci.* 86(3):835-843.
- Reynal, S. M., and G. A. Broderick. 2005. Effect of dietary level of rumen-degraded protein on production and nitrogen metabolism in lactating dairy cows. *J. Dairy Sci.* 88(11):4045-4064.
- Reynolds, C. K. 2005. Nitrogen metabolism by splanchnic tissues of ruminants. Pp. 197-218 in *Biology of Growing Animals*, Vol. 3, D. G. Burrin and H. J. Mersmann, eds. Oxford, UK: Elsevier.
- Reynolds, C. K., and N. B. Kristensen. 2008. Nitrogen recycling through the gut and the nitrogen economy of ruminants: An asynchronous symbiosis. *J. Anim. Sci.* 86(E. Suppl.):E293-E305.
- Rfus, A. G. 2019. Invited review: Adaptations of protein and amino acid metabolism to heat stress in dairy cows and other livestock species. *Appl. Anim. Sci.* 35(1):39-48.
- Robinson, P. H. 2010. Impacts of manipulating ration metabolizable lysine and methionine levels on the performance of lactating dairy cows: A systematic review of the literature. *Livestock Sci.* 127(2-3): 115-126.
- Roets, E., R. Verbeke, G. Peeters, H. Axmann, and G. Proksch. 1979. Metabolism of ornithine in perfused goat udder. *J. Dairy Sci.* 62:259-269.
- Roman-Garcia, Y., R. R. White, and J. L. Firkins. 2016. Meta-analysis of post-ruminal microbial nitrogen flows in dairy cattle: I. Derivation of equations. *J. Dairy Sci.* 99:7918-7931.
- Rubert-Aleman, J., G. Rychen, F. Casseron, F. Laurent, and G. J. Martin. 1999. ¹⁵N enrichment of casein amino acids in the milk from goats given a single intravenous dose of L-[¹⁵N] leucine. *J. Dairy Res.* 66:283-288.
- Ruiz, R., L. O. Tedeschi, J. C. Marini, D. G. Fox, A. N. Pell, G. Jarvis, and J. B. Russell. 2002. The effect of a ruminal nitrogen (N) deficiency in dairy cows: Evaluation of the Cornell Net Carbohydrate and Protein System Ruminal N Deficiency Adjustment. *J. Dairy Sci.* 85(11):2986-2999.
- Rulquin, H., and J. Kowalczyk. 2003. Development of a method for measuring lysine and methionine bioavailability in rumen-protected products for cattle. *J. Anim. Feed Sci.* 12(3):465-474.
- Rulquin, H., J. Guinard, and R. Verite. 1998. Variation of amino acid content in the small intestine digesta of cattle: Development of a prediction model. *Livestock Prod. Sci.* 53(1): 1-13.
- Russell, J. B., J. D. O'Connor, D. G. Fox, P. J. Van Soest, and C. J. Sniffen. 1992. A net carbohydrate and protein system for evaluating cattle diets: I. Ruminal fermentation. *J. Anim. Sci.* 70:3551-3561.
- Sadri, H., F. Giallongo, A. N. Hristov, J. Werner, C. H. Lang, C. Parys, B. Saremi, and H. Sauerwein. 2016. Effects of slow-release urea and rumen-protected methionine and histidine on mammalian target of rapamycin (mTOR) signaling and ubiquitin proteasome-related gene expression in skeletal muscle of dairy cows. *J. Dairy Sci.* 99(81):6702-6713.
- Salter, D. N., and R. H. Smith. 1984. Protein utilization in the young steer: Digestion and nitrogen retention of ¹⁵-labelled rumen bacterial protein. *Br. J. Nutr.* 51:531-539.
- Sandek, A., K. Krawielitzki, J. Kowalczyk, F. Kreienbring, U. Schoenhusen, M. Gabel, T. Zebrowska, H. Hagemeister, and J. Voigt. 2001. Studies on N-metabolism in different gastrointestinal sections of sheep using the digesta exchange technique. 2. Passage of endogenous nitrogen. *J. Anim. Feed Sci.* 10:605-618.
- Santos, F. A. P., J. E. P. Santos, C. B. Theurer, and J. T. Huber. 1998. Effects of rumen-undegradable protein on dairy cow performance: A 12-year literature review. *J. Dairy Sci.* 81:3182-3213.
- Sauvant, D., and P. Nozière. 2016. Quantification of the main digestive processes in ruminants: the equations involved in the renewed energy and protein feed evaluation systems. *Animal* 10(5):755-770.
- Sauvant, D., G. Catalapiedra-Hijar, L. Delaby, J.-B. Daniel, P. Faverdin, and P. Nozière. 2015. Actualisation des besoins protéiques des ruminants et détermination des réponses des femelles laitières aux apports de protéines digestibles dans l'intestin. *INRA Prod. Anim.* 28:347-368.
- Schmidely, P. F., Glasser, M., Doreau, and D. Sauvant. 2008. Digestion of fatty acids in ruminants: A meta-analysis of flows and variation factors: I. Total fatty acids. *Animal* 2(5):677-690.
- Schwab, C. G. 1995. Protected proteins and amino acids for ruminants. *Biotechnol. Anim. Feeds Anim. Feed.* Jun 29:141.
- Schwab, C. G., and G. A. Broderick. 2017. A 100-year review: Protein and amino acid nutrition in dairy cows. *J. Dairy Sci.* 100(12):10094-10112.
- Schwab, C. G., L. D. Satter, and B. Clay. 1976. Response to lactating dairy cows to abomasal infusion of amino acids. *J. Dairy Sci.* 59(71):1254-1270.
- Seo, S., L. O. Tedeschi, C. G. Schwab, B. D. Garthwaite, and D. G. Fox. 2006. Evaluation of the passage rate equations in the 2001 dairy NRC model. *J. Dairy Sci.* 89:2327-2342.
- Simon, R. 2007. Resampling strategies for model assessment and selection. Pp. 173-186 in *Fundamentals of Data Mining in Genomics and Proteomics*, W. Dubitzky, M. Granzow, and D. P. Berrar, eds. New York: Springer.
- Sinclair, K. D., P. C. Gamsworthy, G. E. Mann, and L. A. Sinclair. 2014. Reducing dietary protein in dairy cow diets: Implications for nitrogen utilization, milk production, welfare and fertility. *Animal* 8(2):262-274.
- Sok, M., D. R. Ouellet, J. L. Firkins, D. Pellerin, and H. Lapierre. 2017. Amino acid composition of rumen bacteria and protozoa in cattle. *J. Dairy Sci.* 100:5241-5249.
- Sordillo, L. M. 2016. Nutritional strategies to optimize dairy cattle immunity. *J. Dairy Sci.* 99:4967-4982.
- Spek, J. W., A. Bannink, G. Gort, W. H. Hendriks, and J. Dijkstra. 2013. Interaction between dietary content of protein and sodium chloride on milk urea concentration, urinary urea excretion, renal recycling of urea, and urea transfer to the gastrointestinal tract in dairy cows. *J. Dairy Sci.* 96(9):5734-5745.
- Stem, M. D., A. Bach, and S. Calsamiglia. 1997. Alternative techniques for measuring nutrient digestion in ruminants. *J. Anim. Sci.* 75:2256-2276.
- Storm, E., E. R. Orskov, and R. Smart. 1983. The nutritive value of rumen microorganisms in ruminants. *Br. J. Nutr.* 50(2):471-478.
- St-Pierre, N. R. 2003. Reassessment of biases in predicted nitrogen flows to the duodenum by NRC 2001. *J. Dairy Sci.* 86:344-350.
- Susmel, R., E. Plazzotta, C. R. Mills, and B. Stefanon. 1993. Determination of RNA and ATP in the rumen of cows fed with diets differing in forage to concentrate ratio. *J. Sci. Food Agr.* 63:39-45.
- Swaigood, H. E. 1995. Protein and amino acid composition of bovine milk. Pp. 464-472 in *Handbook of Milk Composition*, R. G. Jensen, ed. London, UK: Academic Press.
- Swanson, E. W. 1977. Factors for computing requirements of protein for maintenance of cattle. *J. Dairy Sci.* 60:1583-1593.
- Swanson, E. W. 1982. Estimation of metabolic protein requirements to cover unavoidable losses of endogenous nitrogen in maintenance of cattle. Pp. 183-197 in *Protein Requirements for Cattle: Symposium*, F. N. Owens, ed. Stillwater: Oklahoma State University.
- Sylvester, J. T., S. K. R. Kamati, Z. Yu, C. J. Newbold, and J. L. Firkins. 2005. Evaluation of a real-time PCR assay for measuring the ruminal pool and duodenal flow of protozoal nitrogen. *J. Dairy Sci.* 88:2083-2095.
- Syvaöja, E.-L., and A. I. Virtanen. 1965. Fractionation of milk proteins, especially of milk produced on protein-free feed with urea and ammonium salt as the sole source of nitrogen. *Enzymologia* 29:205-219.
- Tas, M. V., R. A. Evans, and R. F. E. Axford. 1981. The digestibility of amino acids in the small intestine of the sheep. *Br. J. Nutr.* 45(1):167-174.
- Thatcher, W., J. E. P. Santos, and C. R. Staples. 2011. Dietary manipulations to improve embryonic survival in cattle. *Theriogenology* 76(9):1619-1631.

- Thomley, J. H. M., and J. France. 2007. *Mathematical Models in Agriculture: Quantitative Methods for the Plant, Animal and Ecological Sciences*. 2nd ed. Wallingford, UK: CABI Publishing.
- Titgemeyer, E. C., and N. R. Merchen. 1990. Sulfur-containing amino acid requirement of rapidly growing steers. *J. Anim. Sci.* 68(7):2075-2083.
- Tyrrell, H. F., P. W. Moe, and W. P. Flatt. 1970. Influence of excess protein intake on energy metabolism of the dairy cow. *Eur. Assoc. Anim. Prod.* 16:68-71.
- Valkeners, D., H. Lapierre, U. Schonhosen, P. Junghans, C. C. Metges, and D. R. Ouellet. 2008. Endogenous nitrogen (EN) flows: Effects of metabolizable protein (MP) supply in lactating dairy cows. *J. Dairy Sci.* 91 (E. Suppl. 1):485-486 (Abstr.).
- Van Amburgh, M. E., E. A. Collao-Saenz, R. J. Higgs, D. A. Ross, E. B. Recktenwald, E. Raffrenato, L. E. Chase, T. R. Overton, J. K. Mills, and A. Foskolos. 2015. The Cornell Net Carbohydrate and Protein System: Updates to the model and evaluation of version 6.5. *J. Dairy Sci.* 98(9):6361—6380.
- van Duinkerken, G., M. C. Blok, A. Bannink, J. W. Cone, J. Dijkstra, A. M. van Vuuren, and S. Tamminga. 2011. Update of the Dutch protein evaluation system for ruminants: The DVE/OEB2010 system. *J. Agric. Sci.* 149(3):351—367.
- Vanhatalo, A., P. Huhtanen, V. Toivonen, and T. Varvikko. 1999. Response of dairy cows fed grass silage diets to abomasal infusions of histidine alone or in combinations with methionine and lysine. *J. Dairy Sci.* 82(12):2674-2685.
- Velle, W., O. V. Sjaastad, A. Aulie, D. Gronset, K. Feigenwinter, and T. Framstad. 1997. Rumen escape and apparent degradation of amino acids after individual intaruminal administration to cows. *J. Dairy Sci.* 80:3325-3332.
- Vicini, J. L., J. H. Clark, W. L. Hurley, and J. M. Bahr. 1988. Effects of abomasal or intravenous administration of arginine on milk production, milk composition, and concentrations of somatotropin and insulin in plasma of dairy cows. *J. Dairy Sci.* 71 (3):658—665.
- Viechtbauer, W. 2010. Conducting meta-analyses in R with the metafor package. *J. Stat. Softw.* 36:1-48.
- Voelker, J. A., and M. S. Allen. 2003. Pelleted beet pulp substituted for high-moisture com: 3. Effects on ruminal fermentation, pH, and microbial protein efficiency in lactating dairy cows. *J. Dairy Sci.* 86:3562-3570.
- Volden, H., W. Velle, O. M. Harstad, A. Aulie, and O. V. Sjaastad. 1998. Apparent ruminal degradation and rumen escape of lysine, methionine, and threonine administered intraruminally in mixtures to high-yielding cows. *J. Anim. Sci.* 76:1232-1240.
- Volden, H., O. M. Harstad, and L. T. Mydland. 1999. Amino acid content and profile of protozoal and bacterial fractions isolated from ruminal contents of lactating dairy cows fed diets differing in nitrogen supplementation. *Acta Agric. Scand. Sect. A Anim. Sci.* 49:245-250.
- Volden, H., W. Velle, O. V. Sjaastad, A. Aulie, and O. M. Harstad. 2001. Concentrations and flow of free amino acids in ruminal and duodenal liquid in dairy cows in relation to feed composition, time of feeding and level of feed intake. *Acta Agric. Scand. Sect. A Anim. Sci.* 51:35-45.
- Vrese, M. d., R. Frik, N. Roos, and H. Hagemeister. 2000. Protein-bound D-amino acids, and to a lesser extent lysinoalanine, decrease true ileal protein digestibility in minipigs as determined with I5N-labeling. *J. Nutr.* 130(8):2026—2031.
- Vyas, D., and R. A. Erdman. 2009. Meta-analysis of milk protein yield responses to lysine and methionine supplementation. *J. Dairy Sci.* 92(10):5011-5018.
- Walhain, P., M. Foucart, and A. Thdwis. 1992. Influence of extrusion on ruminal and intestinal disappearance in sacco of pea (*Pisum sativum*) proteins and starch. *Anim. Feed Sci. Technol.* 38(1):43—55.
- Wanderley, R. C., J. T. Huber, Z. Wu, M. Pessarakli, and C. Fontes. 1993. Influence of microbial colonization of feed particles on determination of nitrogen degradability by in situ incubation. *J. Anim. Sci.* 71(11):3073—3077.

- Weekes, T. L., P. H. Luimes, and J. P. Cant. 2006. Responses to amino acid imbalances and deficiencies in lactating dairy cows. *J. Dairy Sci.* 89(6):2177—2187.
- Weiss, W. P., and N. R. St-Pierre. 2009. A method to quantify changes in supply of metabolizable methionine to dairy cows using concentrations of selenium in milk. *J. Dairy Sci.* 92:2835-2842.
- White, R. R., Y. Roman-Garcia, and J. L. Firkins. 2016. Meta-analysis of postprandial microbial nitrogen flows in dairy cattle: II. Approaches to and implications of more mechanistic prediction. *J. Dairy Sci.* 99:7932-7944.
- White, R. R., P. J. Kononoff, and J. L. Firkins. 2017a. Technical note. Methodological and feed factors affecting prediction of ruminal degradability and intestinal digestibility of essential amino acids. *J. Dairy Sci.* 100(3): 1946-1950.
- White, R. R., Y. Roman-Garcia, J. L. Firkins, P. J. Kononoff, M. J. VandeHaar, H. Tran, T. McGill, R. Gamett, and M. D. Hanigan. 2017b. Evaluation of the 2001 dairy NRC and derivation of new prediction equations: 2. Rumen degradable and undegradable protein. *J. Dairy Sci.* 100:3611-3627.
- Whitehouse, N. L. 2016. Using plasma free amino acid dose response curve method to determine metabolizable protein concentrations of lysine and methionine in rumen protected supplements. PhD diss., University of New Hampshire, Durham, NH.
- Whitehouse, N. L., C. G. Schwab, and A. F. Brito. 2017. The plasma free amino acid dose-response technique: A proposed methodology for determining lysine relative bioavailability of rumen protected lysine supplements. *J. Dairy Sci.* 100(12):9585-9601.
- Whitelaw, F. G., J. S. Milne, E. R. Orskov, and J. S. Smith. 1986. The nitrogen and energy metabolism of lactating cows given abomasal infusions of casein. *Br. J. Nutr.* 55(3):537—556.
- Williams, A. G., and G. S. Coleman. 1992. *The Rumen Protozoa*. New York: Springer-Verlag.
- Williams, A. P. 1978. The amino acids, collagen and mineral composition of preruminant calves. *J. Agric. Sci. (Camb)*. 90:617-624.
- Wohlt, J. E., J. H. Clark, R. G. Derrig, and C. L. Davis. 1977. Valine, leucine and isoleucine metabolism by lactating bovine mammary tissue. *J. Dairy Sci.* 60:1875-1882.
- Wood, G. M., P. J. Boettcher, J. Jamrozik, G. B. Jansen, and D. F. Kelton. 2003. Estimation of genetic parameters for concentrations of milk urea nitrogen. *J. Dairy Sci.* 86(7):2462-2469.
- Wood, S. N., N. Pya, and B. Safken. 2016. Smoothing parameter and model selection for general smooth models. *J. Am. Stat. Assoc.* 111(516): 1548—1563.
- Woods, V. B., A. P. Moloney, and F. P. O'Mara. 2003. The nutritive value of concentrate feedstuffs for ruminant animals. Part II: In situ ruminal degradability of crude protein. *Anim. Feed Sci. Technol.* 110(1):131—143.
- Wray-Cahen, D., J. A. Metcalf, F. R. C. Backwell, B. J. Bequette, D. S. Brown, J. D. Sutton, and G. E. Lobley. 1997. Hepatic response to increased exogenous supply of plasma amino acids by infusion into the mesenteric vein of Holstein-Friesian cows in late gestation. *Br. J. Nutr.* 78:913-930.
- Wu, S. H. W., and A. Papas. 1997. Rumen-stable delivery systems. *Adv. Drug Deliv. Rev.* 28:323-334.
- Yang, W. Z., K. A. Beauchemin, and L. M. Rode. 1999. Effects of an enzyme feed additive on extent of digestion and milk production of lactating dairy cows. *J. Dairy Sci.* 82(2):391-403.
- Yoder, P. S., X. Huang, I. A. Teixeira, J. P. Cant, and M. D. Hanigan. 2020. Effects of jugular infused methionine, lysine, and histidine as a group or leucine and isoleucine as a group on production and metabolism in lactating dairy cows. *J. Dairy Sci.* 103(3):2387-2404.
- Zanton, G. I. 2016. Analysis of production responses to changing crude protein levels in lactating dairy cow diets when evaluated in continuous or change-over experimental designs. *J. Dairy Sci.* 99(6):4398—4410.
- Zanton, G. I., G. R. Bowman, M. Vdzquez-Añón, and L. M. Rode. 2014. Meta-analysis of lactation performance in dairy cows receiving supplemental dietary methionine sources or postprandial infusion of methionine. *J. Dairy Sci.* 97(11):7085-7101.